Semiparametric Continuous Difference-in-Differences with Causal Forests and an Application to U.S. Charter Schools*

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Abstract

We develop a difference-in-differences causal forest for estimating heterogeneous treatment effects in settings with continuous and/or time-varying treatment. We then apply this method to the case of U.S. charter schools. While prior research suggested charter school effectiveness was most closely associated with student demographics, we find that this mostly reflects low baseline achievement levels. Higher school funding is the only policy variable that seems to improve charter effectiveness, while two dozen other factors are unrelated to charter effects. Our results suggest that our data-driven methods better capture the many linear and nonlinear heterogeneity dimensions present in common policy settings.

Keywords: Causal forests; effect heterogeneity; charter schools; student achievement

JEL Classification: H75; I21; I28

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1 Introduction

Understanding treatment effect heterogeneity is important for understanding the mechanisms of treatment effects and for allocating interventions to those who are expected to benefit from them. The traditional approach to heterogeneity analysis involves interacting covariates within an OLS estimator of the average treatment effect (ATE or ATT). This approach has well-known weaknesses: the linear functional form may be misspecified, the researcher has to choose which interactions to include (so heterogeneity is only discovered where one decides to look), and estimating many specifications raises multiple hypothesis testing concerns. With enough data, researchers can avoid these problems by estimating semi- or non-parametric models, such as sieves/splines (e.g. Chen et al., 2025), kernel regression, or machine learning estimators like double/debiased machine learning (Chernozhukov et al., 2018) and causal forests (Wager and Athey, 2018; Athey et al., 2019, etc.), among others.

Until recently, the vast majority of the causal machine learning literature was designed only for effect heterogeneity cases where treatment assignment is conditionally random, e.g., randomized controlled trials. But, in most settings, such methods are infeasible and, instead, quasi-experimental methods (e.g. instrumental variables, regression discontinuity (RD), and difference-in-differences (DiD)) are required. A smaller recent wave of papers has begun to modify machine learning effect heterogeneity methods to these quasi-experimental settings. Athey et al. (2019), for example, introduce the "generalized random forest," a modified causal forest designed to function under any identifying local moment condition. This opened the door for causal forests to be further extended into the realm of RD (Reguly, 2021; Alcantara et al., 2025) and DiD (Gavrilova et al., 2025; Kattenberg et al., 2023).[1]

The existing causal forest extensions into DiD, however, are designed for simple, binary treatment settings. Many policy settings involve continuous treatment where units adopt at different times and the intensity of treatment may vary post-adoption. Standard causal forests (and their existing DiD extensions) estimate slope coefficients within covariate neighborhoods. In a continuous treatment DiD setting, these local slope coefficients resemble the TWFE coefficients that Callaway et al. (2025) show do not recover ATT-like parameters and generally do not recover interpretable marginal effect parameters.

In this study, we develop a semiparametric DiD estimator for continuous, staggered, and time-varying treatments that estimates heterogeneous treatment effects with causal forests (Athey et al., 2019). We begin by defining conditional-on-covariate versions of the Callaway et al. (2025) building blocks: the conditional average treatment effect on the treated ($CATT$), which is

[1] A few papers have considered merging Double-ML techniques with quasi-experimental methods as well. For example, Hatamyar et al. (2023) propose the 'MLDID,' Chang (2020) outlines a Double-ML DiD estimator, and Nie and Wager (2021) propose R-learners, among others.

the "level effect" or "dose-response function" of receiving a specific dose relative to receiving no dose, and the conditional average causal response to treatment ($CACRT$), which is the marginal effect of a small increase in that dose. We outline the assumptions necessary to identify these causal parameters in settings of staggered treatment adoption and/or time-varying treatment, which include parallel trends assumptions akin to covariate-conditional versions of Callaway et al. (2025)'s unconditional parallel trends assumption and unconditional strong parallel trends assumption.

Our estimator is a generalization of the Difference-in-Differences Causal Forest (DiDCF) proposed by Gavrilova et al. (2025). With a single treated group and a binary treatment, the DiDCF transforms each unit's outcome into a long difference relative to a base period and grows a separate causal forest in each period. Each forest therefore compares the change in outcomes of treated units to the change in outcomes of untreated units that look similar in terms of their covariates. The authors provide an extension (though they do not implement it empirically) which allows for staggered adoption following the logic of Callaway and Sant'Anna (2021). They break the problem down further into cohort-by-period comparisons: each cohort of first-treated units is compared only to never-treated units, using the change in outcomes since that cohort's pre-treatment base period.

We further generalize this estimator along two dimensions. First, in order to accommodate continuous treatment, we follow the "forward engineering" approach of Mogstad and Torgovitsky (2024); Callaway et al. (2025), where we explicitly specify a partially linear functional form of the $CATT$ and construct a conditional DiD estimator which directly targets it with a causal forest. We show that such an estimator behaves as a "local" (in the sense of covariates) analogue to the $ATT/ACRT$ estimators proposed by Callaway et al. (2025). Second, in order to allow for time-varying treatment intensity, we rely on the insights of De Chaisemartin and D'Haultfœuille (2026) and show how one may still obtain interpretable estimates of $CATT$ parameters with our estimator and, under an additional assumption of path independence, $CACRT$ parameters. Because each cohort-by-period comparison is a generalized random forest problem defined by a local moment condition, the estimates inherit the consistency and asymptotic normality results of Athey et al. (2019).

We then apply our estimator to U.S. charter schools. These schools are publicly governed and funded with over $30 billion annually, but operated by private organizations under contract with government agencies, and they currently enroll about seven percent of U.S. public school students. Research consistently shows that charter schools have positive effects on the achievement of students who attend them (e.g., Sass, 2006; Booker et al., 2007; Hoxby and Rockoff, 2004; Angrist et al., 2016) as well as high school graduation (Chen and Harris, 2023).[2] However, these

[2]See Cohodes and Roy (2025) for a review of lottery studies that include both test scores and high school grad-

charter school participant effects also vary across states (CREDO, 2023) and across cities (Angrist et al., 2016; Harris and Larsen, 2023).

This is a useful application of our DiDCF extension for several reasons. First, charter enrollment share is a continuous, time-varying, staggered treatment (i.e. a complex quasi-experimental design). Second, we can carry out the analysis on a national basis; more than a thousand school districts, across 47 states, have had at least one charter school over the past two decades and the share of students in a district who attend a charter changes annually. Third, the overall distribution of school-level charter performance is wider than that of traditional public schools, i.e. some charter schools perform worse than the lowest-performing traditional public schools (TPS) and, conversely, the best charter schools are better than the highest-performing TPS (Baude et al., 2020). Relatively little is known about the reasons behind this performance variation, and understanding these reasons is critical for determining where future charter schools should locate in order to improve student outcomes. Fourth, what we do know about the sources of this effect variation is difficult to interpret.

The most consistent finding is that charter schools have larger participant effects for disadvantaged students (i.e. racial/ethnic minority and low-income students) and in urban areas (Cohodes and Parham, 2021). Additional analysis in Massachusetts echoes this point and also attributes variation to certain instructional practices (Angrist et al., 2013), though that study is limited to participant effects and to a single state that has particularly effective urban charter schools. Their analysis also suggests that low baseline achievement might be associated with higher charter participant effects but only in that state's urban schools; the opposite pattern arises in non-urban charter schools. Like Angrist et al. (2013), we are able to study and help disentangle the roles of student demographics, urbanicity, and baseline achievement levels–all of which are highly correlated–and do so on a national basis and accounting for an additional charter effect mechanisms.

Our analysis is based on nationwide data from the National Longitudinal School Database (NLSD). Rather than participant effects, we follow Chen and Harris (2023) and estimate market-level effects: the impact on the enrollment-weighted average of TPS and charter school outcomes within a geographic school district. This allows us to incorporate the breadth of impacts additional charter schools may have on the academic environment of the district, including participant effects, competitive effects (spillovers onto TPS performance), closure effects, and match effects.[3]

uation. They report a mix of positive, null, and negative effects from lottery studies. However, this is a selected sub-sample of over-subscribed schools, whereas Chen and Harris (2023) find positive market-level effects when looking at the entire country.

[3]Closure effects arise when low-performing schools are pressed to close, which increases average school quality. Match effects arise if households have information about the types of schools that are the most beneficial for their children.

Our main outcomes are math and English language arts (ELA) test scores in grades 3–8 from the Stanford Education Data Archive (SEDA) for 2009–2018 (Ho, 2020). We also consider average freshman graduation rate (AFGR) from 1995–2010. We incorporate 26 covariates as potential heterogeneity dimensions in our analysis which fall into one of four broad categories: student demographics, market setting, school finance, and state charter policy.

A practical challenge with causal forests is that they output individualized treatment effect estimates, which can implicitly incorporate any number of nonlinearities. In our setting, the DiDCF produces a separate $CATT$ estimate for every treated district (or district-grade) and event time, and these estimates implicitly account for any number of nonlinearities and interactions among covariates. To address these complexities, we propose a framework for interpreting the results that can be used across CF settings. First, we aim to establish variable importance by computing the (depth-weighted) share of trees which split along a given covariate; known as the VIF (Variable Importance Factor). Second, we investigate which covariates exhibit signs of linear trends in their relationship to the $CATT$. Finally, we characterize the nonlinearity in $CATT$ estimates using (1) event studies segmented by covariate quartile, (2) the average profile of district groupings defined via k-means clustering, and (3) shallow depth decision trees in the policy learning framework of Kitagawa and Tetenov (2018) and Athey and Wager (2021) where we search for the optimal charter expansion policy over the set of simple decision rules via the construction of doubly-robust scores (Graham and de Xavier Pinto, 2022).

To illustrate how our flexible, semiparametric forest-based approach differs from a more traditional (though still Callaway et al. (2025) consistent) model, we compare the DiDCF results to its linear parametric counterpart, which we call the CGS-DiD.[4] The CGS-DiD uses the same cohort-by-period comparisons, the same never-treated comparison group, and the same identifying assumptions as our DiDCF, but specifies the $CATT$ as linear in the covariates. This estimator is numerically equivalent to a fully interacted first-differenced TWFE regression estimated within each cohort-by-period cell.

Consistent with Chen and Harris (2023), we find that average market-level effects are suggestively positive but small, based on the DiDCF ATT, which is the effect of charter entry at the charter intensity districts actually experienced relative to no charter schools. On average, treated districts experienced a charter intensity of 13.4% (median 6.9%). We estimate an ATT over the distribution of realized charter intensity of 0.0001 s.d. for math and 0.014 s.d. for ELA. ELA effects in particular seem to grow over time: the event-study $CATT$ reaches nearly 0.07 s.d. four years after initial entry. More to the point, these effects seem to mask substantial heterogeneity. The 5th – 95th percentile range of district $\times$ event-time $CATT$ estimates ranges from $[-0.29, 0.33]$

[4]Note that Callaway et al. (2025) do not necessarily advocate for a parametric model. They propose a parametric model among others. We use this acronym referring to that model.

for math and from $[-0.26, 0.30]$ for ELA, indicating roughly half of all ever-treated districts were *negatively* impacted by charter entry.

We find that the single strongest predictor of effect heterogeneity is baseline achievement. Baseline test scores are more than twice as important for predicting charter school impacts than the next highest variable (which is the share of students on free or reduced-price lunch). Districts with lower baseline achievement benefit more from charter entry, and this pattern is consistent across nearly all heterogeneity measures we consider. This suggests that part of what prior work has attributed to "student disadvantage" may be more accurately attributed to low performance, a variable highly correlated with disadvantage. Demographics still matter, but it appears that they are nonlinearly and/or interactively related to $CATT$ estimates.

The importance of baseline achievement is also consistent with some recent small-scale studies finding that variation in school participant effects are driven substantially by variation in quality of their counterfactuals (Bruhn et al., 2026), which studies do not normally document. Our analysis shows this is true for market-level effects, of which participant effects are only one part. Baseline achievement might also influence the various types of competitive effects, e.g., because students in low-performing schools are more eager to exit or because TPS see more obvious means of improvement when their performance is low to start with.

District resources (per-pupil spending, the student–teacher ratio, and teacher salaries) also matter, but less so than baseline performance and demographics and mainly through interactions. No other state charter policy variables appear to have a meaningful influence on the $CATT$. We also find evidence of diminishing marginal returns by estimating $ACRT(w)$ functions via the nonparametric approach of Chen et al. (2025). Specifically, the marginal effect of additional charter schools falls to zero at around 50–60 percent charter share.

When comparing the forest-based results to those of the linear model, the CGS-DiD corroborates the DiDCF that charter effectiveness is greater in lower performing districts. This is to be expected if the relationship between baseline test scores and the $CATT$ is fairly linear. However, one can see the strength of the flexible DiDCF when considering the other moderators which seem to have more nonlinear relationships with the $CATT$. For example, district enrollment size and percent White are positively associated with charter market effects in the CGS-DiD but negatively associated with charter market effects in our preferred DiDCF. Also, we see indications of a positive role for school spending in our DiDCF but a negative effect in the simpler CGS-DiD.

The paper proceeds as follows. Section 2 discusses our methodological extension of the DiDCF. Section 3 describes the data, with particular attention to the market-level structure of the analysis, and reviews the research motivating the moderator variables. Section 4 presents the results. Section 5 concludes.

2 Methodology

We will outline this section with respect to our charter school application, which studies the market-level heterogeneous effects of the introduction of charter schools on student outcomes: high school graduation rates and (primarily) test scores.

2.1 Notation and Definitions

In this section, we define our treatment effect parameters of interest and discuss identification. We observe school districts (or district-grades), indexed with $i \in \{1, ..., N\}$, over time periods $t \in \{1, ..., T\}$. Our outcomes of interest, $Y_{i,t} \in \mathbb{R}$, include average high school freshman graduation rate (AFGR), standardized elementary math test scores (Math), and standardized elementary English/language arts scores (ELA). Each district has a set of time-invariant observable characteristics/moderators $X_i \in \mathcal{X}$, where $\mathcal{X}$ denotes the support of X_i. Districts are observed with some continuous charter intensity, $W_{i,t} \in [0, 1]$. Some of the characteristics in X_i may be measured confounders of the relationship between $W_{i,t}$ and $Y_{i,t}$, and some may not confound but may nonetheless be related to heterogeneity.

In our setting, charter adoption is staggered over time. Each district belongs to a cohort of districts which are first treated with a positive charter intensity in the same calendar year, or to the group of districts who are never treated. We denote these timing cohorts by $G_i \in \mathcal{G}$, where $G_i = \infty$ for never-treated districts by convention. Also by convention, we exclude always-treated districts so that $\mathcal{G} \subseteq \{2, ..., T, \infty\}$, and $\bar{\mathcal{G}} = \mathcal{G} \setminus \{\infty\}$ is the set of timing cohorts among ever-treated districts. The dose that district i experiences in period t is therefore $W_{i,t} = W_i \cdot \mathbb{1}\{t \geq G_i\}$, where G_i is the first period in which the district is treated.

Denote $Y_{i,t}(g, w)$ as the potential outcome for district i in time period t if that district is first treated in cohort g under dose w. We write $Y_{i,t}(0) = Y_{i,t}(\infty, 0)$ as the untreated potential outcome. In each period, the observed outcome for district i is $Y_{i,t} = Y_{i,t}(G_i, W_i)$. We define the base period for cohort g as the period before first treatment, which we denote $b_g = g - 1$.

Our first causal parameter of interest is a conditional-on-covariates version of what Callaway et al. (2025) refer to as the "level treatment effect." This is defined as the difference in potential outcomes for district i in timing cohort g if that district experienced dose w vs. if that same district had experienced zero dose in period t: $Y_{i,t}(g, w) - Y_{i,t}(0)$. This is a straightforward extension of the binary treatment effect to a continuous "treatment effect function" or "dose-response function." The second causal parameter of interest is a conditional-on-covariates version of what Callaway et al. (2025) call the "causal response." A unit's (conditional) causal response is the derivative of the potential outcome function with respect to dose: $\frac{\partial Y_{i,t}(g,w)}{\partial w}$. That is, what is the marginal impact on district i from cohort g and covariates x if that district increases its

charter intensity slightly above w?

We aim to conduct inference on averages of these two causal parameters. The first is the conditional average treatment effect on the treated ($CATT$) in levels:

$$CATT(g,t,w,x \mid g,w,x) \equiv \mathbb{E}[Y_{i,t}(g,w) - Y_{i,t}(0) \mid G_i = g,\ W_i = w,\ X_i = x] \tag{1}$$

This is the average effect of dose w relative to zero dose in period t, among cohort g districts with covariates x that actually received dose w. This parameter serves as a building block for more easily interpretable aggregates which we will define later in this section. The second building block is the conditional average causal response to treatment. It is the derivative of the $CATT(\cdot)$ with respect to dose, evaluated at the dose the district received:

$$\begin{aligned} CACRT(g,t,w,x \mid g,w,x) &\equiv \frac{\partial\, CATT(g,t,l,x \mid g,w,x)}{\partial l}\bigg|_{l=w} \\ &= \frac{\partial\, \mathbb{E}[Y_{i,t}(g,l) \mid G_i = g,\ W_i = w,\ X_i = x]}{\partial l}\bigg|_{l=w} \end{aligned} \tag{2}$$

The $CACRT$ is interpreted as the average effect of a marginal increase in the dose from w among units that actually experienced dose w (for a district in period t, among cohort g, with covariates x). The second equality holds because the untreated potential outcome $Y_{i,t}(0)$ does not depend on l. Equations 1 and 2 are, notably, just conditional-on-covariates versions of the staggered-adoption building blocks of Callaway et al. (2025). We also consider versions of these building block parameters that are averages over the entire dose distribution:

$$CATT(g,t,w,x) \equiv \mathbb{E}[Y_{i,t}(g,w) - Y_{i,t}(0) \mid G_i = g,\ W_i > 0,\ X_i = x] \tag{3}$$

$$CACRT(g,t,w,x) \equiv \frac{\partial\, CATT(g,t,w,x)}{\partial w} = \frac{\partial\, \mathbb{E}[Y_{i,t}(g,w) \mid G_i = g,\ W_i > 0,\ X_i = x]}{\partial w} \tag{4}$$

Equations 3 and 4 average over all treated districts in cohort g with covariates x, rather than just those districts that received dose w.

The above causal functions are high dimensional objects which are difficult to interpret and report (though we will estimate them directly). Therefore, we also consider aggregations of Equations 1 – 4 into event study-like parameters. Denote $e = t - g$ as the number of periods since initial treatment for units in cohort g. We first consider an average over event-times and within dose groups:

$$\widetilde{CATT}^{dose,es}(w,x \mid w,e,x) = \sum_{g \in \bar{\mathcal{G}}} \sum_{t=2}^{T} \phi^{dose,es}(g,t,w,x \mid e)\ CATT(g,t,w,x \mid g,w,x) \tag{5}$$

$$\widetilde{CACRT}^{dose,es}(w,x \mid w,e,x) = \sum_{g\in\bar{\mathcal{G}}}\sum_{t=2}^{T} \phi^{dose,es}(g,t,w,x \mid e)\, CACRT(g,t,w,x \mid g,w,x) \quad (6)$$

where the weight on cohort g is the share of districts with dose w that belong to that cohort:

$$\begin{aligned}\phi^{dose,es}(g,t,w,x \mid e) = \mathbb{1}\{g+e \in [2,T]\}\ \mathbb{1}\{g+e=t\} \\ \times \mathbb{P}(G_i = g \mid W_i = w,\ X_i = x,\ G_i + e \in [2,T],\ G_i \leq T)\end{aligned}$$

We then consider a second average over dose in order to report conditional event-study parameters which are interpreted as the average effect of treatment for districts with covariates x that have been exposed to treatment for e periods:

$$CATT^{es}(e,x) = \mathbb{E}\left[\widetilde{CATT}^{dose,es}(W_i,x \mid W_i,e,x) \,\middle|\, X_i = x, G_i + e \in [2,T],\ G_i \leq T\right] \quad (7)$$

$$CACRT^{es}(e,x) = \mathbb{E}\left[\widetilde{CACRT}^{dose,es}(W_i,x \mid W_i,e,x) \,\middle|\, X_i = x, G_i + e \in [2,T],\ G_i \leq T\right] \quad (8)$$

2.1.1 Assumptions

Going forward, we adopt the following assumptions:

Assumption 1 (Random Sampling). *The observed data is drawn from an i.i.d. distribution and consists of* $\{Y_{i,1},...,Y_{i,T},W_i,G_i,X_i\}_{i=1}^N$.

Assumption 2 (Overlap). *Let* $\mathcal{W}_+ \subseteq (0,1]$ *denote the set of positive doses in the support of* W_i. *There exists an open interval* $\mathcal{W}_+^c = (w_L, w_U) \subseteq \mathcal{W}_+$ *and a constant* $\kappa > 0$ *such that, for all* $g \in \bar{\mathcal{G}}$ *and* $x \in \mathcal{X}$*: (a) the conditional density of dose among cohort-*g *districts is bounded away from zero on* $\mathcal{W}_+^c$, $f_{W_i|G_i,X_i}(w \mid g,x) \geq \kappa$ *for all* $w \in \mathcal{W}_+^c$*; and (b)* $\kappa \leq \mathbb{P}(G_i = g \mid X_i = x,\ G_i \in \{g,\infty\}) \leq 1-\kappa$.

Assumption 3 (No Anticipation). *For all* $g \in \bar{\mathcal{G}}$, $w \in \mathcal{W}_+$, *and* $t \leq b_g$: $Y_{i,t}(g,w) = Y_{i,t}(0)$.

Assumption 4 (Dose Support and Smoothness). $\mathbb{P}(W_i = 0) > 0$. *In addition, for all* $g \in \bar{\mathcal{G}}$, $t = 2,...,T$, *and* $x \in \mathcal{X}$, $\mathbb{E}[Y_{i,t} - Y_{i,b_g} \mid G_i = g, W_i = w, X_i = x]$ *is continuously differentiable in* w *on* $\mathcal{W}_+^c$, *the interval defined in Assumption 2.*

We also adopt one of two conditional parallel trends assumptions depending on which causal function we are attempting to estimate:

Assumption 5 (Conditional Parallel Trends). *For all* $g \in \bar{\mathcal{G}}$, $t = 2,...,T$, $w \in \mathcal{W}_+$, *and* $x \in \mathcal{X}$*:*

$$\begin{aligned}\mathbb{E}[Y_{i,t}(0) - Y_{i,b_g}(0) \mid G_i = g,\ W_i = w,\ X_i = x] \\ = \mathbb{E}[Y_{i,t}(0) - Y_{i,b_g}(0) \mid G_i = \infty,\ W_i = 0,\ X_i = x]\end{aligned} \quad (9)$$

In other words, the expected counterfactual change in outcomes between period t and b_g for districts in cohort g who experienced dose w with covariates x, had they not been treated, is equal to the observed change in outcomes among never-treated districts, conditional on X. This implies that all districts have the same counterfactual trends under $w = 0$ which are equal to those of the never-treated cohort.

Assumption 6 (Conditional Strong Parallel Trends)**.** *Both of the following are true. (a) For all $g \in \bar{\mathcal{G}}$, $t = 2, ..., T$, $w, l \in \mathcal{W}_+$, and $x \in \mathcal{X}$:*

$$\begin{aligned}&\mathbb{E}[Y_{i,t}(g,w) - Y_{i,b_g}(0) \mid G_i = g,\ W_i = l,\ X_i = x] \\ &\quad = \mathbb{E}[Y_{i,t}(g,w) - Y_{i,b_g}(0) \mid G_i = g,\ W_i = w,\ X_i = x]\end{aligned}$$

(b) Assumption 5 holds.

Assumption 6 states that districts in the same cohort and covariate cell share a common path under any given dose. Assumption 6 is a stronger parallel trends assumption than Assumption 5, and requires that every dose group (within x) follow the same causal function with respect to dose, which implies that districts with the same covariate values have the same treatment effects at all dose levels. Assumptions 5 and 6 are relaxations of Assumptions PT-MP and SPT-MP in Callaway et al. (2025), respectively, such that they must only hold within groups of districts with the same covariate values. It is worth noting that only the subset of X_i variables which are confounders are required in order to satisfy Assumptions 5 and 6. The non-confounding variables in X_i are not required to obtain consistent estimates of causal parameters, but can nonetheless be included in order to further explore heterogeneity and because the set of confounders is unknown.

2.1.2 Identification Under Parallel Trends vs. Strong Parallel Trends

Our first result is that under Assumptions 1 – 5, the $CATT$ is identified via a difference-in-differences estimator which compares the change in outcomes for districts in dose group w, cohort g, period t, and covariates x to districts with covariates x who were never treated:

Lemma 1 (Identification of the $CATT$). *Under Assumptions 1 – 5, for all $g \in \bar{\mathcal{G}}$, $t \geq g$, $w \in \mathcal{W}_+$, and $x \in \mathcal{X}$:*

$$\begin{aligned}CATT(g,t,w,x \mid g,w,x) = \mathbb{E}[Y_{i,t} - Y_{i,b_g} \mid G_i = g,\ W_i = w,\ X_i = x] - \\ \mathbb{E}[Y_{i,t} - Y_{i,b_g} \mid G_i = \infty,\ W_i = 0,\ X_i = x]\end{aligned} \tag{10}$$

For proof, see Appendix D.

If Assumption 6 also holds, then the $CACRT$ is identified as the derivative of the $CATT$ function evaluated at w:

Lemma 2 (Identification of the $CACRT$). *Under Assumptions 1 – 6, for all $g \in \bar{\mathcal{G}}$, $t \geq g$, $w \in \mathcal{W}_+^c$, and $x \in \mathcal{X}$:*

$$CACRT(g,t,w,x \mid g,w,x) = \frac{\partial\, \mathbb{E}[Y_{i,t} - Y_{i,b_g} \mid G_i = g,\ W_i = l,\ X_i = x]}{\partial l}\bigg|_{l=w} \tag{11}$$

For proof, see Appendix D. Equation 11 has no never-treated term. The never-treated comparison in Equation 10 does not vary with l, so it vanishes when we differentiate. This is the conditional analogue of Theorem C.1 of Callaway et al. (2025).

2.1.3 Time-Varying Treatment Extension

To this point, we have assumed the treatment dose remains fixed after adoption, $W_{i,t} = W_i \cdot \mathbb{1}\{t \geq G_i\}$, while in many continuous-dose policy settings (including our charter school setting in Section 3), the realized dose may vary within unit over time. In such a case, a unit may be impacted by its prior treatment history in addition to its current dose. In order to extend our absorbing state framework into one which allows unit-specific dose paths, we rely on the insights of De Chaisemartin and D'Haultfœuille (2026), hereafter dCDH. Throughout, W_i in Assumptions 1 – 6 is now denoted as $W_{i,t}$.

Updated notation. For $t \geq g$, let $\mathbf{W}_{i,g:t} = (W_{i,g}, \ldots, W_{i,t})$ denote the realized dose path of district i between periods g and t, and let $\mathbf{w}_{g:t} = (w_g, \ldots, w_t) \in [0,1]^{t-g+1}$ denote any generic path a district may undertake. Denote $Y_{i,t}(g, \mathbf{w}_{g:t})$ as the potential outcome in period t of a district first treated in g that follows path $\mathbf{w}_{g:t}$. We impose dCDH's assumption that units have no anticipation of future doses, which can be easily merged with our framework by adjusting Assumption 3 to state $Y_{i,t}(g, \mathbf{w}_{g:T}) = Y_{i,t}(0)$ for all $g \in \bar{\mathcal{G}}$, $\mathbf{w}_{g:T} \in [0,1]^{T-g+1}$, and $t \leq b_g$. The untreated outcome is $Y_{i,t}(0) = Y_{i,t}(\infty, 0)$, the observed outcome is $Y_{i,t} = Y_{i,t}(G_i, \mathbf{W}_{i,G_i:t})$ for $t \geq G_i$, and $Y_{i,t} = Y_{i,t}(0)$ otherwise.

Our target building block parameters are defined as:

$$CATT^{path}(g,t,w,x \mid g,w,x) \equiv \mathbb{E}[Y_{i,t}(g, \mathbf{W}_{i,g:t}) - Y_{i,t}(0) \mid G_i = g,\ W_{i,t} = w,\ X_i = x] \tag{12}$$

$$CACRT^{path}(g,t,w,x \mid g,w,x) \equiv \frac{\partial \mathbb{E}\big[Y_{i,t}\big(g, (\mathbf{W}_{i,g:t-1}, l)\big) \;\big|\; G_i = g,\ W_{i,t} = w,\ X_i = x\big]}{\partial l}\bigg|_{l=w} \tag{13}$$

$CATT^{path}$ is interpreted as the average impact of dose-path $\mathbf{W}_{i,g:t}$ for a unit with current

dose w in cohort g with covariates x relative to zero dose for that same unit. Similarly, a unit's $CACRT^{path}$ is interpreted as the marginal impact of additional dose in period t for a unit with dose-history $\mathbf{W}_{i,g:t-1}$, current dose w, who is in cohort g with covariates x.

Lemma 3 (Identification of the Path-$CATT$). *Under Assumptions 1 – 5 (with W_i as $W_{i,t}$), the same difference-in-differences estimator as in Lemma 1 identifies the Path-$CATT$:*

$$CATT^{path}(g,t,w,x \mid g,w,x) = \mathbb{E}[Y_{i,t} - Y_{i,b_g} \mid G_i = g,\ W_{i,t} = w,\ X_i = x] - \mathbb{E}[Y_{i,t} - Y_{i,b_g} \mid G_i = \infty,\ W_{i,t} = 0,\ X_i = x] \tag{14}$$

For proof, see Appendix D.

The right-hand side of Equation 14 is the same comparison underlying the $DID_{g,\ell}$ estimator of dCDH: the change in outcomes for cohort-g districts since the year before charter entry, minus the comparison group's change over the same time periods. The implication of this result is that the $CATT$ is identified by the same difference-in-differences estimator outlined by Lemma 1 regardless of if the treatment dose is an absorbing state or time-varying. The only difference is the interpretation of the $CATT$: under an absorbing state, the $CATT$ represents the average effect of being treated by one's current dose for a given time horizon relative to zero treatment; under a time-varying dose, the $CATT$ represents the average effect of following a specific dose path relative to zero treatment.

Identifying the Path-$CACRT$ is somewhat more difficult than the Path-$CATT$. We first extend Assumption 6 (Conditional Strong Parallel Trends) to refer to dose paths rather than just current period dose.

Assumption 7 (Conditional Strong Parallel Trends for Dose Paths). *(a) For all $g \in \bar{\mathcal{G}}$, $t \geq g$, $x \in \mathcal{X}$, and paths $\mathbf{w}_{g:t}$ and $\mathbf{l}_{g:t}$:*

$$\begin{aligned}&\mathbb{E}[Y_{i,t}(g,\mathbf{w}_{g:t}) - Y_{i,b_g}(0) \mid G_i = g,\ \mathbf{W}_{i,g:t} = \mathbf{l}_{g:t},\ X_i = x] \\ &\quad = \mathbb{E}[Y_{i,t}(g,\mathbf{w}_{g:t}) - Y_{i,b_g}(0) \mid G_i = g,\ \mathbf{W}_{i,g:t} = \mathbf{w}_{g:t},\ X_i = x]\end{aligned}$$

(b) Assumption 5 holds.

Assumption 7 says that within a $(g \times x)$ cell, the average counterfactual change in outcomes for districts that experienced dose path $\mathbf{l}_{g:t}$ had they experienced dose path $\mathbf{w}_{g:t}$ is equal to the average change in outcomes for all units that actually experienced dose path $\mathbf{w}_{g:t}$. Assumption 7 reduces to Assumption 6 in the special case where treatment is an absorbing state as in Section 2.1.

The marginal effect we seek holds the dose history fixed and varies only the current dose. We therefore define the Path-$CACRT$ at a given history, reusing the same notation with the dose path in place of the current dose:

$$\begin{aligned} &CACRT^{path}(g,t,\mathbf{w}_{g:t},x \mid g,\mathbf{w}_{g:t},x) \\ &\quad\equiv \frac{\partial}{\partial l}\,\mathbb{E}\big[Y_{i,t}\big(g,(\mathbf{w}_{g:t-1},l)\big) \;\big|\; G_i=g,\ \mathbf{W}_{i,g:t}=\mathbf{w}_{g:t},\ X_i=x\big]\Big|_{l=w_t} \end{aligned} \tag{15}$$

Averaging over the histories of districts with current dose w recovers Equation 13:

$$\begin{aligned} &CACRT^{path}(g,t,w,x \mid g,w,x) \\ &\quad= \mathbb{E}\big[CACRT^{path}(g,t,\mathbf{W}_{i,g:t},x \mid g,\mathbf{W}_{i,g:t},x) \;\big|\; G_i=g,\ W_{i,t}=w,\ X_i=x\big] \end{aligned} \tag{16}$$

Lemma 4 (Identification of the Path-$CACRT$). *Under Assumptions 1 – 5 and 7, and if the conditional density of $W_{i,t}$ given $G_i=g$, $\mathbf{W}_{i,g:t-1}=\mathbf{w}_{g:t-1}$, and $X_i=x$ is positive in a neighborhood of w_t, a difference-in-differences estimator which compares outcome changes from districts that are identical on cohort, covariates, and dose history, differing only in the dose in period t, identifies the marginal effect of the period t dose:*

$$\begin{aligned} &CACRT^{path}(g,t,\mathbf{w}_{g:t},x \mid g,\mathbf{w}_{g:t},x) \\ &\quad= \frac{\partial}{\partial l}\,\mathbb{E}[Y_{i,t}-Y_{i,b_g} \mid G_i=g,\ \mathbf{W}_{i,g:t-1}=\mathbf{w}_{g:t-1},\ W_{i,t}=l,\ X_i=x]\Big|_{l=w_t} \end{aligned} \tag{17}$$

For proof, see Appendix D.

In the fixed-dose case, Lemma 2 obtains the $CACRT$ by comparing outcome changes across districts with slightly different doses. When one allows for dose paths, the same comparison gives the slope of the Path-$CATT$ in the current dose. However, in practice it is unlikely that any two districts will share the exact same dose path, which makes a comparison within dose paths infeasible. The derivative of the Path-$CATT$ (a feasible comparison) instead produces a parameter which mixes two things assuming Assumption 7 holds. Dropping $G_i=g$ and $X_i=x$

from the conditioning sets for simplicity,

$$\frac{\partial\, CATT^{path}(g,t,l,x \mid g,l,x)}{\partial l} = \underbrace{\mathbb{E}\left[\frac{\partial\, \mathbb{E}[Y_{i,t} - Y_{i,b_g} \mid \mathbf{W}_{i,g:t-1},\ W_{i,t} = l]}{\partial l} \,\middle|\, W_{i,t} = l\right]}_{CACRT^{path}(g,t,l,x|g,l,x)} + \underbrace{\frac{\partial}{\partial l'}\, \mathbb{E}\big[\mathbb{E}[Y_{i,t} - Y_{i,b_g} \mid \mathbf{W}_{i,g:t-1},\ W_{i,t} = l] \,\big|\, W_{i,t} = l'\big]\Big|_{l'=l}}_{\text{history composition}} \tag{18}$$

For the derivation, see Appendix D. The first term is the Path-$CACRT$ we desire (Equation 16): the effect of a marginally higher current dose holding fixed the dose history. The second term is the difference in marginal effects that arises only from alternative dose histories. We introduce an assumption which imposes path-independence of the outcome, a sufficient condition to identify the Path-$CACRT$:

Assumption 8 (Path Independence). *For all $g \in \bar{\mathcal{G}}$, $t > g$, $x \in \mathcal{X}$, and paths $\mathbf{w}_{g:t}$ and $\mathbf{l}_{g:t}$ with the same current dose, $w_t = l_t$, whose histories $\mathbf{w}_{g:t-1}$ and $\mathbf{l}_{g:t-1}$ lie in the support of $\mathbf{W}_{i,g:t-1}$ given $G_i = g$ and $X_i = x$:*

$$\mathbb{E}[Y_{i,t}(g, \mathbf{w}_{g:t}) \mid G_i = g,\ X_i = x] = \mathbb{E}[Y_{i,t}(g, \mathbf{l}_{g:t}) \mid G_i = g,\ X_i = x]$$

In other words, within a $(g \times x)$ cell, two dose paths that end at the same current dose produce the same expected outcome in period t regardless of the paths themselves.

Under Assumption 8, the "history composition" term is zero. This is also shown in Appendix D. In our charter school application in the following sections, we will be targeting $CATT^{path}(g,t,w,x \mid g,w,x)$ and $CACRT^{path}(g,t,w,x \mid g,w,x)$ parameters, imposing Assumption 8 for $CACRT^{path}(\cdot)$.

2.2 Generalized Difference-in-Differences (CGS-DiD) Identification and Estimation

Callaway et al. (2025) propose estimators which identify the unconditional $ATT(g,t,w)$ and $ACRT(g,t,w)$ under unconditional versions of our Assumptions 1 – 6 (Assumption 6 only required for $ACRT(\cdot)$). Their estimators are constructed to ensure the regression weights implicit in the selected model match the true dose distribution. We extend their estimators to identify the $CATT(g,t,w,x)$ and $CACRT(g,t,w,x)$ under Assumptions 1 – 6 (we discuss this section in

the context of the absorbing state scenario, though all results extend to the time-varying treatment scenario of our charter school application without loss of generality) which condition on covariates and are therefore relaxations of their unconditional parallel trends and strong parallel trends assumptions. First, recall from Lemma 1 that for some positive dose w, between period t and b_g for a district in cohort g we have: $CATT(g,t,w,x \mid g,w,x) = \mathbb{E}[\Delta Y_{i,g,t} \mid G_i = g, W_i = w, X_i = x] - \mathbb{E}[\Delta Y_{i,g,t} \mid G_i = \infty, W_i = 0, X_i = x]$, where $\Delta Y_{i,g,t} \equiv Y_{i,t} - Y_{i,b_g}$ denotes the long difference in outcomes between period t and cohort g's base period. With an estimate of the baseline conditional mean function $m_{g,t}(x) = \mathbb{E}[\Delta Y_{i,g,t} \mid G_i = \infty, W_i = 0, X_i = x]$, the general form of their estimators can be expressed as the following:

$$\Delta\widetilde{Y}_{i,g,t} = CATT(g,t,W_i,X_i \mid g,W_i,X_i) + \varepsilon_{i,g,t}$$

Where $\Delta\widetilde{Y}_{i,g,t} = \Delta Y_{i,g,t} - \hat{m}_{g,t}(X_i)$ and $CATT(g,t,W_i,X_i \mid g,W_i,X_i)$ can be any researcher-selected functional form of the $CATT$. We will propose both parametric and semi-parametric functional forms in our case. In our parametric model, we specify a linear function akin to a generalized difference-in-differences (which we abbreviate as CGS-DiD), where the following regression is estimated on treated districts separately within cohort g and time t:

$$\Delta\widetilde{Y}_{i,g,t} = \alpha_0 + X_i'\alpha_1 + \tau_0 W_i + W_i X_i'\tau_1 + \varepsilon_{i,g,t} \tag{19}$$

Where $\hat{\alpha}(x) = \hat{\alpha}_0 + x'\hat{\alpha}_1$ and $\hat{\tau}(x) = \hat{\tau}_0 + x'\hat{\tau}_1$. This implies that $\widehat{CATT}(g,t,w,x) = \hat{\alpha}(x) + \hat{\tau}(x)w$ and, under Assumption 6, $\widehat{CACRT}(g,t,w,x) = \hat{\tau}(x)$.[5] Our estimate of the baseline conditional mean in this model comes from a weighted least squares regression of $\hat{m}_{g,t}(x) = \hat{\gamma}_0 + x'\hat{\gamma}_1$, estimated from never treated units.

To see that this model closely matches a traditional two-way fixed effects (TWFE) specification, consider a 2-period regression between period t and b_g estimated on all units in cohort g and never-treated units $\mathcal{S}_g = \{i : G_i = g\} \cup \{i : G_i = \infty\}$:

$$Y_{i,t} = \mu_i + \lambda_t + Post_t\Big[\Big(\gamma_0 + X_i'\gamma_1\Big) + \mathbb{1}\{G_i = g\}(\alpha_0 + X_i'\alpha_1) + W_{i,t}(\tau_0 + X_i'\tau_1)\Big] + \varepsilon_{i,g,t}$$

Where μ_i and λ_t are unit and time fixed effects, respectively, and $Post_t$ is an indicator for a unit being in a post-treatment period. In period b_g, $Post_t = 0$, so taking the first difference between periods t and b_g yields:

$$\Delta Y_{i,g,t} = \underbrace{(\lambda_t - \lambda_{b_g}) + \gamma_0 + X_i'\gamma_1}_{m_{g,t}(X_i)} + \mathbb{1}\{G_i = g\}(\alpha_0 + X_i'\alpha_1) + W_{i,t}(\tau_0 + X_i'\tau_1) + \Delta\varepsilon_{i,g,t}$$

[5] All $\hat{\alpha}$ and $\hat{\tau}$ are specific to a $(g \times t)$ cell. Going forward, we omit this notation to reduce clutter.

Rearranging yields Equation 19.

One potential challenge with so many different dimensions of effect heterogeneity (see Section 3) is that the factors in X_i may be correlated with each other. For example, state policies may be correlated with household demographics. Also, states with more accountability might also have distinctive funding formulas. We are able to overcome this challenge because the richness of our data allows us to include essentially all of these factors at the same time to isolate the role of each.

The standard errors are clustered at the district level. The estimates are weighted by the district size, so that the estimates are nationally representative (among the estimation sample which excludes mature charter markets); specifically, we weight by graduation rate denominator in the high school sample and by grade-level enrollment for math and ELA. Following Chen and Harris (2023), we measure treatment with a one-year lag of charter enrollment share, assuming charter entry takes at least one year to affect outcomes; cohorts G_i are defined by the first year of positive lagged share, so the base period b_g is the first year in which charter schools operate.

The above derivation of Equation 19 illustrates that our method accounts for the time-invariant unobserved differences across districts that one would want to control for in a TWFE specification. Identification of the $CATT$ requires that, conditional on observed covariates, districts receiving a given level of charter school exposure would, in the absence of treatment, have experienced the same average change in outcomes as never-treated districts. This assumption would be violated if unobserved time-varying factors jointly affected charter-school expansion and student outcomes. We note that, even if charter market share changed based on time-invariant unobserved district characteristics that affected student outcomes, this would not introduce bias if the timing of charter entry were conditionally random, which is possible given all the steps involved in opening a charter school (e.g., putting in an application, getting it approved, gaining access to a building, recruiting students, and hiring staff).

Still, it is worth considering specific scenarios under which this might be violated. First, charter schools are more likely to locate in districts with low contemporaneous student outcomes (Glomm et al., 2005), signaling that we also might expect charter schools to locate where *expected* future TPS performance is lower, though this is unobservable in our analysis.[6] This would lead to a downward bias in our estimates because TPS outcomes comprise the majority of the weighted

[6]Relatedly, charter schools might open where districts experience idiosyncratic shocks, in which case future outcomes regress toward the mean. If charter schools prefer to locate near low-performing schools, as noted above, then this would yield an upward bias (i.e., charter schools enter because of the negative shock in existing schools, but then those outcomes bounce back in the next period). We do not see this scenario as very likely because it takes several years to create an organization that can assemble a charter application, submit the application and gain approval, hire personnel, purchase necessary capital, and recruit students. Also, with such a long-term investment, charter organizations are likely to consider multiple years of information in making their entry decisions, which reduces the chance of regression to the mean.

average outcomes.

Alternatively, local governments might change non-charter education policies at the same time they introduce charter schools, as part of a package of reforms. Since school districts are often charter authorizers, it may be that a change in school board politics leads both to an increase in charter schools and, for example, new reading and math curricula in TPS. In this case, we might falsely attribute outcome changes to charter schools that were actually caused by other policy changes. We discuss various tests for these and related types of violations in the next section, after describing our main results.

2.3 Causal Forest Identification and Estimation

The CGS-DiD identifies the $CATT/CACRT$ within a linear parametric model. However, there exists a large literature which considers partially linear models of conditional average treatment effects (e.g. Robinson, 1988), which can be written generally as:

$$Y_i = \alpha(X_i) + \tau(X_i)W_i + \varepsilon_i \tag{20}$$

Where the coefficient on W_i, $\tau(X_i)$, is an unknown function of X_i. Under a conditional independence condition (i.e. $\mathbb{E}[\varepsilon_i \mid X_i] = \mathbb{E}[W_i\varepsilon_i \mid X_i] = 0$), $\tau(X_i)$ identifies the conditional average treatment effect ($CATE$)[7]. This translates into a generalization of the CGS-DiD framework under our Assumptions 1 – 6, where $\tau(X_i)$ now identifies the $CACRT$[8]:

$$Y_{i,t} = \mu_i + \lambda_t(X_i) + \mathbb{1}\{W_{i,t} > 0\} \underbrace{\left(\alpha_{g,t}(X_i) + \tau_{g,t}(X_i)W_{i,t}\right)}_{CATT} + \varepsilon_{i,t} \tag{21}$$

It is worth noting that Equation 21 is not a structural model of potential outcomes, as interpreting it as such would require X_i to contain all moderators that contribute to heterogeneity (Frölich, 2007), some of which may be unobservable in practice. Nonetheless, extrapolating the result of Abadie (2003) allows us to conduct inference on the $CATT/CACRT$ without additional assumptions beyond Assumptions 1 – 6.

In order to conduct inference on the $CATT/CACRT$, our task is to non-parametrically estimate $\left\{\alpha(X_i), \tau(X_i)\right\}$ in Equation 21. We begin by briefly reviewing $CATE$ estimation using

[7] Abadie (2003) takes this further and shows that even if W_i is endogenous, if one obtains a valid instrument then $\tau(X_i)$ identifies the conditional *local* average treatment effect ($CLATE$) within the same partially linear model without any further assumptions beyond those of a standard IV estimator.

[8] Traditionally, parallel trend assumptions like Assumption 5 only assume that counterfactual trends of *treated* units are parallel to the observed trends of control units. PTA's do not assume that the counterfactual trends of control units had they been treated are parallel to the observed outcomes of units that were actually treated. This is why we do not attempt to identify the CATE: because doing so would require a stronger PTA that stipulates the observed outcomes of treated districts are representative of the counterfactual outcomes of control districts.

causal forests in order to build intuition for the extensions we employ, and refer the reader to other studies which provide more detailed expositions of $CATE$ estimation using causal forests (e.g. Knaus et al., 2021; Caron et al., 2022; Strittmatter, 2023). Next, we discuss the method we employ to operationalize the causal forest in DiD scenarios: the DiDCF of Gavrilova et al. (2025).

2.3.1 Causal Forest Overview

To discuss the basics of causal forest estimation, consider the case of a completely randomized experiment with a binary treatment. In this case, we seek to estimate the $CATE$ for a set of pre-specified covariates, which corresponds to $\tau(X_i)$ in Equation 20. A *causal forest* (Wager and Athey, 2018) is built from many *causal trees* (Athey and Imbens, 2016). Causal trees group observations by recursively splitting the data into finer and finer subgroups. With each split, the causal tree aims to partition units such that the $ATEs$ within each partition are as different as possible. Splits occur at "nodes," and the resulting partitions of the data which exist after the initial split are called "children nodes." With continuously distributed covariates, splitting can in principle continue until each unit is contained in its own partition. To prevent this level of overfitting, hyperparameters are used which define stopping rules such that a tree knows when to cease splitting along a branch. These may include (1) a minimum number of treatment/control units in a child node, (2) a minimum share of a parent node that must be contained in a child node, (3) imbalance penalty parameters, etc. The final partition along a branch is called a "leaf" or "terminal node," $L(x)$, and results in a small set of units defined by a similar vector of covariate values, x. Within any given leaf, the $CATE$ is then estimated as the difference in outcomes between treated vs. untreated units in that leaf. Causal forests then provide an estimate of each unit's individualized $CATE$ by averaging estimates across causal trees.[9] The purpose of growing many trees is to improve accuracy, as singular trees do not produce smooth estimates of $CATE$ functions (Wager and Athey, 2018).

Athey et al. (2019) extend the causal forest approach such that it can be applied to more general settings of causal effect estimation such as observational studies with non-randomized treatment, which they refer to as *generalized random forests* (GRF). They do this by re-framing the causal forest algorithm such that it no longer views forests as an average of causal trees, but rather as an adaptive nearest-neighbor estimator of any causal parameter that can be identified with a local moment condition. In the case of Equation 20, $\tau(X_i)$ is identified via the (conditional)

[9] All units will fall into some leaf within each tree, and thus will receive a treatment effect estimate from each tree.

local moment condition corresponding to conditional independence of W_i and Y_i:

$$\begin{aligned} &\mathbb{E}[(1, W_i)'\varepsilon_i | X_i = x] = 0 \quad \forall x \in \mathcal{X} \\ \implies &\mathbb{E}[(1, W_i)'(Y_i - \alpha(x) - \tau(x)W_i) | X_i = x] = 0 \quad \forall x \in \mathcal{X} \end{aligned} \tag{22}$$

Based on Equation 22, the GRF estimates of $(\hat{\alpha}(x), \hat{\tau}(x))$ are given by:

$$\big(\hat{\alpha}(x), \hat{\tau}(x)\big) \in \arg\min_{\alpha,\tau} \Big\{ \Big|\Big| \sum_{i=1}^{N} \omega_i(x)[(1, W_i)'(Y_i - \alpha - \tau W_i)] \Big|\Big|_2 \Big\} \tag{23}$$

Where $\omega_i(x)$ is the GRF-defined similarity weight which captures the relevance of the i-th observation for estimating $\hat{\alpha}(x)$ and $\hat{\tau}(x)$; which is how often observation i appears in the same terminal leaf as x across all causal trees. It is straightforward to see that Equation 23 is simply a GRF-weighted OLS estimator. The task is therefore to derive the similarity weights such that the resulting $CATE$ estimates from Equation 23 are consistent and asymptotically normal. We will outline the GRF algorithmic approach to growing causal trees that accomplishes this.

There are two important concepts to note before explaining the GRF-training algorithm. First, GRF performs Neyman orthogonalization (i.e. local recentering) (Robinson, 1988; Chernozhukov et al., 2018; Nie and Wager, 2021) by differencing out the propensity score and conditional mean from W_i and Y_i, respectively. These propensity score functions are estimated via auxiliary regression trees with leave-one-out estimates. The purpose of this is to improve finite sample performance of the model, as GRF estimates can be biased in finite samples if the conditioning moderators in X_i are not fully partialled out because of the small sample size. Athey et al. (2019) show that this reduces finite sample bias in these situations. Second, trees are estimated based on the concept of "honesty" developed by Athey and Imbens (2016). The consistency and asymptotic normality of non-parametric estimators relies on cross-training: using a different sample for model estimation than the one used for model training. The reason is that using the same sample for training and estimation introduces dependence between the structure of the model and the estimates themselves. Therefore, an honest causal tree takes the full data sample and extracts a random sub-sample to grow the tree, then further divides this sample into a training sample and an estimation sample. The training sample is used to perform splitting and define the tree structure, the estimation sample is used to populate the leaves and provide $CATE$ estimates. Appendix E summarizes the process for growing trees and obtaining the GRF-weights.

2.3.2 DiDCF Overview

The above description of the GRF algorithm is designed for settings where treatment is credibly conditionally random, as in the case of a randomized experiment. Our study is quasi-experimental

and relies on the conditional parallel trends assumptions stated above as Assumption 5 (for the $CATT$) and Assumption 6 (for the $CACRT$). We therefore require an operationalization of the GRF algorithm that identifies that $CATT/CACRT$ under Assumptions 5 and 6 rather than conditional independence. One such approach is the Difference-in-Differences Causal Forest (DiDCF) of Gavrilova et al. (2025). In this subsection, we will provide a brief overview of the DiDCF, which is designed for the most straightforward types of event-study: cases with a single treatment group which is affected by a binary treatment (the authors also provide an extension for staggered treatment adoption settings). The next section will discuss our further extension of this approach to settings of staggered treatment adoption by multiple treatment groups under continuous treatment.

The standard DiDCF assumes that all treated units are treated in the same period, and that treatment is both binary and an absorbing state. The target estimand in this case is the canonical binary $CATT$ within a given period t, which is the $CATT$ corresponding to Equation 1 in the case of binary treatment (and only 2 groups):

$$CATT(t,x) \equiv \mathbb{E}[Y_{i,t}(1) \mid W_i = 1,\ X_i = x] - \mathbb{E}[Y_{i,t}(0) \mid W_i = 1,\ X_i = x] \tag{24}$$

The DiDCF estimator proceeds in the following sequence:

Standard DiDCF Algorithm:

1. Transform the outcome variable by differencing it relative to a base year, b:

$$\Delta Y_{i,t} = Y_{i,t} - Y_{i,b}$$

   Conducting analysis on this transformed variable ensures that any forests trained on it are DiD-based estimators. Gavrilova et al. (2025) show that under a similar set of assumptions to our Assumptions 1 – 5 (and some standard regularity assumptions), that a DiD-based estimator identifies the $CATT$ from Equation 24.

2. For each period $t \neq b$, estimate a separate (standard) causal forest (GRF) in the manner of Athey et al. (2019). This produces an estimate of the $CATT$ function in each period, $\widehat{CATT}(t,x)$. Intuitively, this approach is equivalent to training a single causal forest on the full data, but forcing the first split to be on time period. This is necessary because not doing so would lead to cross-period contamination within leaves (i.e. units from different time periods would fall in the same leaf). If the treatment effect is not uniform across time periods, heterogeneity bias in the same spirit as that highlighted by Sun and Abraham (2021) in the case of staggered adoption may enter the estimates. It is also important to

note that this approach allows estimation of pre-treatment placebo tests of parallel trends by estimating forests in periods $t < b$, where the outcomes of not-yet-treated units are compared to the outcomes of never-treated units.

3. Predict the treatment effect for unit i in period t among only treated units, using trees not trained on unit i.

2.3.3 Extending DiDCF to Staggered and Continuous Treatment

Gavrilova et al. (2025) show that their estimator is consistent and asymptotically normal under Assumptions[10] 1 – 5. They provide (though they do not implement) an approach to extend their estimator to cases of staggered treatment adoption. Here, we outline an approach to adapt the staggered-adoption DiDCF to a setting with continuous treatment by blending it with the estimators of Callaway et al. (2025).

Recall from Section 2.1 that $\mathcal{G} \subseteq \{2, ..., T, \infty\}$ is the set of cohorts of initial treatment we consider (after dropping always-treated units), and $\bar{\mathcal{G}} = \mathcal{G} \setminus \{\infty\}$ is the set of ever-treated cohorts. Units in the same cohort have the same base period, which is the period before initial treatment, denoted b_g. The first adjustment to the DiDCF algorithm is that we begin by splitting the data into $|\bar{\mathcal{G}}|$ separate sub-samples, where each subsample contains only units from cohort g or never-treated units: $\mathcal{S}_g$.

In the case of a binary treatment, one could then estimate a standard DiDCF within each subsample, yielding consistent estimates of the cohort $\times$ time $CATT$ with a corresponding cohort $\times$ time variance. However, in the case of a continuous treatment, this approach would fail to identify the $CATT(g, t, w, x)$ or the $CACRT(g, t, w, x)$ targets of Equations 3 and 4. The reason is because DiDCF calls the standard GRF algorithm within each subsample $\mathcal{S}_g$, which estimates conditional average partial effect parameters for continuous treatments. Conditional average partial effects are akin to the two-way fixed effects estimator which Callaway et al. (2025) decompose into weighted sums of the $CATT$ and $CACRT$ parameters *plus* possible selection bias terms, where the weights generally do not match the true dose distribution.

The selection bias is easiest to see with two dose groups. Take two groups of districts in cohort g and covariate cell x, one with dose h and one with dose l, both positive. A difference-

[10] Their version of the parallel trends assumption is less strict than our Assumption 5 because they consider only one treated group. Satisfaction of our Assumption 5 implies satisfaction of their parallel trends assumption.

in-differences comparison of the two gives

$$\begin{aligned}
&\mathbb{E}[\Delta Y_{i,g,t} \mid W_i = h, G_i = g, X_i = x] - \mathbb{E}[\Delta Y_{i,g,t} \mid W_i = l, G_i = g, X_i = x] \\
&\qquad = \underbrace{CATT(g,t,h,x \mid g,h,x)}_{\text{level effect of dose } h} - \underbrace{CATT(g,t,l,x \mid g,l,x)}_{\text{level effect of dose } l} \\
&\qquad = \underbrace{\mathbb{E}[Y_{i,t}(g,h) - Y_{i,t}(g,l) \mid W_i = h, G_i = g, X_i = x]}_{\text{causal response}} \\
&\qquad\quad + \underbrace{\Big(CATT(g,t,l,x \mid g,h,x) - CATT(g,t,l,x \mid g,l,x)\Big)}_{\text{selection bias}}
\end{aligned} \tag{25}$$

In order to restructure the DiDCF estimand to return $CATT/CACRT$ parameters, notice that within cohort g, we can construct $\Delta Y_{i,g,t}$ in terms of Equation 21:

$$\begin{aligned}
\Delta Y_{i,g,t} &= Y_{i,t} - Y_{i,b_g} \\
&= \big(\lambda_t(X_i) - \lambda_{b_g}(X_i)\big) + \mathbb{1}\{W_i > 0\}\big(\alpha_{g,t}(X_i) + \tau_{g,t}(X_i)W_i\big) + \varepsilon_{i,g,t}
\end{aligned} \tag{26}$$

Recall that the $CATT$ for any positive dose w is given as $CATT(g,t,w,x \mid g,w,x) = \mathbb{E}[\Delta Y_{i,g,t} \mid G_i = g, W_i = w, X_i = x] - \mathbb{E}[\Delta Y_{i,g,t} \mid G_i = \infty, W_i = 0, X_i = x]$. Estimating the baseline conditional mean function within $\mathcal{S}_g$, denoted

$$\begin{aligned}
m_{g,t}(x) &= \lambda_t(x) - \lambda_{b_g}(x) \\
&= \mathbb{E}[\Delta Y_{i,g,t} \mid W_i = 0, G_i = \infty, X_i = x]
\end{aligned} \tag{27}$$

can be done in a number of ways (in the CGS-DiD we do this via a weighted least squares regression, one may also consider LASSO, random forests, neural networks, or other semi or nonparametric approaches). Then define the partially linear model to be estimated only among treated districts:

$$\Delta\widetilde{Y}_{i,g,t} = \alpha_{g,t}(X_i) + \tau_{g,t}(X_i)\, W_i + \varepsilon_{i,g,t} \tag{28}$$

Where $\Delta\widetilde{Y}_{i,g,t} = \Delta Y_{i,g,t} - \hat{m}_{g,t}(X_i)$. Equation 28 can now be estimated with the staggered DiDCF procedure outlined above and will produce estimates $\widehat{CATT}(g,t,W_i,X_i) = \hat{\alpha}_{g,t}(X_i) + \hat{\tau}_{g,t}(X_i)W_i$ and $\widehat{CACRT}(g,t,W_i,X_i) = \hat{\tau}_{g,t}(X_i)$ which are consistent for the corresponding targets (the $CACRT(\cdot)$ additionally requiring Assumption 6). One may also consider a 1-step equation of the form $\Delta Y_{i,g,t} = m_{g,t}(X_i) + \mathbf{D}_i'\boldsymbol{\beta}_{g,t}(X_i) + \varepsilon_{i,g,t}, \quad i \in \mathcal{S}_g$ where $\mathbf{D}_i = (\mathbb{1}\{W_i > 0\}, W_i)'$ and $\hat{\boldsymbol{\beta}}_{g,t}(x) = \big(\hat{\alpha}_{g,t}(x), \hat{\tau}_{g,t}(x)\big)'$. This alternative specification simultaneously estimates the base-

line conditional mean function and the treatment effect function among all units in $\mathcal{S}_g$, rather than just treated units. The two approaches are equivalent so long as $\hat{m}_{g,t}(x)$ is constant within x groups (a condition likely to be true when the number of untreated units is large). In our application, we favor the latter specification.

Before formally stating the consistency result, we introduce the regularity assumptions needed to invoke the proof of Proposition 1 in Gavrilova et al. (2025):

Assumption 9 (Regularity). *For each cohort $g \in \bar{\mathcal{G}}$ and period $t \geq g$, within the cohort-g subsample $\mathcal{S}_g = \{i : G_i = g\} \cup \{i : G_i = \infty\}$, let $\mathbf{D}_i = (\mathbb{1}\{W_i > 0\}, W_i)'$. (a) The covariates X_i have a density on $[0,1]^k$ that is bounded away from zero and infinity. (b) $\mathbb{E}[\Delta Y_{i,g,t} \mid X_i = x]$, $\mathbb{E}[\mathbf{D}_i \mid X_i = x]$, $\mathrm{Cov}(\Delta Y_{i,g,t}, \mathbf{D}_i \mid X_i = x)$, and $\mathrm{Var}(\mathbf{D}_i \mid X_i = x)$ are Lipschitz continuous in x. (c) $\mathrm{Var}(\mathbf{D}_i \mid X_i = x)$ is invertible and $\mathrm{Var}(\Delta Y_{i,g,t} \mid X_i = x) > 0$ for all x. (d) For some $\nu, M > 0$, $\mathbb{E}\big[|\Delta Y_{i,g,t} - \mathbb{E}[\Delta Y_{i,g,t} \mid X_i = x]|^{2+\nu} \mid X_i = x\big] \leq M$ uniformly in x. (e) Observations are independent across districts. These are the conditions under which Theorems 3 and 5 of Athey et al. (2019) apply to the conditional linear model of their Section 6.*

Assumption 10 (Local Linearity in Dose). *For each $g \in \bar{\mathcal{G}}$, $t \geq g$, and $x \in \mathcal{X}$, there exist $\alpha_{g,t}(x)$ and $\tau_{g,t}(x)$ such that, for all w in the support of W_i given $G_i = g$, $W_i > 0$, and $X_i = x$:*

$$CATT(g, t, w, x \mid g, w, x) = \alpha_{g,t}(x) + \tau_{g,t}(x)\, w$$

We impose Assumption 10 in order to specify a partially linear model of the $CATT$. A partially linear model implies that within a $(g \times t \times x)$ neighborhood, there exists a locally linear model of the $CATT = \alpha + \tau w$ where the marginal effect of additional dose, τ, is constant across doses (though it may vary within x and across cohorts and time periods). This assumption is optional from an identification standpoint as it would not be required under a fully nonparametric functional form of the $CATT$. However, the partially linear model improves interpretability and requires sacrificing less statistical power than a fully nonparametric specification.[11]

Lemma 5 (Consistency of the Continuous DiDCF). *Let $\hat{\boldsymbol{\beta}}_{g,t}(x) = \big(\hat{\alpha}_{g,t}(x),\, \hat{\tau}_{g,t}(x)\big)'$ be the lm_forest estimates of*

$$\Delta Y_{i,g,t} = m_{g,t}(X_i) + \mathbf{D}_i' \boldsymbol{\beta}_{g,t}(X_i) + \varepsilon_{i,g,t}, \qquad i \in \mathcal{S}_g,$$

[11] If Assumption 10 fails, the forest produces the best linear projection in dose within each $(g \times t \times x)$ neighborhood. This may miss the true $CATT$ at individual doses, but averages of the fitted $CATT$s, including the event-study aggregates in Equation 30, still estimate the average effect of the doses districts actually received. Similarly, under Assumption 6, $\tau_{g,t}(x)$ is still an average of the true $CACRT$s across doses, with nonnegative weights (Theorem 3.4 of Callaway et al., 2025). Assumption 10 matters only for interpreting $\hat{\alpha}(x) + \hat{\tau}(x)w$ as the effect of a specific dose w.

$$\mathbf{D}_i = \big(\mathbb{1}\{W_i > 0\},\, W_i\big)'.$$

Under Assumptions 1 – 5 and 9 – 10, for each $g \in \bar{\mathcal{G}}$, $t \geq g$, $w \in \mathcal{W}_+$, and $x \in \mathcal{X}$, $\widehat{CATT}(g,t,w,x) = \hat{\alpha}_{g,t}(x) + \hat{\tau}_{g,t}(x)\, w$ is consistent and asymptotically normal for $CATT(g,t,w,x \mid g,w,x)$. If Assumption 6 also holds, $\hat{\tau}_{g,t}(x)$ is consistent and asymptotically normal for $CACRT(g,t,w,x)$ for all $w \in \mathcal{W}_+^c$.

For proof, see Appendix D.

Aggregation. Each cohort $\times$ period (g,t) cell is estimated separately, so each cohort has its own base period b_g and its own never-treated comparison group. Evaluating the fitted functions at each treated district's observed dose yields the district-level building block estimates:

$$\widehat{CATT}(g,t,W_i,X_i) = \hat{\alpha}_{g,t}(X_i) + \hat{\tau}_{g,t}(X_i)\, W_i, \qquad \widehat{CACRT}(g,t,W_i,X_i) = \hat{\tau}_{g,t}(X_i)$$

We report averages of these estimates over treated districts whose covariates fall in a set $\mathcal{A} \subseteq \mathcal{X}$, often referred to as a "Group Average Treatment Effect" ($GATE/GATT$) (Chernozhukov et al., 2025). For example, one may be interested in plotting event-studies for districts in certain covariate groupings (e.g. within quartiles of particular covariates or within other user-defined groupings of interest). In this case we define the target parameter as the $GATT$ among treated districts with $X_i \in \mathcal{A}$, which integrates the $CATT^{es}(e,x)$ of Equation 7 over $x \in \mathcal{A}$:

$$\begin{aligned} GATT^{es}(e,\mathcal{A}) &\equiv \mathbb{E}[CATT^{es}(e,X_i) \mid X_i \in \mathcal{A},\ G_i + e \in [2,T],\ G_i \leq T] \\ &= \sum_{g \in \bar{\mathcal{G}}} \phi_g(e,\mathcal{A})\, \mathbb{E}[CATT(g,g+e,W_i,X_i \mid g,W_i,X_i) \mid G_i = g,\ X_i \in \mathcal{A}] \end{aligned} \tag{29}$$

where $\phi_g(e,\mathcal{A}) = \mathbb{1}\{g+e \in [2,T]\}\, \mathbb{P}(G_i = g \mid X_i \in \mathcal{A},\ G_i + e \in [2,T],\ G_i \leq T)$ is the share of treated districts in $\mathcal{A}$ observed at event time e who are in cohort g. Following Callaway and Sant'Anna (2021), we estimate Equation 29 by averaging the district-level estimates within cohort and then across cohorts:

$$\widehat{GATT}^{es}(e,\mathcal{A}) = \sum_{g \in \bar{\mathcal{G}}} \hat{\phi}_g(e,\mathcal{A})\, \widehat{CATT}_g(e,\mathcal{A}), \qquad \hat{\phi}_g(e,\mathcal{A}) = \frac{n_g(\mathcal{A})\, \mathbb{1}\{g+e \in [2,T]\}}{\sum_{g' \in \bar{\mathcal{G}}} n_{g'}(\mathcal{A})\, \mathbb{1}\{g'+e \in [2,T]\}} \tag{30}$$

where $n_g(\mathcal{A})$ is the number of cohort-g districts in $\mathcal{A}$, and $\widehat{CATT}_g(e,\mathcal{A})$ averages the district-level estimates over the set $\mathcal{I}_{g,e}(\mathcal{A})$ of those districts observed in period $g+e$, of size $n_{g,e}(\mathcal{A}) = |\mathcal{I}_{g,e}(\mathcal{A})|$:

$$\widehat{CATT}_g(e,\mathcal{A}) = \frac{1}{n_{g,e}(\mathcal{A})} \sum_{i \in \mathcal{I}_{g,e}(\mathcal{A})} \widehat{CATT}(g,g+e,W_i,X_i)$$

The event-study $GACRT^{es}(e, \mathcal{A})$ is defined and estimated in the same way, where $\widehat{CACRT}(g, g+e, W_i, X_i) = \hat{\tau}_{g,g+e}(X_i)$.

Event-study standard errors and uniform confidence bands follow the multiplier bootstrap of Callaway and Sant'Anna (2021), clustered by district. The influence functions are the augmented inverse-probability weight (AIPW) scores of the binary causal forests in placebo cells (Sant'Anna and Zhao, 2020), and the deviations of $\Delta\widetilde{Y}_{i,g,t}$ from the cell's treated average in post-treatment cells, which treat $\hat{m}_{g,t}$ as known. See Callaway and Sant'Anna (2021) for the construction of the uniform bands. [12]

We close with some implementation details. In practice, we estimate the following regression with the lm_forest() algorithm from the grf package:

$$\Delta Y_{i,g,t} = m_{g,t}(X_i) + \mathbb{1}\{W_i > 0\}\alpha_{g,t}(X_i) + \tau_{g,t}(X_i)W_i + \varepsilon_{i,g,t}, \qquad i \in \mathcal{S}_g$$

Rather than being estimated in a separate first stage, $\hat{m}_{g,t}(x)$ is the local intercept of this regression, $\hat{m}_{g,t}(x) = \hat{\mathbb{E}}[\Delta Y_{i,g,t} \mid X_i = x] - \hat{\alpha}_{g,t}(x)\,\hat{\mathbb{E}}[\mathbb{1}\{W_i > 0\} \mid X_i = x] - \hat{\tau}_{g,t}(x)\,\hat{\mathbb{E}}[W_i \mid X_i = x]$, where the conditional means are out-of-bag predictions from auxiliary regression forests (375 trees) fit on all units in $\mathcal{S}_g$. This process provides more statistical power to the causal forests in the splitting process because using the full sample (rather than just treated units) allows us to overcome minimum node size constraints, and therefore allows for deeper trees to be grown. In our setting, this is helpful because each cohort $\times$ event-time cell has a relatively small number of treated units.

Each causal forest is estimated with 1,500 trees. For each tree, we hold out half of the subsample for honest estimation, and require at least 20 observations in a resulting child node for a split to occur. We use the default rule-of-thumb for determining the number of variables considered at each split point: if k is the number of covariates, select the minimum of $\sqrt{k} + 20$ or k. When training each tree, units are weighted by enrollment share for population representativeness, and we perform clustering at the district level. Variances of individualized $\left\{\hat{\alpha}(x), \hat{\tau}(x)\right\}$ are constructed using the bootstrap of little bags, which is accomplished by training sets of trees and comparing estimates within and across those sets (Athey et al., 2019).

[12] Each aggregate is a weighted average of district-level $CATT$ estimates. Because leaves are finite in size, they will contain units with slightly different covariate values (i.e. they cannot partition to homogeneous neighborhoods) which may have slightly different counterfactual trends (Gavrilova et al., 2025). This causes bias in individual leaves that dissipates as the sample grows, but more slowly than $n^{-1/2}$. When considering aggregations of leaves, the bias comes mainly from uncertainty in the estimated counterfactual trend $\hat{m}_{g,t}$. In our case, $\hat{m}_{g,t}$ is estimated from a large never-treated sample in any given cohort $\times$ year cell. So we expect this bias to be small.

2.4 Policy Learning via Optimal Decision Trees

A potentially useful exercise which applies our semiparametric estimates of heterogeneous treatment effects is to understand alternative treatment assignment regimes for targeting districts towards charter school expansion. For school districts considering the decision to expand charter schools or not, what would a simple decision rule rooted in our $CACRT$ estimates look like in order to provide guidance for that question? The question we pose is: how should a social planner optimally choose which districts to expand charter schools to maximize the post-treatment outcome defined as either AFGR, Math test scores, or ELA test scores.

Formally, we seek to estimate a decision rule, $\pi \in \Pi$, that maps an individual's observed covariates to a binary decision of treatment assignment. That is, $\pi : \mathcal{X} \to \{0, 1\}$, where $\pi(X_i) = 1$ indicates that the district should marginally expand charter school intensity and $\pi(X_i) = 0$ indicates that the district should not. Kitagawa and Tetenov (2018) specify this optimization problem as one which finds the policy that minimizes regret relative to the best policy, $R(\pi)$, defined as:

$$R(\pi) = \max_{\pi' \in \Pi} \{V(\pi') : \pi' \in \Pi\} - V(\pi) \tag{31}$$

Where $V(\pi) = \frac{d}{dv}\mathbb{E}\big[Y_{i,t}\big(G_i,\, W_i + \upsilon\,\pi(X_i)\big) \,\big|\, G_i \leq t\big]\Big|_{\upsilon=0}$ is the average impact of marginally increasing the charter intensity of the districts selected for expansion under π (Athey and Wager, 2021). Under regularity conditions, $V(\pi) = \mathbb{E}\big[\pi(X_i)\, CACRT(G_i, t, W_i, X_i \mid G_i, W_i, X_i) \,\big|\, G_i \leq t\big]$. Kitagawa and Tetenov (2018) solve this problem with inverse propensity score weighting, but Athey and Wager (2021) have proposed writing Equation 31 in terms of the $CACRT$ (in their case the $CATE$) and estimate the policy by solving:

$$\hat{\pi}_t = \arg\max_{\pi \in \Pi} \left\{ \frac{1}{N_t} \sum_{i:G_i \leq t} \big(2\pi(X_i) - 1\big)\, \hat{\Gamma}_{i,t} : \pi \in \Pi \right\} \tag{32}$$

Where $\hat{\Gamma}_{i,t}$ is the doubly robust score obtained via the AIPW estimator and N_t is the number of districts treated by period t. Chernozhukov et al. (2018) show that AIPW scores are more efficient when aggregating $CATE$ (or $CACRT$ in our case) estimates to higher-level averages. For instance, if we wanted to estimate the $ACRT$ of charter expansion via the causal forest estimators, doing so by averaging AIPW scores of the $CACRT$s (i.e. via $\widehat{ACRT}(t) = \frac{1}{N_t}\sum_{i:G_i \leq t} \hat{\Gamma}_{i,t}$) is more efficient than simply averaging over $CACRT$ estimates. With a single binary treatment, these scores take the familiar form of Nie and Wager (2021):

$$\hat{\Gamma}_{i,t} = \hat{\tau}(X_i) + \frac{W_i - \hat{p}(X_i)}{\hat{p}(X_i)\big(1 - \hat{p}(X_i)\big)} \Big[Y_{i,t} - \hat{\mu}(X_i) - \big(W_i - \hat{p}(X_i)\big)\hat{\tau}(X_i) \Big],$$

where $\hat{\mu}(x) = \hat{\mathbb{E}}[Y_{i,t} \mid X_i = x]$ and $\hat{p}(x) = \hat{\mathbb{E}}[W_i \mid X_i = x]$. This formula does not apply directly to our estimator because lm_forest estimates two coefficients simultaneously: the jump $\alpha_{g,t}(x)$ and the dose slope $\tau_{g,t}(x)$, on two treatment columns $\mathbf{D}_i = (\mathbb{1}\{W_i > 0\}, W_i)'$. We therefore construct the scores using the general form of the semiparametric efficient influence function that Graham and de Xavier Pinto (2022) derive for regressions whose coefficients vary with covariates:

$$\hat{\mathbf{\Gamma}}_{i,t} = \hat{\boldsymbol{\beta}}_{g,t}(X_i) + \hat{\Sigma}(X_i)^{-1}\big(\mathbf{D}_i - \hat{\mathbf{D}}(X_i)\big)\Big[\Delta Y_{i,g,t} - \hat{\mu}(X_i) - \hat{\boldsymbol{\beta}}_{g,t}(X_i)'\big(\mathbf{D}_i - \hat{\mathbf{D}}(X_i)\big)\Big] \tag{33}$$

Here $\hat{\boldsymbol{\beta}}_{g,t}(x) = (\hat{\alpha}_{g,t}(x), \hat{\tau}_{g,t}(x))'$; $\hat{\mu}(x) = \hat{\mathbb{E}}[\Delta Y_{i,g,t} \mid X_i = x]$ and $\hat{\mathbf{D}}(x) = \hat{\mathbb{E}}[\mathbf{D}_i \mid X_i = x]$ are the centering estimates from lm_forest; and $\hat{\Sigma}(x) = \sum_j \omega_j(x)\big(\mathbf{D}_j - \hat{\mathbf{D}}(x)\big)\big(\mathbf{D}_j - \hat{\mathbf{D}}(x)\big)'$ is the covariance of the two treatment columns among districts similar to x, weighted by the forest weights $\omega_j(x)$. The policy trees use the dose element of $\hat{\mathbf{\Gamma}}_{i,t}$, which we denote $\hat{\Gamma}_{i,t}$: the doubly robust counterpart of $\widehat{CACRT}$.

It is important to note that the solution to Equation 32 is the optimal charter expansion policy under an extremely strict set of assumptions. Namely, the solution to Equation 32 assumes that the cost to expand charter schools is equal across all districts and charter intensities. It also assumes that the social welfare function is Utilitarian and perfectly proxied by a single outcome. For this reason, we use the solution policies only to better understand the $CACRT$ function estimates from the causal forests, and do not interpret them as the true optimal treatment allocation policies.

Equation 32 can be solved easily (though inefficiently) by performing an exhaustive search over a limited class of possible policies. Athey and Wager (2021) show that doing this will yield the regret-minimizing policy within that limited class. Here, we search over the space of policies representing depth-2 and depth-3 decision trees. For each treated district, we average scores across all post-treatment event times to obtain a single welfare score $\bar{\Gamma}_i$ (rather than estimating a separate policy tree for each period) which yields a single optimal policy, π, over all districts with $G_i \leq T$. We restrict the covariate vector X_i to the ten moderators with the highest variable importance at $e = 0$. The policytree package (Sverdrup et al., 2020) performs the search to solve Equation 32.

3 Application to Charter Schools: Data

3.1 Outcome Data and Structure

Charter schools are a useful application of this new method because of their continuous, time-varying changes in market share, prior evidence of wide variation in charter treatment effects, the large number of theoretically and empirically motivated moderators, and the wide availability of data with which to test them.

Most treatment effect studies, especially in the charter school arena, focus on participant effects, i.e., the effect of attending a charter school compared with the counterfactual (usually nearby) TPS. This is less useful with charter schools, however, where the rationale for the policy relies not only on participant effects but a range of other market mechanisms whereby charter schools may affect non-participants, too (Chen and Harris, 2023). The entry of a charter school in a given location has been shown to have: *near competitive effects* on the closest public schools (Hoxby, 2003; Bettinger, 2005; Bifulco and Ladd, 2006; Sass, 2006; Zimmer and Buddin, 2009; Imberman, 2011; Linick, 2014; Ridley and Terrier, 2025; Griffith, 2019; Slungaard Mumma, 2022), *far competitive effects* on other traditional public schools located within the geographic school district (Hess, 2002; Holley et al., 2013; Figlio et al., 2023),[13] *closure effects* on low-performing traditional public schools (Engberg et al., 2012; Bross et al., 2023; Carlson and Lavertu, 2016), and *match effects* (Abdulkadiroğlu et al., 2017; Campos and Kearns, 2024). Given the evidence on all of these mechanisms, it is important to capture all of them — what we call the "market-level" effect — to understand charter school policy.

Our focus on market-level effects also shapes our data structure. The unit of analysis is the geographic district, which generally follows the boundaries of the local administrative school districts. This is for two reasons: first, distance is a key factor affecting the choice sets of families and therefore the geographic scope of competitive processes; second, most schools are traditional public schools (TPSs) that are governed and led by elected school boards whose authority resides only within the district's geographic boundaries.[14]

The shape files and other data for these geographic districts come from the National Longitudinal School Database (NLSD) and contain a near-census of all TPSs, charter schools, and private schools in the U.S. from 1991 to the most recent academic year.[15] The NLSD merges school and district-level data from Common Core of Data (CCD), Stanford Education Data Archive (SEDA), Census Small Area Income and Poverty Estimates (SAIPE), and other sources. We are most in-

[13] There is very little evidence on far competitive effects. These citations are mainly case studies of individual districts. Also, the Figlio et al. citation refers to district responses to private schools.

[14] We recognize that the size of these districts varies regionally, e.g., the geographic districts in the U.S. South are much larger geographically than those in the North and some other states.

[15] All the school years mentioned in this paper are spring school years unless specifically stated otherwise.

terested in the student outcomes, test scores and graduation rates, but also include a vector of district-level covariates in some models. The street addresses of charter schools also allow us to place the location of each charter school within its geographic school district. These districts define the scope of the market, i.e., which schools are assumed to compete with one another.[16]

We focus mainly on test score outcomes, i.e., student math and English Language Arts (ELA) test scores in grades 3-8 from 2009-2018 from the Stanford Education Data Archive (SEDA). The SEDA provides nationally comparable, publicly available test score data for U.S. public school districts (Ho, 2020).[17] We normalize test score levels to a mean of zero and standard deviation of unity within the grade, year, and subject.

The AFGR is defined by the U.S. Department of Education and is calculated from aggregate student enrollment data to estimate the size of an incoming freshman class and aggregate counts of the number of diplomas awarded four years later.[18] For example, the AFGR for a school year in 2006 is the total number of diploma recipients in 2006 divided by the average enrollment of the 8th-grade class in 2002, the 9th-grade class in 2003, and the 10th-grade class in 2004. The graduation rate sample includes schools covering grades 9-12 annually from 1995 to 2010.[19]

Treatment status is defined as the charter enrollment share, or the percentage of students attending publicly funded schools who are enrolled in charter schools. These market share measures are created separately by grade level under the theory that charter schools only affect the geographic district outcomes in the specific grades being served (Jinnai, 2014; Slungaard Mumma, 2022).

For both outcomes, data are aggregated to the geographic district level by averaging TPS

[16]The SEDA team uses the 2019 Elementary and Unified School District Boundaries (https://nces.ed.gov/programs/edge/Geographic/DistrictBoundaries) to define the Geographic School District for test score sample (except for special education or virtual schools). We use the same boundaries to define the Geographic School District for the graduation sample.

[17]To make estimates comparable across states, grades, and years. The SEDA research team took the following steps: (1) estimate the location of each state's proficiency "thresholds" in the distribution of scores; (2) place the proficiency thresholds on the same scale using the National Assessment of Educational Progress (NAEP), a test taken by a representative sample of students in each state; (3) estimate the mean test scores in each school, district, county, metropolitan statistical area, commuting zone, and state from the raw data and the threshold estimates, and (4) create estimates of average scale scores and achievement growth measures. See details in SEDA website https://edopportunity.org/methods/. While this method is of course based on assumptions (e.g., that the NAEP sample is truly state representative and the distribution of state scores is the same as the NAEP distribution despite the differing content of these various tests), this could only result in bias if test error in charter-entering districts differs from never-charter districts (within grades and states and conditional on other included covariates) and changes over time in conjunction with charter school entry. We could not come up with potential examples where this might be the case.

[18]The AFGR was the standard federal metric in the years before student-level data systems were prevalent. It was not possible to see where students moved to when they exited districts in these earlier years that are important to our analysis.

[19]The underlying diploma and enrollment data contain missing values, which we address through linear interpolation. Although this may generate estimates for observations with no actual enrollment, such cases are excluded when constructing the final analysis sample using other available information.

outcomes with those of charter schools located within their boundaries[20] (weighted by sector enrollment share). This is crucial to the analysis as it allows us to capture charter effects operating through all the various market mechanisms. In contrast, prior studies of charter schools have focused on the outcomes of individual schools in the examination of school participant effects or the competitive effects of individual charter schools on nearby TPS. Here, we are interested in the total effect of all the mechanisms, which means we need to account for the outcomes of all traditional and charter schools. Aggregating outcomes to the geographic district level also allows us to sidestep the main problem with prior studies, i.e., selection bias from students moving across schools (within districts).

The test score (graduation) data are available for 11,399 (10,658) districts, including 1,597 (1,073) districts that have at least one charter school. The final samples for the test scores (graduation) analyses include 97 (98) percent of the nation's publicly funded schools and 99 (97) percent of the nation's charter schools.[21] Table 1 reports weighted summary statistics for the graduation-rate and test-score samples, including the outcome variables, district demographics, financial variables, and charter-school policy measures.

We estimate total charter effects leveraging the timing of charter entry. One key factor affecting this is the timing of the passage of charter laws that allow them to enter. The first law allowing the establishment of public charter schools was passed in Minnesota in 1991. As of fall 2022, charter school legislation had been passed in 47 states and the District of Columbia.[22] As a result of the growing number of states allowing charter schools, the percentage of publicly funded schools that are charter schools increased from 0 to 7.3 percent, and their percentage of enrollment increased from 0 to 6.2 percent.

3.2 Moderator Variables

Pre-specification is important in effect heterogeneity analysis because there are usually large numbers of potential dimensions that can be studied and the odds of finding some statistically significant patterns by chance are high, i.e., the multiple comparisons problem (Athey and Imbens, 2019). It is also easy to "fish" for some statistically significant pattern. Selecting variables based on theory and prior research makes this less likely as it directs analysis to dimensions where we have good reasons to expect meaningful patterns. In this section, we summarize theory and research that led us to focus on the moderators used later. This section also highlights some of the potential

[20] SEDA provides test score data at the geographic district level. We converted the graduation rate data to the geographic district level ourselves. Please refer to Appendix B for the description of how the sample was created.

[21] Since geographic districts are the unit of analysis, an observation is only missing if all the schools in the unit-by-school-type are missing. There are some missing schools within non-missing districts.

[22] The states in which public charter school legislation had not been passed by that time were Montana, Nebraska, North Dakota, South Dakota, and Vermont.

complexities in the interpretation of results regarding these moderators, setting the stage for the later discussion of results. We organize the moderator variables into four categories: student demographics, market characteristics, school district financing, and charter-related policies.[23]

3.2.1 Student Demographics

Race/ethnicity and parent income. Education is meant to be an equalizer across student background, so such measures are easily the most widely studied moderators. The fact that it is commonly studied also partially explains why this is a commonly identified dimension of effect heterogeneity for charter schools and other education policies. We also note that student background is closely correlated with other moderators discussed below and one objective of this analysis is to help disentangle these potential interpretations.[24]

Charter schools also tend to locate in areas where public school students are racial minorities and come from families with lower incomes. These locations also tend to be more densely populated urban areas that are different from suburban and rural areas on a variety of dimensions that are theoretically related to school market functioning (e.g., families have more schooling choices in more densely populated areas).

These demographic variables are captured in essentially all K-12 education data sets. The income measures are usually in the form of the percent of students eligible for free or reduced-price lunches. The underlying source for these variables in our analysis is the federal Common Core of Data (CCD).

3.2.2 Market Setting Moderators

Baseline school quality. As noted earlier, the common finding that charter schools are more effective for racial/ethnic minorities and low-income students might reflect that their traditional public schools are low-performing; in other words, these students have worse counterfactual options, which increases the odds that alternatives like charter schools will improve their outcomes. To test this at the market level, we use the baseline test scores and graduation rates (from two periods before initial charter adoption to avoid mean reversion concerns).

Market enrollment size. Prior studies have found larger effects in urban districts and urbanicity is related to district size. It may be, for example, that school districts become inefficient once they reach a certain size. Management may become more bureaucratic or centralized administrative units may become poorly informed about school conditions or the needs of specific

[23]These variables were pre-specified in a dated grant proposal written for this study, which is available upon request.

[24]We could also describe this as a "diminishing returns" explanation but we use that term later to refer to the non-linearities in charter market share effects.

neighborhoods.

Teacher supply. The importance of teachers is now widely recognized (Chetty et al., 2014). This is unsurprising given that education is a labor-intensive service and the vast majority of resources go to funding and supporting classroom teachers.

In the charter sector, the strongest evidence of effectiveness also comes from large cities that are attractive to young college graduates (e.g., Boston, Denver, and New Orleans). Also, charter schools tend to rely on young, highly educated teachers, often from elite colleges and universities and/or teachers from outside the local area. Therefore, the variation in charter effectiveness might partially reflect the availability of productive teachers who fit their educational and organizational models. We measure teacher supply in two ways: by whether there is a major university located within the geographic district and the share of the local population aged 25-34.

3.2.3 State Policy Variables

Effect heterogeneity also depends on variation in the treatment itself. In the case of charter schools, this might depend on the state policies that fund, restrict, and guide the work of charter operators.

Following Harris and McKenzie (2027), we also relied on assessments of state policies by charter advocacy organizations. Specifically, we gathered state ratings of charter school policies published in 2014 by the National Alliance for Public Charter Schools (NAPCS). These rankings are useful for understanding regulation because these organizations are well-informed about charter laws and are likely to focus their rankings on policies that are genuinely important, at least to their charter constituents.[25]

NAPCS evaluates state charter laws based on charter school laws that promote the creation of "high-quality charter schools while holding underperforming schools and authorizers accountable."[26] The six most heavily weighted components used for our study include state ratings by performance-based contract required, clear processes for renewal/nonrenewal, automatic exemption of charter schools from laws applied to traditional public schools (TPS), no caps on the growth of public charter schools, and equitable operational funding.

The NAPCS performance-based contract requirement component provides higher ratings to state laws that have requirements for contracts that provide academic performance expectations, operational performance expectations, and school and authorizer rights and duties. The process for renewal component examines whether there are "clear processes" for renewal, nonrenewal, and revocation decisions, including school closure and dissolution procedures. The no

[25]We selected the highest weighted components from each sub-index that are likely to be relevant for school performance, openings, and closures.

[26]https://www.publiccharters.org/publications/model-law-supporting-high-quality-charter-public-schools

charter caps category determines whether there are caps on growth for public charter schools in a state. Equitable operational funding compares general operational, transportation, and other categorical funding of charter schools to TPS.

3.2.4 School Funding

We include additional funding variables beyond NAPCS's equitable school funding. Funding is one of the most widely studied elements in education policy and seems to affect student outcomes (Jackson et al., 2016). However, the role of funding with market-level charter effects is theoretically complex.

If we were studying only participant effects, then we would hypothesize that higher per-pupil charter funding (absolute or relative to nearby traditional public schools (TPS)) would increase charter effectiveness. But the market-level effects include various competitive effects, which means additional resources for TPS should also increase our estimated effects. Which effect dominates depends on the school funding formula. TPS revenue is usually partially buffered from enrollment losses to charter schools either because: (a) a large share of TPS funding comes from local property taxation, irrespective of enrollment; or (b) state funding includes hold-harmless provisions. In the extreme, where TPS funding is completely independent of enrollment, increased charter funding might have no competitive effect. But the market-level effect would still be positive because higher charter spending increases the participant effect. The upshot is that we need to include multiple school funding variables.

Finally, we note that the available funding measures are subject to considerable measurement error. Since the measures are not available at the market-level, we have to construct them from school-level data (in the case of charter schools); while such data are available, they are not widely trusted. One reason is that accurate accounting requires knowing how charter funding "passes through" school districts. The resulting measurement error likely attenuates the estimated moderator role. Taken together, the above points suggest that we should still expect charter effects to be larger when charter spending and TPS spending are both higher.

4 Application to Charter Schools: Results

We focus this section on the test score outcomes—Math and ELA—because of the measurement issues with graduation mentioned in Section 3 and the fact that graduation standards are more subjective and controlled by schools themselves. But we do briefly discuss the graduation rate results below and provide analogous analysis in Appendix Section B.

4.1 Replicating the Average Treatment Effects

Table 1 reports descriptive statistics for the test-score sample, separately for districts that are ever treated by charter school expansion and those that are never treated. Ever-treated districts differ substantially from never-treated districts along several dimensions. They have lower average math and ELA scores, smaller shares of White students, larger shares of Black, Hispanic, and low-income students, and substantially larger enrollments. They are also much more likely to be urban and much less likely to be rural. In contrast, average per-pupil spending and revenue are relatively similar across the two groups. These differences reinforce the importance of accounting for observed district characteristics and fixed unobserved differences across districts in the treatment-effect analysis.

We next estimate average treatment effects of charter-school expansion using the CGS-DiD. Table 2 reports both the average treatment effect on the treated (ATT) and the average causal response to treatment ($ACRT$) for math and ELA achievement. We obtain these by estimating Equation 19 via the two-stage procedure outlined in Section 2.2. Within each cohort g by calendar-year t, we compare cohort-g units with never-treated units over the long difference ($Y_{i,t} - Y_{i,b_g}$). We first regress the outcome change of never-treated units on $(1, X)$ by weighted least squares to obtain $\hat{m}_{g,t}(x)$, then regress the residualized change of treated units on $(1, X, W, WX)$ (Columns 1 and 3 use $(1, W)$ only). Each treated unit's $\widehat{CATT}$ estimate is its fitted value $\hat{\alpha}_0 + X_i'\hat{\alpha}_1 + W_i(\hat{\tau}_0 + X_i'\hat{\tau}_1)$ and its $\widehat{CACRT}$ is $\hat{\tau}_0 + X_i'\hat{\tau}_1$. The ATT and $ACRT$ are averaged over all treated unit-by-year observations, which is equivalent to weighting the cell-level estimates by the share of all treated observations contained in that cell. Both stages are weighted by enrollment. Standard errors are clustered by district, including the first-stage estimation of $\hat{m}_{g,t}$.

As in Chen and Harris (2023), our estimates suggest that charter schools have small, positive market-level effects. At the charter shares treated districts actually experienced (mean 13.4%, median 6.9%), charter entry raised Math and ELA scores by 0.029 and 0.014 s.d. on average relative to no charter schools (ATT), though our estimates are usually imprecise because the two-stage procedure sacrifices statistical power. The $ACRT$ estimates (which, under certain assumptions, are directly comparable to the estimates of Chen and Harris (2023) because their TWFE specifications produced slope coefficients) imply that a 10pp increase in charter market share raises Math and ELA scores by about 0.018 and 0.008 s.d., respectively. For reference, Chen and Harris (2023) estimated slope coefficients of 0.0021 and 0.01 s.d. for Math and ELA, respectively.

4.2 Distribution of Effects

The slightly positive ATT and $ACRT$ estimates reported above may mask substantial treatment-effect heterogeneity, which is the central focus of our study. We therefore examine how both the level and causal response components of treatment effects vary with observed district characteristics, using the linear CGS-DiD and our more flexible semiparametric DiDCF extension. We focus primarily on the DiDCF estimates, with the linear CGS-DiD results serving as a benchmark for comparison.

The first set of results regarding market-level effect heterogeneity are the distributions of effects for each school district. We estimate and plot $CATTs$ for each geographic district $\times$ event-time with the DiDCF (Figures 1 and 2 for math and ELA, and B1 for AFGR). These distributions aim to answer the first primary question of our analysis: is there notable heterogeneity in charter school effects?

For test scores, the effect distributions point to meaningful heterogeneity. The range of math score $CATT$ estimates is from [-0.29, 0.33] for both outcomes, with the distribution appearing Gaussian centered near zero (the $ATT \approx 0.0001$ according to DiDCF). These results suggest that more than half of district $\times$ event-time estimates are negative, with the most negatively impacted districts having lost about 0.3 s.d. in math performance under the charter intensity they adopted relative to what their math performance would have been without charters. However, about 45% of district $\times$ event-time estimates were positive, the largest of which experienced a 0.3 s.d. increase to math scores at their chosen charter intensity. The results are nearly identical for ELA scores: the distribution again ranges from about [-0.26, 0.3] with an $ATT = 0.014$.

As noted earlier, these estimates are the average of each district's effect at their realized dose. That means the variation in the dose itself partially explains the variation in estimates documented in these figures.

Districts with a high $\widehat{CATT}$ estimate for Math scores also tend to have high estimates for ELA scores. Figure C9 displays a binscatter of $\widehat{CATT}$ estimates across the Math and ELA outcomes within district. The correlation is 0.52 across these two outcomes.

4.3 Analyzing Effect Heterogeneity by District Traits

Most of the rest of the study focuses on explaining the above variation across districts, using both the CGS-DiD and DiDCF methods, with priority placed on the DiDCF. The factors are broken into four categories: student demographics (e.g., race and district size), market context (e.g., TPS baseline outcomes and teacher supply metrics), school finance (e.g., charter spending, TPS-charter funding gaps, and total average spending), and other state-level charter school policies. We organize the analysis topic by topic, so that we can directly compare the results across meth-

ods and summarize the results for each set. We also generally focus our attention on the $CATT$ (except for the policy trees where the $CACRT$ is particularly relevant) because Assumption 5 is more easily satisfied than Assumption 6, and the $CATT$ is therefore more likely to be identified.

4.3.1 Our Framework for Interpreting Results

Interpreting heterogeneous treatment effect estimates in the presence of (possibly) complex nonlinearity is a challenging task. The individualized $CATT$ estimates produced by the DiDCF implicitly incorporate any number of possible higher-order polynomials and/or covariate interactions. We therefore developed a protocol for understanding the $CATT$ functionals: (1) establish variable importance, (2) investigate linear patterns in $CATT$ estimates within important individual covariates, and (3) investigate nonlinear patterns within quantiles of important individual covariates and across important covariate interactions. Our interpretation framework focuses on the DiDCF, and the CGS-DiD estimates from the linear parametric model (Table 3) are discussed as a point of comparison.

For the first step, establishing variable importance, we rely on the "Variable Importance Factor (VIF)" produced by causal forests within the DiDCF. The VIF is a simple measure of variable importance and is calculated by computing the share of total trees within each forest that split along a given variable, weighted by the depth at which the split occurred so that earlier splits are weighted more heavily. With a large enough sample and forests with a sufficient number of trees, these scores roughly indicate how much of the heterogeneity is explained by each moderator. This is a more reliable indicator of importance than simply considering the statistical significance of CGS-DiD coefficients for two reasons. First, because CGS-DiD coefficients test only for significant *linear* relationships, they could miss important nonlinear relationships which are accounted for by the VIF. Second, because each split in a CF is chosen conditionally on past splits, a moderator whose information is already captured by another tends to receive fewer splits. In our framework, we focus exclusively on variables that score in the top 10 of VIF. We treat every variable in this list as important in some manner, whether it be linearly or nonlinearly. Because the DiDCF estimates separate forests across event-times, we report these VIF scores in matrix form as Figures 3 and 4 (B2 for AFGR). This allows us to evaluate which variables are important for determining short-run vs. long-run impacts.

The second step of our interpretation framework seeks to establish a "general direction" of the heterogeneity within each important moderator. That is, we look for evidence that a high-VIF moderator exhibits a linear relationship with the $CATT$. We specifically consider two different statistics: (1) Tables 4 and 5 (B5 for AFGR) which report differences in the mean values of each moderator for districts with positive $CATT$ estimates vs. districts with negative $CATT$ estimates, and (2) Tables 6 and 7 (B6 for AFGR) report the Best Linear Predictors (BLP) of the $CATT$

for each outcome among the top 10 highest VIF moderators for DiDCF.[27] The difference-in-means values examine the "profile" of an average positively/negatively impacted district $\times$ event-time. This provides some additional insight into the relative linearity of the relationship between a particular moderator and the $CATT$ estimate. For example, if the mean positively-impacted district for a particular (high VIF) moderator is approximately equal to the mean of the average negatively-impacted district, this is evidence for a non-linear relationship. Conversely, if the two means are notably different, this provides evidence that there is a level difference in moderator values across positively vs. negatively impacted districts. The coefficients in the BLP tables represent the linear relationship between each moderator and the $CATT$ and are obtained by regressing the moderators on the $CATT$ estimate itself. A significant coefficient provides additional evidence that a linear approximation of the relationship between a particular moderator and the $CATT$ is a useful summary parameter.

The third and final step in our interpretation framework seeks to establish nonlinear relationships between high-VIF moderators and the $CATT$, particularly among those moderators that failed to exhibit any clear linear patterns. To do this, we consider three additional sets of results:

1. We report event studies for each outcome in Figures 5 and 6 for DiDCF (we also report these for $CACRT$ using the DiDCF in Figures C11 and C12). These event studies segment units by their moderator value (for the top 5 highest VIF moderators at event-time 0) such that districts falling in each quartile are plotted separately. This allows us to simultaneously assess how each moderator drives heterogeneity across its own distribution in both an overall sense and dynamically. The event-studies also allow us to provide evidence towards the validity of the conditional parallel trends assumption (particularly Assumption 5) by estimating $CATT$s in pre-treatment periods via a placebo indicator for not-yet-treated districts. The DiDCF finds little evidence of diverging trends in pre-treatment periods across ever-treated vs. never-treated units for math scores, but hints that they may be present (if we had more statistical power) for AFGR and (possibly, but less likely) ELA scores.[28][29] We provide a deeper discussion of the consequences to our heterogeneity inferences in the event of linear trend divergences in Appendix A. The key insight is that regardless of the pres-

[27] The causal forest also allows us to consider missingness as a valid dimension of heterogeneity by splitting on missingness itself as a covariate value. We therefore include an indicator for missingness in the BLP for cases where missingness was in the top-10 VIF.

[28] Because these pre-treatment placebo estimates are computed via a binary indicator for "not-yet-treated," we can only estimate pre-treatment $\widehat{CATT}$s, not $\widehat{CACRT}$s ($CACRT$s cannot be estimated without variation in dose). We include these in the $CACRT$ event-studies, but these are notably the same estimates as in the $CATT$ event studies.

[29] We also report the results of an integrated moments test for the conditional parallel trends assumption proposed by Callaway and Sant'Anna (2021) in Table A1. We find no evidence that parallel trends conditional on covariates fails to hold in our data.

ence of district-specific linear time trends, the heterogeneity results remain valid under a plausible set of conditions which appear to be satisfied in our setting.

2. We report shallow-depth decision trees from the policy learning exercise of Athey and Wager (2021) discussed in Section 2.4 in Figures 7 – 10 (B6 and B7 for AFGR). These trees represent the optimal treatment assignment rule among the class of depth-2 (or depth-3) decision trees to maximize the expected outcome in question. These assignment rules are informative of the interactive relationships between moderators within the $CACRT$ function estimate. However, they are not intended to represent the true optimal treatment assignment policy.[30]

3. Finally, in order to further explore non-linearities in treatment effects, we group districts into 10 subgroups which are defined via k-means clustering on X_i, weighted by VIF. The objective is to identify neighborhoods of districts with similar qualities along the important heterogeneity dimensions, and evaluate the characteristics of districts that perform well vs. those which do not. Figure C10 plots separate event studies for each of these 10 groups, and Tables C8 and C9 (Table B7 for AFGR) report the covariate means within each group, sorted by ATT. In these tables, the best performing group of districts is Group 1 and the lowest performing group is Group 10. We view these tables primarily as adding context and nuance to the quartile event study figures. Therefore, we place them in the appendix.

The following subsections will apply this framework to interpret the various figures and tables that arise from CF analysis and to discern the roles of the various moderators. This method of interpretation is one contribution of the present study.

4.3.2 Demographic Moderators

Prior research has consistently found that charter schools are more effective for disadvantaged students, broadly defined (Cohodes and Parham, 2021). We consider five demographic measures: the shares of students who are Black, Hispanic, White, on Free/Reduced Price Lunch (FRL), and in Special Education.

Consistent with prior research, we find that Demographics are among the most important explainers of charter school effects on Math scores. All six measures rank in the top 10 by VIF: FRL (2), Black (3), Hispanic (6), White (8), and Special Ed (9), and FRL generally grows in importance at longer time horizons. As a family of moderators, they explain 45.8% (43.1%) of the heterogeneity

[30]These learned policies are welfare-maximizing under a very strict set of assumptions: the social welfare function must be perfectly proxied by the outcome in question, welfare weights must be Utilitarian, and charter expansion costs must be equivalent across units. We refer the reader to Kitagawa and Tetenov (2018) for a complete discussion of these assumptions.

in Math (ELA) test score effects and 52.9% of the heterogeneity in AFGR effects. The diff-in-means tables show that the districts benefiting from charter schools had higher shares of FRL and Black students, lower shares of White students, and smaller enrollments. The BLP reinforces this only for the last two patterns.

For FRL, the diff-in-means shows that positive-$CATT$ districts had more low-income students, and the event study shows larger impacts in the top quartile of FRL. The BLP, however, detects no linear pattern, and the k-means profiles show that while the districts with the largest CATTs tend to have higher FRL shares, the moderator appears interactive. The policy trees, in which FRL is a key splitting variable at both depth-2 and depth-3, clarify this interaction: low-FRL (more affluent) districts should still expand charter schools if Total Spending and Black are both low, or if Special Ed is low, while high-FRL districts should expand if baseline Math performance is sufficiently low. This suggests that charter schools are more effective in districts at both ends of the income and minority-share distributions (higher-income districts with few minority students and lower-income districts with many minority students), as well as in districts with lower baseline performance.

The race and ethnicity measures (Black, Hispanic, White) follow a similar pattern. The diff-in-means and k-means profiles agree that more Black and/or fewer White students are associated with positively impacted districts. The event study for Black hints that the bottom quartile of Black share had smaller impacts than the rest but shows no clear linear pattern. The depth-3 policy tree elaborates: districts with a high Black share benefit if they also have low spending and high enrollment, while districts with a low Black share can benefit if they also have higher incomes (low FRL) and low spending. White appears in neither the policy trees nor the event studies (it is not in the top 5 by VIF), but the BLP indicates that the $CATT$ declines as the White share rises.

We draw the same general conclusions with ELA. Demographics are again among the most important explainers of heterogeneity. However, the evidence that more non-White students are associated with more positive $CATT$s is somewhat weaker for ELA: the Black share does not differ between positive- and negative-$CATT$ districts. Unlike Math, the policy trees do not split on these measures.

Demographics are also important explainers of effects on the average freshman graduation rate (AFGR): all six measures rank in the top 10 by VIF, led by Log Enrollment (1) and FRL (2), followed by Hispanic (4), Special Ed (7), White (8), and Black (10). The relationships are also even more interactive and nonlinear than for test scores. Only Hispanic and Special Ed suggest a linear relationship with AFGR effects, with the diff-in-means and BLP agreeing in sign and at least one precisely estimated (the Special Education relationship is negatively related to the AFGR effect). See Appendix B for a more thorough discussion of AFGR results.

4.3.3 Market Context Moderators

We include five types of market context variables: baseline academic performance in traditional public schools, region type (urban, suburban, town, and rural), district enrollment size, attractiveness to young teachers (proxied by whether a university is located in the city), the share of magnet schools, and the city population aged 25-34 (as a measure of teacher supply). The last four in this list seem to have little predictive power, but quite the opposite is true for baseline performance.

Baseline achievement has the top-ranked VIF by a wide margin for test scores, across all 26 measures in the analysis, explaining 22.2% (26.3%) of the Math (ELA) test score heterogeneity by itself. It remains at the top in most periods, though its importance diminishes somewhat at longer horizons.

The direction of the relationship is also clear: charter schools are more effective in districts where students had low initial achievement. The diff-in-means, event study, k-means profiles, and policy tree all point this way, although the BLP coefficient is not statistically significant. Positive-$CATT$ districts have notably lower baseline Math scores than negative-$CATT$ districts in the diff-in-means; the bottom quartile of baseline performers is impacted more in nearly every period of the event study; the most positively impacted k-means groups are all lower performing at baseline; and the depth-2 policy tree recommends expansion in low-performing districts with a high FRL share.

The results are similar for ELA where baseline achievement, again, has the highest VIF and the diff-in-means, event study, and k-means profiles each indicate that lower initial performance is associated with larger $CATTs$, pointing to a linear relationship. Unlike Math, the policy trees do not split on baseline ELA.

The baseline outcome is also a consistent, but weaker, explainer of effect heterogeneity when we switch to AFGR as the outcome (VIF rank: 6). No market context variable, including baseline performance, shows a clear linear relationship with AFGR effects. The BLP weakly suggests, and the k-means profiles show more clearly, that districts with lower baseline AFGR are more often positively impacted, and the depth-3 policy tree recommends expansion in districts with low baseline AFGR that have few Black students and are not predominantly White, and so likely have larger Hispanic shares. As with test scores, lower-performing districts benefit more, but the pattern is less clear.

Our takeaway is that the prior literature has focused too much on the demographics of school districts and too little on factors correlated with demographics that have a clear theoretical connection to charter effectiveness. Low achievement levels are one such signal. Disadvantaged students are more likely to attend low-performing traditional public schools, so that the entry of an alternative (charter) schools is more likely to improve district-level outcomes.

4.3.4 School Funding Moderators

We include six main measures of school finance: Total Revenue, Total Spending, Charter Spending, the TPS-Charter Spending Gap, district average Teacher Salaries, and Student-Teacher Ratios. (The first four in this list are per-pupil.)

School finance variables drive heterogeneity in Math effects less than demographics or baseline performance, though they still matter and as a family of variables explain 28.3% (26.3%) of the Math (ELA) heterogeneity (30.7% for AFGR). Student-Teacher Ratio (4), Total Per-Pupil Spending (7), and the TPS-Charter Spending Gap (10) rank in the top 10 by VIF, and Total Per-Pupil Revenue (11) and Teacher Salary (12) just miss, with VIF scores of at least 0.04. Teacher salaries are lower, not higher, in positive-$CATT$ districts for Math.

The diff-in-means suggests that districts more positively impacted by charter schools spend more per student and have lower Student-Teacher Ratios; for both of these moderators, the diff-in-means and k-means profiles show that the districts with the highest $CATTs$ are higher spenders. However, the BLP finds no significant linear relationship for either moderator, and the k-means profile for Student-Teacher Ratio suggests nonlinearity, since the cluster with the highest Student-Teacher Ratio has a positive ATT. Both moderators are key splitting variables in the depth-3 policy tree, so their interactions are informative, and Total Spending is especially so. High-spending districts should expand charter schools when they also have a low Student-Teacher Ratio and low enrollment, or a high Student-Teacher Ratio and a low Special Ed share. Low-spending districts may still benefit if they have higher income families, or high enrollment and a large share of Black students.

These school finance variables are about equally, or perhaps slightly more, important for ELA. The main difference is that variable interactions are much more prominent for ELA. For example, the most positively impacted k-means cluster had the lowest Student-Teacher Ratios, and the event study also suggests the largest $CATTs$ are those districts in the lowest quartile, but neither shows a broader linear relationship. The depth-3 policy tree suggests spending interacts with Student-Teacher Ratio and Log Enrollment: charter schools should generally be expanded in districts with high spending and low Student-Teacher Ratios. Among high-spending districts, expansion is recommended for smaller districts with low Student-Teacher Ratios; among low-spending districts, it is recommended mainly for larger districts with the lowest spending, while the smallest low-spending districts are not recommended for expansion (Figure 10). In contrast to Math, the most positively impacted k-means group has the highest salaries (Table C9).

School finance variables also drive heterogeneity in AFGR effects, but less than for test scores. There are, however, some important interactions. The TPS-Charter Spending Gap (3), Student-Teacher Ratio (5), and Per-Pupil Revenue (9) rank in the top 10 by VIF, and all have largely nonlinear relationships with the $CATT$. But no finance variable shows a clear linear

relationship with AFGR effects. See Appendix B for a deeper discussion.

4.3.5 State Charter Policy Moderators

We consider six policy dimensions drawn from the charter industry groups. These pertain to the extent to which the state has equitable funding, clear charter school startup and contract renewal policies, automatic charter exemption from state/district regulations, caps on charter school growth, and performance-based contracts.

These variables all have extremely low VIF scores with the highest being Exemptions from State/Local Regulations (13). Clear Renewal Policies, and Equitable Funding score above zero, but none exceeds 0.01 at event time 0 (Figure 3). As a family, they determine only 3.6% (4.4%) of the splits for Math (ELA) effects. Perhaps not coincidentally, they reveal almost nothing about the $CATT$. The DiDCF diff-in-means and k-means tables suggest that positive-$CATT$ districts are slightly more likely to be exempt from state/local regulations and slightly less likely to have clear renewal policies, but little else. Because of their low VIFs, these variables are not included in the BLP or quartile event studies, and they do not appear in either policy tree. The results are the same for ELA and AFGR.

One reason that the policy variables might matter less is that this is the only category that is at the state level. We know from other research that, for example, authorizers vary in their priorities and approach (Harris et al., 2026). Most authorizers are local school districts but it is difficult to characterize their specific charter policies and practices. This would also explain why the state-level Equitable Funding measure does not show up as an important variable while the district/market-level funding measures usually do (see section 4.3.4 above).

4.3.6 Diminishing Returns

Another important heterogeneity question is estimating the dose-response functions themselves: $ATT(w)$ and $ACRT(w)$. In other words, one may be interested in understanding how the marginal impact of charter schools changes as we vary the charter intensity, holding constant all other covariates. For this type of heterogeneity, the CGS-DiD and DiDCF are the wrong estimators. Recall from Lemma 5 that our DiDCF procedure specifies the $CATT(\cdot)$ as a partially linear model: $\hat{\alpha}(x) + \hat{\tau}(x)w$. This functional form is meant to flexibly model heterogeneity in treatment effects across covariates x but *within* dose levels, w. It is therefore locally linear within doses rather than a smooth estimator of dose-responses. However, one can still use data-driven non-parametric estimators to obtain estimates of $ATT(w)$ and $ACRT(w)$ by following the approach of Chen et al. (2025), which is the preferred approach of Callaway et al. (2025) as well. Their method involves regressing $\Delta\widetilde{Y}_{i,g,t}$ on flexible K-dimensional transformations of W, which take

the form of cubic B-splines, $\psi^K(W)$, among treated units:

$$\Delta\widetilde{Y}_{i,g,t} = \psi^K(W)'\beta_K + \varepsilon_{i,g,t}$$

The nonparametric estimators of $ATT(w)$ and $ACRT(w)$ are then formed as:

$$\widehat{ATT}(w) = \left(\psi^K(w)\right)'\hat{\beta}_K, \quad \widehat{ACRT}(w) = \left(\partial\psi^K(w)\right)'\hat{\beta}_K$$

Where $\partial\psi^K(s) = [d\psi_{K1}(s)/ds, ..., d\psi_{KK}(s)/ds]$ and $\hat{\beta}_K$ is the K-dimensional vector of OLS estimates. We refer the reader to Chen et al. (2025) and Callaway et al. (2025) for further details on construction of the optimal sieve dimension, K, and construction of confidence bands.[31] We focus on the $\widehat{ACRT}(w)$ function which emphasizes how the marginal effect of additional charter schools changes as the dose increases. For test scores, these are displayed in Figure 11. We find evidence for diminishing marginal returns in both outcomes. While the estimates are expectedly noisy at high dose levels due to lack of power, the marginal impacts of additional charter schools diminish to near-zero at around 50-60% charter share. One of these turns positive again after that, but we have much less confidence in the estimates at these very high charter share levels given the limited density of districts in this range.

4.4 Comparing CGS-DiD to DiDCF

Table 3 reports the results for the CGS-DiD. This is an important point of comparison and shows how conventional methods might miss some of the conclusions we find with our causal forest method.

The CGS-DiD corroborates our earlier finding that districts with lower baseline achievement experience more positive charter school effects, in both Math and ELA. But there are also major differences. The CGS-DiD is inconsistent with the idea that charter schools are more effective for disadvantaged students. Rather, those results suggest more positive effects exist where the White share is higher. The FRL and Hispanic coefficients are in the expected direction, but are imprecise.

Similarly, we see little indication that school spending increases charter school effects according to the linear model. The level coefficients do not even point in the expected direction for five of the 14 estimates and are only in the expected direction and precise for four of them.

[31] It is worth mentioning that identification of the dose-response functions in this manner requires the assumptions undertaken by Callaway et al. (2025), one of which is an unconditional parallel trends assumption that is stronger than the conditional parallel trends assumption we invoke in our main analysis.

4.5 Limitations

While we see our method as an important advance, we note a few important limitations to our charter analysis. First, we chose to exclude districts that are always observed as having charter schools in our sample ("early adopters") because including them would create a cohort of districts with different charter exposure durations. This also has the advantage of greater relevance to current policy decisions, which focus on whether to continue expanding the charter sector. However, we also recognize that this limits generalizability and may be less informative to the early history of charter schools.

This choice also limits the number of districts we observe with high charter shares. After removing always-treated districts, we are left with 495 unique ever-treated districts in our test score sample (855 unique ever-treated districts for the AFGR sample). Because we observe district-grades in the test score sample, we have only 1,718 unique ever-treated district-grades, observed for an average of 3.5 years each for a total of 7,547 ever-treated observations that are spread across cohort $\times$ year estimation cells (an average of about 210 treated units per cell). The main consequence of this is that forests may understate heterogeneity and have large confidence intervals.

Second, the state policy variables are ordinal ratings that take only four or five distinct values and vary only across states. They are therefore less likely to rank highly on the VIF than continuous, district-level variables, partly because of their structure rather than their actual importance (Grömping, 2009). The VIF is a type of "algorithmic" variable importance measure (Williamson and Feng, 2020) because it is associated with the modeling strategy itself. A model-agnostic alternative is the treatment effect variable importance measure (TE-VIM) of Hines et al. (2025): the share of treatment effect heterogeneity lost when a covariate set X_s is removed from the conditioning set, $\big(\operatorname{Var}\{\theta(X)\} - \operatorname{Var}\{\theta_s(X)\}\big)/\operatorname{Var}\{\theta(X)\}$ with $\theta_s(x) = \mathbb{E}[\theta(X) \mid X_{-s} = x_{-s}]$, an R^2-type measure of the heterogeneity uniquely attributable to X_s. They propose estimating this with an influence-function-based approach built on doubly robust pseudo-outcomes, which yields standard errors and does not depend on the share of split points associated with a given covariate. The estimator's guarantees, however, require accurate estimates of individualized treatment effects. As we discussed above, we expect noisy estimates of individualized treatment effects in our sample (that aggregate efficiently into useful heterogeneity conclusions of the $CATT$). In Monte Carlo simulations with 300 treated units per cell, we compute TE-VIMs from the DiDCF doubly-robust scores. In general, the TE-VIMs did not converge until the number of treated units was at least 3,000 in a cell and ranked the most important moderator first only about half the time in a sample of 300. We therefore continue to rely on the VIF as our measure of importance, noting the caveats discussed above and that the Hines et al. (2025) approach is likely better in many settings.

Third, two of our charter funding variables, Charter Expenditure and the TPS-Charter Spending Gap, are set to zero for all never-treated districts (some districts with charter schools are missing for district finance variables, and in the forest these missing values are retained as missing). This is potentially an issue for the validity of the Overlap assumption (Assumption 2) since these values largely separate treated from control units. We estimated all models without these two covariates (results omitted but available upon request) and find that the VIF and broad heterogeneity conclusions discussed in this section are largely unchanged.

Finally, it is worth noting that the $CATT$ we attempt to identify/estimate is consistent *at the realized dose* of treatment the unit adopted. The implication of this is that moderators which predict charter share will also predict the $CATT$ to some extent, making interpretation of heterogeneity along X_i dimensions more difficult. For example, if the $CATT$ tends to increase with dose, as we would expect, and lower performing districts tend to adopt more charter schools, then the heterogeneity we estimate along the baseline performance dimension may partially be capturing the contribution of additional dose to the $CATT$ rather than purely the relationship of baseline performance and the $CATT$. To address this, we estimate the event-studies from the main analysis by segmenting within dose groups in Figures C13 and C14. We find that the core findings still appear to hold: districts with lower baseline performance and/or more FRL students appear to benefit more even within dose groups. We also include BLP specifications in the main BLP tables which control for dose, and find the estimates change very little.

5 Conclusion

We generalize the DiDCF of Gavrilova et al. (2025) such that it may be operationalized in complex quasi-experimental settings, particularly settings of continuous (Callaway et al., 2025) and/or time-varying treatment (De Chaisemartin and D'Haultfœuille, 2026). Within our framework, causal forests can be implemented to discover heterogeneity in treatment effects in a large set of DiD designs. We also outline the assumptions necessary to identify conditional causal parameters under our framework.

A limitation of our method, as with most CF analysis, is that one must generate a large number of statistics in order to understand the heterogeneity. In our case, we consider VIF, diff-in-means, BLP, event studies, policy trees, and k-means plots; all of which we view as necessary to draw conclusions about aspects of how the moderator variables interact with the causal effect of interest. One contribution of our study is to outline an approach to make sense of all this information so that the results generate genuinely useful research conclusions.

Our application to charter schools is a good policy setting for our method and yields important new insights. We broadly ask three questions of charter school heterogeneity. First, from

the perspective of targeting charter school expansion, are certain school districts better/worse candidates for charter schools than others, and is this estimable? While we affirm a role for demographics, our analysis suggests it is not the primary explainer of heterogeneity. Instead, we find that the best predictor for both test score outcomes is the district's baseline achievement level (roughly speaking, the outcome of traditional public schools), which is highly correlated with demographics. This is important. Pointing to demographics alone could easily lead policymakers and practitioners down the wrong road, especially as some urban districts serving disadvantaged students are already high-performing and some non-urban and more affluent districts are low-performing. Our finding also provides further evidence about the importance of counterfactual conditions in treatment effects and extends it beyond just participant effects to include market-level effects. Future research is needed to understand whether charter competitive effects also depend on the counterfactual condition.

Second, from an optimal policy perspective, do funding, accountability, authorizers, or regulations improve the environment for charter school success? State policy effects are conspicuous in their absence. Only school spending seems important and only when we measure at the district/market level. None of the policy variables measured at the state level, including charter accountability and transparency, play an important role. This may be because the real variation in decisions and resources is more localized. A basic premise of the charter concept is to give autonomy to school leaders and the government agencies that authorize them, which leaves considerable room for local discretion, reducing the role of state charter policy variation.

Third, from an optimal concentration perspective, are there diminishing returns to charter schools and at what point is the marginal charter school harmful to student outcomes? While such estimates are naturally imprecise, we find evidence that the effects are most positive at low market shares (i.e., the first one or two charter entrants do the most good) and that the effects diminish from there, hitting zero somewhere in the 50-60 percent range for test scores (and even lower charter shares with AFGR). This is consistent with other research on market competition that the first competitor has a greater impact on the market than additional ones (Bresnahan and Reiss, 1991).

Our analysis also provides insights about additional forms of non-linearity. We compare our results to the linear parametric CGS-DiD, which uses the same cohort-by-period comparisons, the same never-treated comparison group, and the same identifying assumptions as our DiDCF, but specifies the $CATT$ as linear in the covariates. The CGS-DiD only identifies the broadly linear conclusion that baseline achievement is negatively related to charter effectiveness, and has little to say about the other covariates (including demographics). Meanwhile, the DiDCF provides numerous useful insights about, for example, how demographics interact with district financing variables to predict charter effectiveness.

All of this suggests that our method, and related recent advances in the literature, represent a promising new direction for econometric analysis.

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Figures

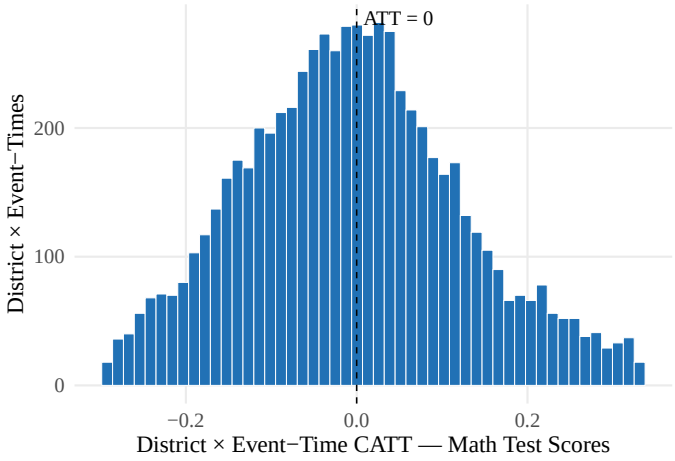


Figure 1. DiDCF CATT Distribution: Math Scores

Notes: Distribution of district × event-time CATT estimates from the DiDCF applied to math test scores (Math). The vertical dashed line marks the ATT after fully aggregating across doses, event-times, and covariates. We drop the top and bottom 5% of CATT estimates from this figure.

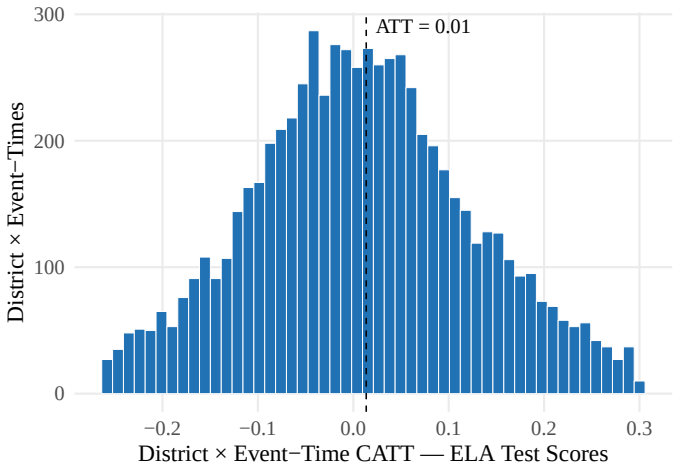


Figure 2. DiDCF CATT Distribution: ELA Scores

Notes: Distribution of district × event-time CATT estimates from the DiDCF applied to ELA test scores (ELA). The vertical dashed line marks the ATT after fully aggregating across doses, event-times, and covariates. We drop the top and bottom 5% of CATT estimates from this figure.



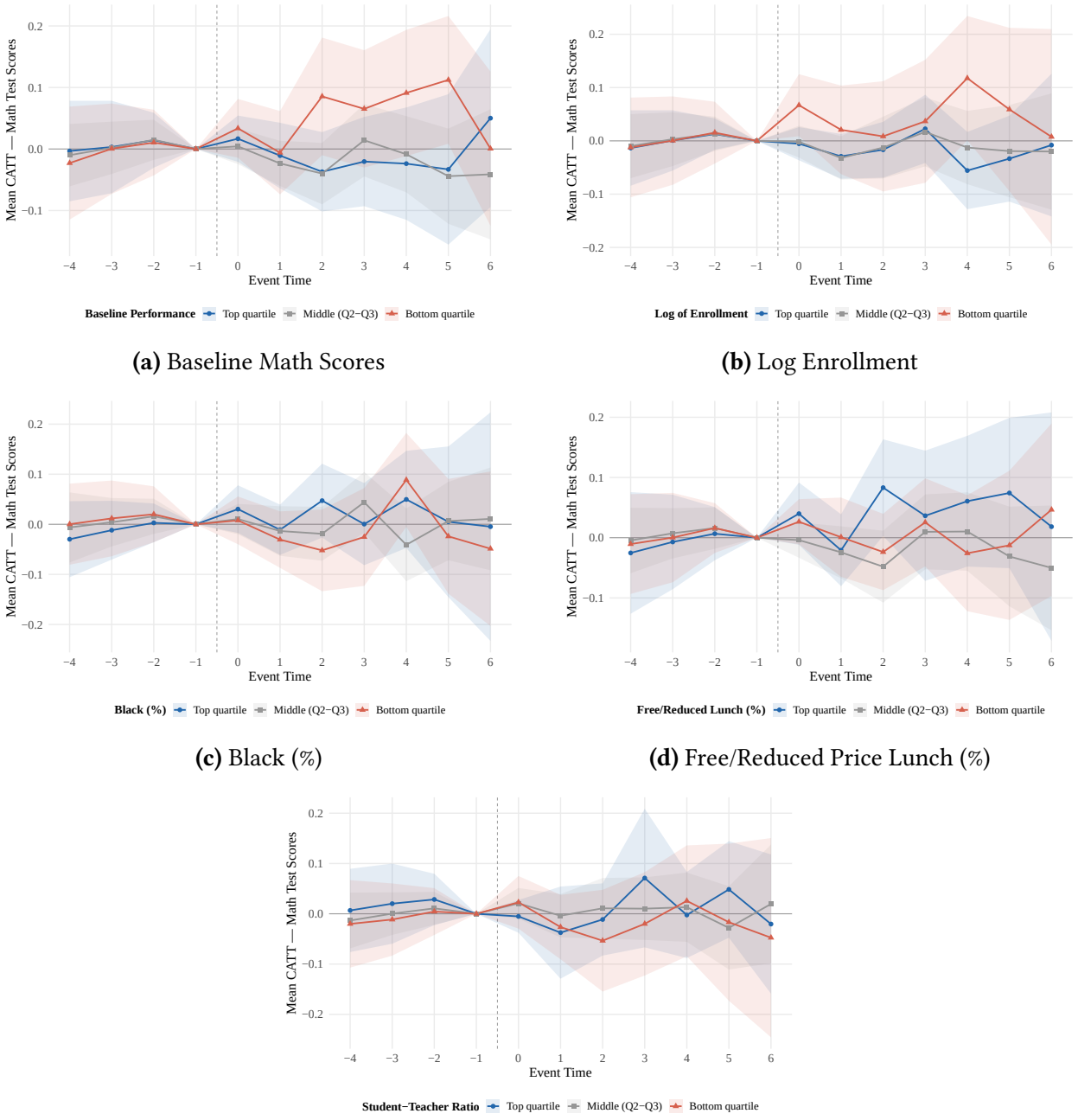
Figure 3. DiDCF Variable Importance Factors: Math Scores

Notes: Heatmap of variable importance factors (VIF) across all covariates and post-treatment event-times ($e = 0, \ldots, 6$) from the DiDCF applied to math test scores. Cell values show the (depth-weighted) share of splits on each covariate at each event-time. Grey cells indicate covariates absent from the data.



Figure 4. DiDCF Variable Importance Factors: ELA Scores

Notes: Heatmap of variable importance factors (VIF) across all covariates and post-treatment event-times ($e = 0, \ldots, 6$) from the DiDCF applied to ELA test scores. Cell values show the (depth-weighted) share of splits on each covariate at each event-time. Grey cells indicate covariates absent from the data.



(a) Baseline Math Scores

(b) Log Enrollment

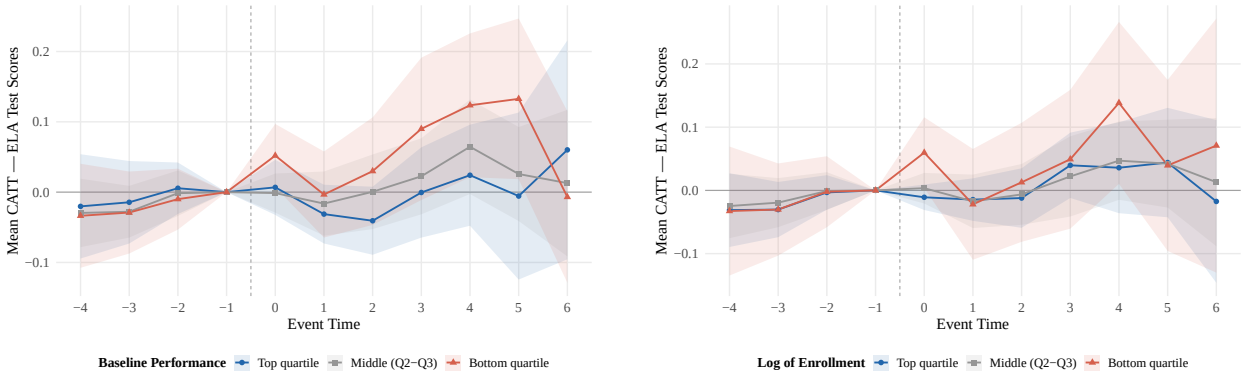
(c) Black (%)

(d) Free/Reduced Price Lunch (%)

(e) Student-Teacher Ratio

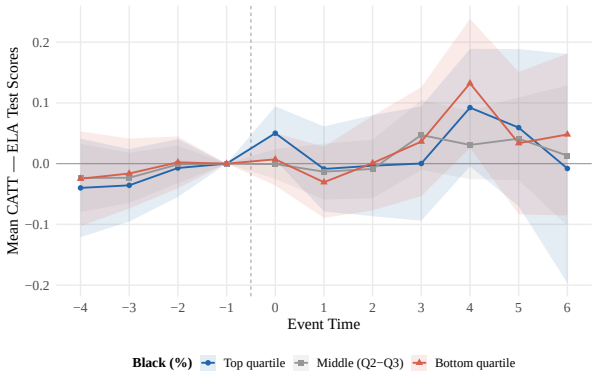
Figure 5. DiDCF Quartile Event Studies (CATT): Math Scores

Notes: Each panel shows $\widehat{GATT}^{es}(e, \mathcal{A})$ estimates (Equation 30). These are group average treatment effects in each event-time, where the groups are districts in the top quartile (blue), middle two quartiles (grey), and bottom quartile (red) of the named moderator. Panels correspond to the five highest-ranked covariates by VIF at $e = 0$ for math scores.

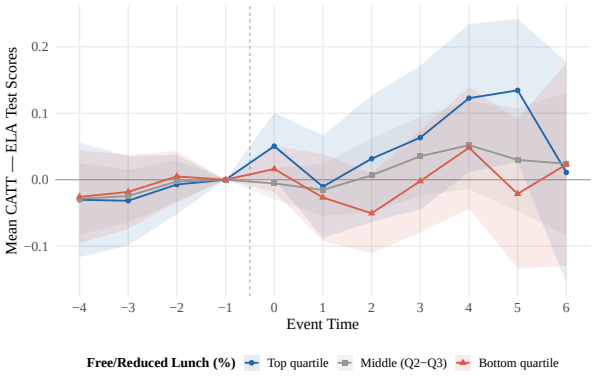


(a) Baseline ELA Scores

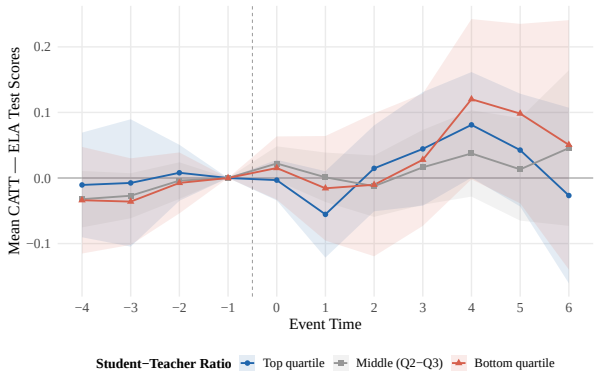
(b) Log Enrollment



(c) Black (%)



(d) Free/Reduced Price Lunch (%)



(e) Student-Teacher Ratio

Figure 6. DiDCF Quartile Event Studies (CATT): ELA Scores

Notes: Each panel shows $\widehat{GATT}^{es}(e, \mathcal{A})$ estimates (Equation 30). These are group average treatment effects in each event-time, where the groups are districts in the top quartile (blue), middle two quartiles (grey), and bottom quartile (red) of the named moderator. Panels correspond to the five highest-ranked covariates by VIF at $e = 0$ for ELA scores.

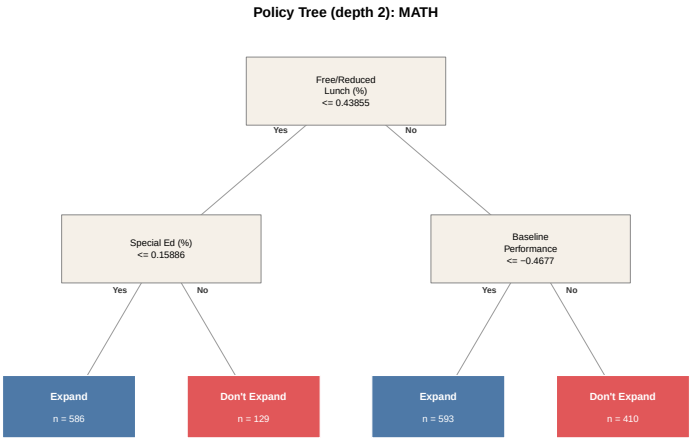


Figure 7. Optimal Policy Tree (Depth 2): Math Scores

Notes: Depth-2 optimal policy tree estimated via the Athey and Wager (2021) framework applied to DiDCF doubly robust scores for the math test scores outcome (Math). Each internal node shows the split variable and threshold; leaf colors indicate the recommended policy action (blue = expand charter market share; red = do not expand). The sample consists of treated districts with non-missing DR scores averaged across post-treatment event times, restricted to the ten highest-VIF moderators at $e = 0$. The doubly-robust scores in question refer to those associated with the $CACRT$ such that the policy prescription refers to the binary decision to "marginally increase" charter intensity or not.

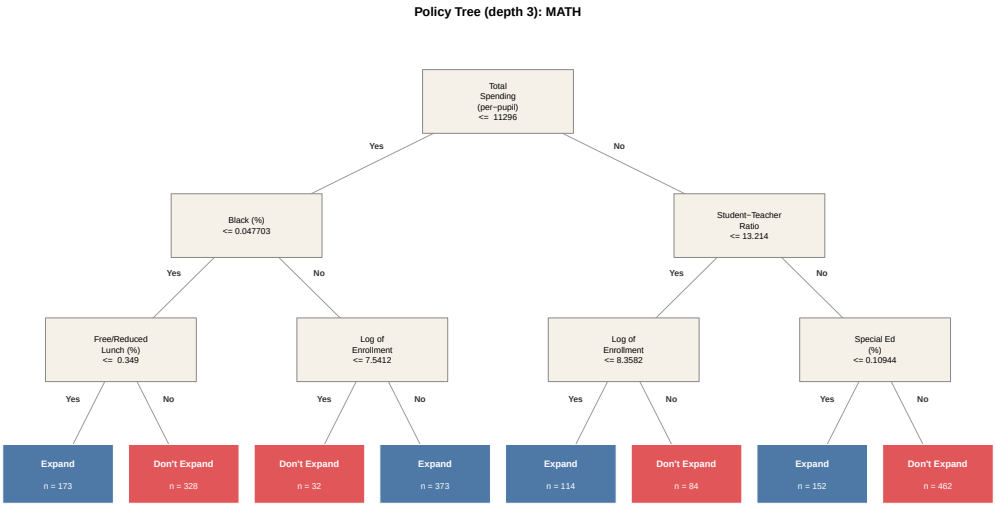


Figure 8. Optimal Policy Tree (Depth 3): Math Scores

Notes: Depth-3 optimal policy tree estimated via the Athey and Wager (2021) framework applied to DiDCF doubly robust scores for the math test scores outcome (Math). Each internal node shows the split variable and threshold; leaf colors indicate the recommended policy action (blue = expand charter market share; red = do not expand). The sample consists of treated districts with non-missing DR scores averaged across post-treatment event times, restricted to the ten highest-VIF moderators at $e = 0$. The doubly-robust scores in question refer to those associated with the $CACRT$ such that the policy prescription refers to the binary decision to "marginally increase" charter intensity or not.

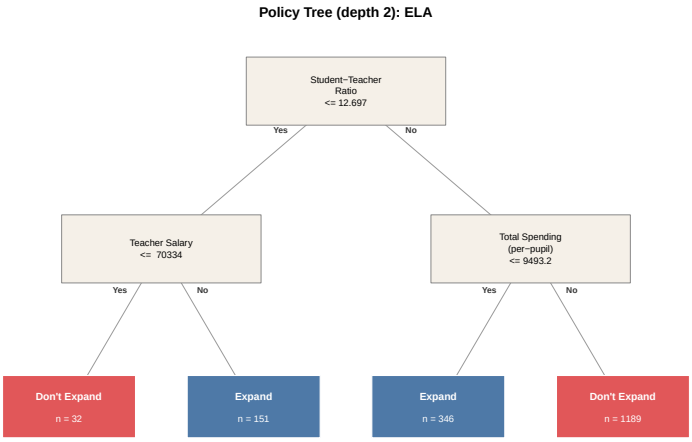


Figure 9. Optimal Policy Tree (Depth 2): ELA Scores

Notes: Depth-2 optimal policy tree estimated via the Athey and Wager (2021) framework applied to DiDCF doubly robust scores for the ELA test scores outcome (ELA). Each internal node shows the split variable and threshold; leaf colors indicate the recommended policy action (blue = expand charter market share; red = do not expand). The sample consists of treated districts with non-missing DR scores averaged across post-treatment event times, restricted to the ten highest-VIF moderators at $e = 0$. The doubly-robust scores in question refer to those associated with the $CACRT$ such that the policy prescription refers to the binary decision to "marginally increase" charter intensity or not.

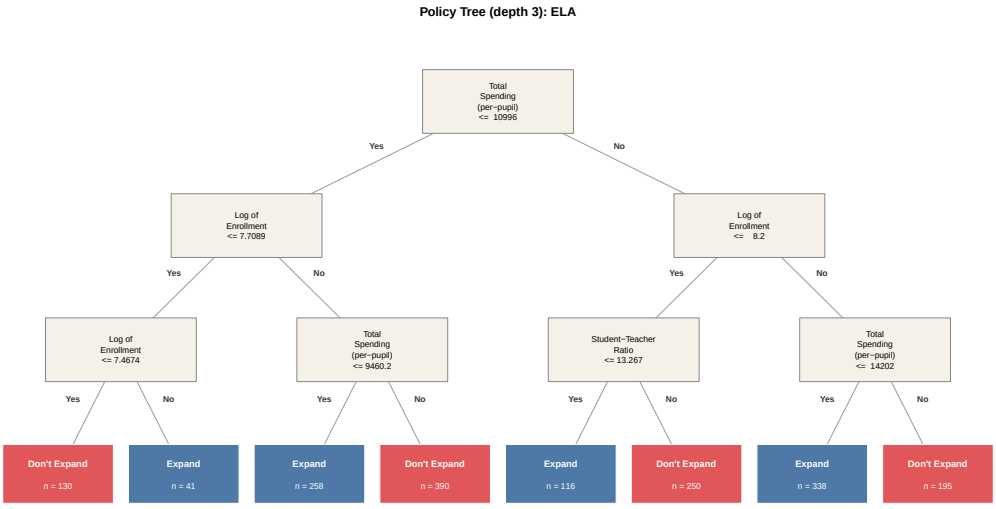
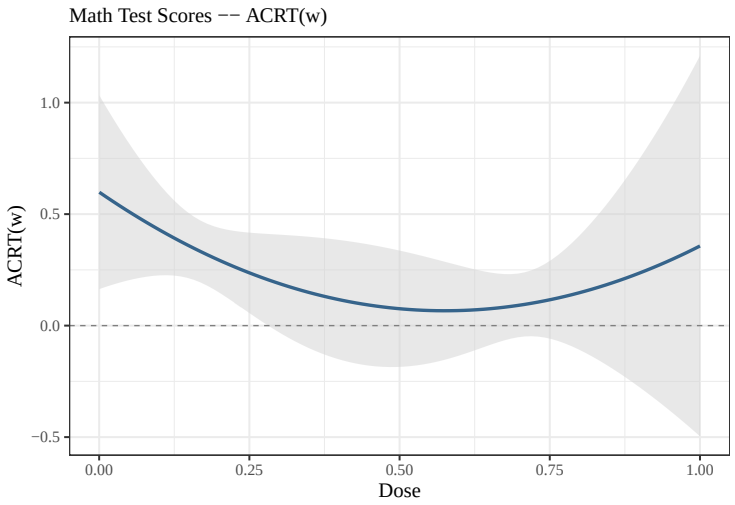
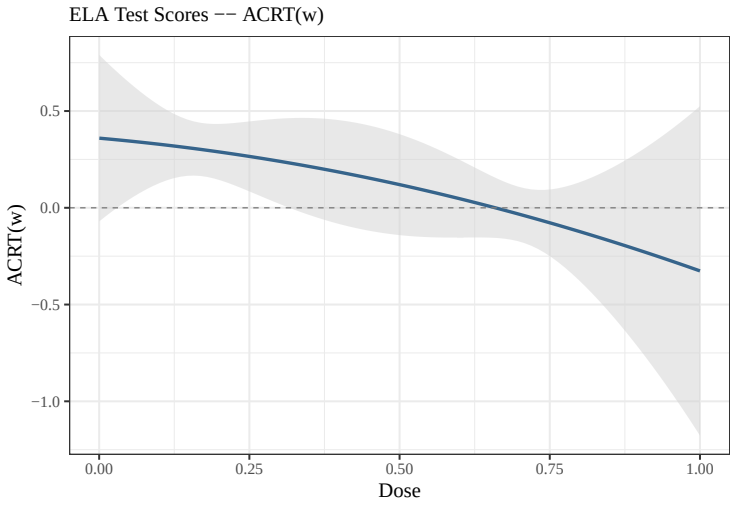


Figure 10. Optimal Policy Tree (Depth 3): ELA Scores

Notes: Depth-3 optimal policy tree estimated via the Athey and Wager (2021) framework applied to DiDCF doubly robust scores for the ELA test scores outcome (ELA). Each internal node shows the split variable and threshold; leaf colors indicate the recommended policy action (blue = expand charter market share; red = do not expand). The sample consists of treated districts with non-missing DR scores averaged across post-treatment event times, restricted to the ten highest-VIF moderators at $e = 0$. The doubly-robust scores in question refer to those associated with the $CACRT$ such that the policy prescription refers to the binary decision to "marginally increase" charter intensity or not.



(a) Math Achievement



(b) ELA Achievement

Figure 11. Diminishing Returns to additional Charter Intensity

Notes: This figure reports $ACRT(w)$ parameters across dose levels for both test score outcomes. These figures are constructed using the Chen et al. (2025) regression approach, where $\Delta\widetilde{Y}_{i,g,t}$ is regressed on cubic B-splines among treated districts. 95% confidence intervals are displayed. These figures are interpreted as the average marginal effect of additional charter schools evaluated at dose w.

Tables

Table 1. Descriptive Statistics for the Test-Score Sample

Variable	Ever-Treated	Never-Treated	Difference	Range
Panel A: Outcomes				
Math test score	-0.253	0.278	-0.531^{***}	(-10.328, 7.533)
ELA test score	-0.277	0.285	-0.563^{***}	(-7.424, 6.256)
Panel B: Student Demographics				
White student share	0.408	0.690	-0.281^{***}	(0.000, 1.000)
Black student share	0.238	0.112	0.126^{***}	(0.000, 1.000)
Hispanic student share	0.285	0.136	0.149^{***}	(0.000, 1.000)
Free or reduced-price lunch share	0.557	0.405	0.152^{***}	(0.000, 1.000)
Special education share	0.125	0.129	-0.005^{***}	(0.000, 0.568)
Panel C: Market Setting				
Urban district	0.483	0.110	0.373^{***}	(0.000, 1.000)
Suburban district	0.400	0.432	-0.032^{***}	(0.000, 1.000)
Town district	0.050	0.173	-0.124^{***}	(0.000, 1.000)
Rural district	0.068	0.285	-0.217^{***}	(0.000, 1.000)
City population (standardized)	-0.019	-0.009	-0.009^{***}	(-1.873, 3.129)
District enrollment size (log)	10.165	8.390	1.775^{***}	(2.639, 13.441)
Number of magnet schools	16.620	0.466	16.154^{***}	(0.000, 266.000)
University presence (standardized)	-0.072	-0.002	-0.070^{***}	(-0.572, 6.346)
Panel D: School Revenue and Spending				
Per-pupil spending ($1,000s)	12.3	12.2	0.1^{***}	(8.3, 23.2)
Per-pupil revenue ($1,000s)	12.0	12.1	-0.1^{***}	(8.5, 22.8)
TPS–charter spending gap	0.124	0.003	0.121^{***}	(-0.657, 1.927)
Equitable operational funding	0.538	0.463	0.076^{***}	(0.000, 1.000)
Per-pupil charter-school spending ($1,000s)	8.3	0.3	8.0^{***}	(0.0, 29.8)
Teacher salary ($1,000s)	99.7	94.7	5.1^{***}	(63.8, 139.6)
Student–teacher ratio	16.504	15.529	0.974^{***}	(10.162, 21.815)
Panel E: State Charter-School Policies				
Regulatory exemptions	0.635	0.560	0.075^{***}	(0.000, 1.000)
No caps on charter-school growth	0.830	0.677	0.153^{***}	(0.000, 1.000)
Clear (non)renewal procedures	0.673	0.608	0.065^{***}	(0.000, 1.000)
Transparency	0.567	0.497	0.070^{***}	(0.000, 1.000)

Notes: Ever-treated districts have positive charter enrollment at some point in the sample; never-treated districts do not. The Difference column reports ever-treated minus never-treated means. Estimates are weighted by grade-level enrollment. $^{*}p < 0.10$, $^{**}p < 0.05$, and $^{***}p < 0.01$.

Table 2. Average Effects of Charter-School Expansion: CGS-DiD

	Math		ELA	
	(1)	(2)	(3)	(4)
ATT	0.004	0.029	0.035**	0.014
	(0.019)	(0.023)	(0.017)	(0.019)
ACRT	0.419***	0.179	0.260**	0.076
	(0.123)	(0.228)	(0.105)	(0.191)
Treated observations	7,547	7,547	7,547	7,547
District controls	No	Yes	No	Yes

Notes: This table reports the ATT and ACRT of charter-school enrollment share using a two-stage staggered CGS-DiD estimator. The first stage uses never-treated districts to estimate counterfactual outcome changes, while the second stage uses treated observations to estimate treatment effects. Columns (1) and (3) exclude moderators, while Columns (2) and (4) include them. The observation counts refer to treated observations. Standard errors are in parentheses. $^{*}p < 0.10$, $^{**}p < 0.05$, and $^{***}p < 0.01$.

Table 3. Heterogeneity in the Effects of Charter-School Expansion: CGS-DiD

Moderator	Component	Math (1)	ELA (2)
Baseline Treatment Effects			
Positive charter exposure	Level	−0.045	−0.219***
		(0.097)	(0.084)
Charter enrollment share	CACRT	−0.283	1.419
		(1.879)	(1.727)
Student Demographics			
White student share	Level	0.020	0.624**
		(0.300)	(0.284)
	CACRT	−0.767	−7.454**
		(3.265)	(3.304)
Black student share	Level	−0.101	0.291
		(0.314)	(0.301)
	CACRT	−2.332	−6.157*
		(3.426)	(3.339)
Hispanic student share	Level	0.043	0.327
		(0.297)	(0.297)
	CACRT	−1.145	−4.380
		(3.215)	(3.342)
Free or reduced-price lunch share	Level	0.071	0.173
		(0.118)	(0.134)
	CACRT	1.018	−0.788
		(1.255)	(1.247)
Special education share	Level	−0.529	−0.744
		(0.601)	(0.559)
	CACRT	2.294	6.733
		(5.212)	(4.754)
Market Setting			
Baseline outcome	Level	−0.046**	−0.049*
		(0.021)	(0.026)
	CACRT	0.048	0.050
		(0.134)	(0.160)
Urban district	Level	−0.084	0.005
		(0.057)	(0.055)
	CACRT	1.032*	−0.801
		(0.623)	(0.614)
Suburban district	Level	−0.202***	−0.050
		(0.057)	(0.053)
	CACRT	2.472***	0.060
		(0.534)	(0.493)
Town district	Level	−0.275***	−0.109**
		(0.057)	(0.053)
	CACRT	3.176***	1.140**
		(0.472)	(0.467)
Log enrollment	Level	0.051**	0.067***
		(0.022)	(0.022)
	CACRT	−1.055***	−1.026***
		(0.206)	(0.201)
Magnet schools	Level	0.013	−0.013
		(0.010)	(0.009)
	CACRT	−0.369	0.272
		(0.292)	(0.275)
School Revenue and Spending			

Continued on next page

Table 3 continued

Moderator	Component	Math (1)	ELA (2)
Per-pupil spending ($1,000s)	Level	0.003	0.002
		(0.012)	(0.011)
	CACRT	−0.022	−0.071
		(0.103)	(0.093)
Per-pupil revenue ($1,000s)	Level	0.026*	0.006
		(0.014)	(0.014)
	CACRT	−0.043	−0.027
		(0.138)	(0.140)
TPS–charter spending gap	Level	0.032	0.182*
		(0.123)	(0.110)
	CACRT	−3.336***	−2.170**
		(0.872)	(0.957)
Equitable operational funding	Level	−0.089	−0.063
		(0.129)	(0.121)
	CACRT	1.308	−1.455
		(1.357)	(1.272)
Charter-school spending ($1,000s)	Level	0.021**	0.019**
		(0.010)	(0.010)
	CACRT	−0.240**	−0.027
		(0.096)	(0.082)
Teacher salary ($1,000s)	Level	−0.004**	−0.000
		(0.002)	(0.002)
	CACRT	0.051***	0.031*
		(0.016)	(0.018)
Student–teacher ratio	Level	0.021**	−0.004
		(0.010)	(0.009)
	CACRT	−0.055	0.143*
		(0.090)	(0.085)
Other State Charter-School Policies			
Regulatory exemptions	Level	−0.019	−0.083
		(0.102)	(0.088)
	CACRT	0.191	0.320
		(0.890)	(0.778)
No caps on charter-school growth	Level	−0.163	0.109
		(0.132)	(0.116)
	CACRT	3.615**	1.245
		(1.466)	(1.292)
Clear renewal and nonrenewal procedures	Level	−0.047	−0.074
		(0.138)	(0.126)
	CACRT	2.205	2.237*
		(1.477)	(1.359)
Transparency	Level	0.252**	0.081
		(0.128)	(0.115)
	CACRT	−2.242*	−0.484
		(1.277)	(1.189)
Performance-based contracts	Level	0.251	0.165
		(0.178)	(0.148)
	CACRT	−4.237**	1.139
		(1.984)	(1.882)
Treated observations		7,547	7,547

Notes: The table reports heterogeneity estimates from the staggered, continuous-dose CGS-DiD, estimated separately in each cohort g by calendar-year t cell using charter enrollment share W and $D = 1\{W > 0\}$. Within a cell, the estimator is equivalent to the first-differenced regression $Y_t - Y_{g-1} = \gamma_0 + X'\gamma + D(\alpha_0 + X'\alpha) + W(\tau_0 + X'\tau) + \varepsilon$ on cohort-g and never-treated district-grades. It is estimated in two stages: the outcome change of never-treated district-grades is regressed on the moderators X, which removes the counterfactual trend $\gamma_0 + X'\gamma$ (not reported), and the residualized change of treated district-grades is then regressed on $(1, X, W, WX)$. Hence $CATT(w, X) = \alpha_0 + X'\alpha + w(\tau_0 + X'\tau)$ and $CACRT(X) = \tau_0 + X'\tau$. Reported coefficients average the cell estimates over post-treatment cells, weighted by each cell's number of treated observations. Moderators are centered at their grade-level-enrollment-weighted sample means, so the baseline rows report the level component α_0 and CACRT τ_0 at the mean moderator vector. For each moderator k, the Level row reports α_k, the coefficient on $D \times X_k$, which is the treatment-by-moderator interaction reported in a binary-treatment difference-in-differences; the CACRT row reports τ_k, the coefficient on $W \times X_k$. Because the dose is continuous, the gradient of the CATT with respect to X_k at dose w is $\alpha_k + \tau_k w$, so α_k is that gradient as $w \to 0^+$. An alternative summary evaluates the gradient at the mean treated dose, $\alpha_k + \tau_k \bar{w}$; we do not report it. Urban, suburban, and town coefficients are interpreted relative to rural districts. Moderators are measured in the first observed year for each district-grade, except the baseline outcome, which is Y_{g-2} centered within each cell. State policy measures are rescaled to $[0, 1]$ before centering. Missing moderator values are set to the sample mean and flagged by missing-value indicators, which are included in the regressions but not reported. Per-pupil spending, per-pupil revenue, charter-school spending, and teacher salary are expressed in thousands of dollars. Both stages are weighted by grade-level enrollment. Standard errors are clustered at the district level and account for estimation of the first stage. Standard errors are shown in parentheses below point estimates. $^{*}p < 0.10$, $^{**}p < 0.05$, and $^{***}p < 0.01$.

Table 4. DiDCF Differences-in-Means: Math Scores

Variable	Mean (pos. CATT)	Mean (neg. CATT)	Difference	p-value
Baseline Performance	-0.31	-0.05	-0.26	0.00
Free/Reduced Lunch (%)	0.53	0.46	0.06	0.00
Black (%)	0.20	0.16	0.04	0.00
Student-Teacher Ratio	15.97	16.36	-0.40	0.01
Log of Enrollment	8.43	8.65	-0.22	0.00
Hispanic (%)	0.21	0.18	0.03	0.02
Total Spending (per-pupil)	12182.03	11781.46	400.58	0.02
White (%)	0.55	0.61	-0.06	0.00
Special Ed (%)	0.13	0.13	0.00	0.39
TPS-Charter Spending (% diff)	0.16	0.22	-0.06	0.01

Notes: For each of the ten moderators with the highest variable importance factor, the table reports the mean value among districts with positive CATT estimates (Mean^{+}), the mean among districts with negative CATT estimates (Mean^{-}), their difference, and the associated p-value of the difference. Rows sorted by variable importance factor.

Table 5. DiDCF Differences-in-Means: ELA Scores

Variable	Mean (pos. CATT)	Mean (neg. CATT)	Difference	p-value
Baseline Performance	-0.33	-0.14	-0.19	0.00
Free/Reduced Lunch (%)	0.51	0.47	0.04	0.00
Student-Teacher Ratio	16.13	16.24	-0.11	0.46
Log of Enrollment	8.46	8.65	-0.19	0.00
Black (%)	0.18	0.17	0.01	0.40
Hispanic (%)	0.21	0.18	0.03	0.01
White (%)	0.56	0.61	-0.05	0.00
Total Spending (per-pupil)	12305.87	11585.57	720.30	0.00
Teacher Salary	94969.76	93290.98	1678.78	0.07
Special Ed (%)	0.13	0.13	0.00	0.21

Notes: For each of the ten moderators with the highest variable importance factor, the table reports the mean value among districts with positive CATT estimates (Mean^{+}), the mean among districts with negative CATT estimates (Mean^{-}), their difference, and the associated p-value of the difference. Rows sorted by variable importance factor.

Table 6. DiDCF Best Linear Predictor: Math Scores

	(1) No dose control			(2) Linear dose			(3) Dose-quintile FE		
	Est.	SE	p	Est.	SE	p	Est.	SE	p
(Intercept)	0.42	0.09	0.00	0.46	0.10	0.00	0.40	0.10	0.00
Baseline Performance	-0.01	0.01	0.31	-0.01	0.01	0.27	-0.01	0.01	0.39
Black (%)	-0.01	0.05	0.80	0.00	0.05	0.93	-0.02	0.05	0.70
Hispanic (%)	-0.09	0.05	0.07	-0.08	0.05	0.10	-0.09	0.05	0.06
Free/Reduced Lunch (%)	0.01	0.04	0.87	0.00	0.04	0.99	0.01	0.04	0.78
White (%)	-0.11	0.05	0.03	-0.11	0.05	0.03	-0.11	0.05	0.03
Special Ed (%)	-0.16	0.14	0.25	-0.17	0.14	0.22	-0.18	0.14	0.22
Log of Enrollment	-0.03	0.01	0.00	-0.03	0.01	0.00	-0.03	0.01	0.00
Total Spending (per-pupil) ($1,000s)	0.00	0.00	0.15	0.00	0.00	0.14	0.00	0.00	0.12
Student-Teacher Ratio	0.00	0.00	0.90	0.00	0.00	0.90	0.00	0.00	0.89
TPS-Charter Spending (% diff)	-0.01	0.02	0.77	-0.01	0.02	0.78	-0.01	0.02	0.73
TPS-Charter Spending (% diff) (missing)	0.06	0.02	0.01	0.06	0.02	0.01	0.05	0.02	0.01
Mean charter share				-0.03	0.03	0.23			

Notes: Best linear predictor (BLP) regression of each district's mean post-treatment $\widehat{CATT}$ estimates on district moderators, applied to math test scores. Each row reports the OLS coefficient of each covariate from regressing the mean post-treatment $\widehat{CATT}$ on the full covariate set. The "missing" indicator on the second instance of Charter Spending refers to a binary indicator for if the district was missing data for Charter Spending. Group (1) reports estimates without dose controls, Group (2) controls for dose linearly, Group (3) includes dose-quintile fixed effects.

Table 7. DiDCF Best Linear Predictor: ELA Scores

	(1) No dose control			(2) Linear dose			(3) Dose-quintile FE		
	Est.	SE	p	Est.	SE	p	Est.	SE	p
(Intercept)	0.39	0.08	0.00	0.27	0.08	0.00	0.25	0.08	0.00
Baseline Performance	0.00	0.01	0.43	0.00	0.01	0.88	0.00	0.01	0.81
Black (%)	-0.03	0.04	0.45	-0.07	0.04	0.11	-0.07	0.04	0.11
Hispanic (%)	0.01	0.04	0.88	-0.02	0.04	0.61	-0.02	0.04	0.58
Free/Reduced Lunch (%)	0.04	0.03	0.21	0.07	0.03	0.03	0.07	0.03	0.03
White (%)	-0.07	0.04	0.07	-0.08	0.04	0.04	-0.08	0.04	0.06
Special Ed (%)	0.06	0.12	0.60	0.11	0.12	0.36	0.04	0.12	0.72
Log of Enrollment	-0.03	0.00	0.00	-0.02	0.00	0.00	-0.02	0.01	0.00
Total Spending (per-pupil) ($1,000s)	0.00	0.00	0.70	0.00	0.00	0.79	0.00	0.00	0.91
Student-Teacher Ratio	0.00	0.00	0.03	0.00	0.00	0.03	0.00	0.00	0.02
Teacher Salary ($1,000s)	0.00	0.00	0.79	0.00	0.00	0.84	0.00	0.00	0.97
Mean charter share				0.13	0.02	0.00			

Notes: Best linear predictor (BLP) regression of each district's mean post-treatment $\widehat{CATT}$ estimates on district moderators, applied to ELA test scores. Each row reports the OLS coefficient of each covariate from regressing the mean post-treatment $\widehat{CATT}$ on the full covariate set. The "missing" indicator on the second instance of Charter Spending refers to a binary indicator for if the district was missing data for Charter Spending. Group (1) reports estimates without dose controls, Group (2) controls for dose linearly, Group (3) includes dose-quintile fixed effects.

Appendix

A Tests of Identifying Assumptions

Table A1. Conditional Parallel Trends Pre-Test

	p-value, by random sample							
Outcome	(1)	(2)	(3)	(4)	(5)	Median	Reject at 5%	Treated
Graduation (AFGR)	0.33	0.67	0.28	0.50	0.56	0.50	0 of 5	171
Math	0.38	0.78	0.58	0.79	0.49	0.58	0 of 5	267
ELA	0.31	0.74	0.48	0.60	0.46	0.48	0 of 5	267

Notes: The table reports the integrated moments test of Callaway and Sant'Anna (2021) for the conditional parallel trends assumption holding in all pre-treatment time periods for all groups. For tractability, we conduct this exercise 5 times with 5 random subsamples of the data. In no draw for any outcome do we reject the null hypothesis that parallel trends holds conditional on covariates in our data.

A.1 District-Specific Time Trends

One important possibility is the existence of district-specific time trends differentially affecting districts that adopt charter schools vs. districts that do not. The pre-trend placebo tests we estimate for the DiDCF suggest this may be present for graduation rates (and possibly ELA scores).[32] However, trend violations which correlate with binary treatment status, but not with overall dose, will not bias our estimates of heterogeneity. Specifically, they will not bias our "ranking" of districts most affected vs. least affected by charter entry nor will they bias our estimates of how the $CATT/CACRT$ varies with x.

To illustrate this, recall the DiDCF's partially linear specification of the $CATT$ from Equation 28:

$$\mathbb{E}[\Delta\widetilde{Y}_{i,g,t} \mid X_i = x] = \alpha_{g,t}(x) + \tau_{g,t}(x)\mathbb{E}[W_i \mid X_i = x] \tag{A1}$$

Going forward we drop the (g,t) subscripts to reduce clutter, noting that everything occurs within a (g,t) cell. Now, imagine we introduce an unobserved random variable within the set of treated districts, δ_i, into the data-generating process which creates differential trends that are uncorrelated with dose:

$$\Delta\widetilde{Y}_i = \alpha(X_i) + \tau(X_i)W_i + \delta_i + \varepsilon_i; \quad \text{Cov}(\delta_i, W_i \mid X_i) = 0; \quad \mathbb{E}[\varepsilon_i \mid W_i, X_i] = 0 \tag{A2}$$

Lemma 6. *Under Equation A2, the $CACRT$ bias equals:*

$$\text{plim } \hat{\tau}(x) - \tau(x) = \frac{\text{Cov}(\delta_i, W_i \mid X_i = x)}{\text{Var}(W_i \mid X_i = x)} = 0 \tag{A3}$$

The CATT bias equals:

$$\begin{aligned} \text{plim } \hat{\alpha}(x) - \alpha(x) &= \mathbb{E}[\delta_i \mid X_i = x] - \big(\text{plim}\,\hat{\tau}(x) - \tau(x)\big)\,\mathbb{E}[W_i \mid X_i = x] \\ &= \mathbb{E}[\delta_i \mid X_i = x] \end{aligned} \tag{A4}$$

[32] It is worth noting that the mechanics of the DiDCF estimator do not allow us to explicitly control for district-specific time trends (linear or otherwise) as is possible in the parametric model.

For proof, see Appendix D.

Lemma 6 states that under the existence of an unobservable covariate causing differential trends across treated districts relative to untreated districts, so long as the unobservable is otherwise uncorrelated with dose (and Conditional Strong Parallel Trends holds), the estimates of the $CACRT$ will be unaffected and all heterogeneity conclusions one may draw from them will remain correct. The unobservable *will*, however, bias the $CATT$ estimate by a scalar equal to the conditional mean of the unobservable.

Given that the $CATT$ will suffer bias under this type of contamination, under which conditions can we be confident that heterogeneity in the $CATT$ will remain valid?

Lemma 7. *Define $\hat{\theta}_i = \theta_i + \eta_i$ as the estimated $CATT$ for district i, which is the sum of the true $CATT$ (θ_i) and the prediction error (η_i). The correlation between the true $CATT$ and the $CATT$ prediction can be stated as:*

$$\text{Corr}(\hat{\theta}, \theta) = \frac{1 + rs}{\sqrt{1 + 2rs + s^2}} \tag{A5}$$

Where $r = \text{Corr}(\theta, \eta)$ and $s = \frac{\text{sd}(\eta)}{\text{sd}(\theta)}$.

Proof. Let $\sigma_\theta^2 = \text{Var}(\theta)$. Then $\text{Cov}(\theta, \hat{\theta}) = \text{Var}(\theta) + \text{Cov}(\theta, \eta) = \sigma_\theta^2(1 + rs)$ and $\text{Var}(\hat{\theta}) = \text{Var}(\theta) + 2\,\text{Cov}(\theta, \eta) + \text{Var}(\eta) = \sigma_\theta^2(1 + 2rs + s^2)$. Divide: $\text{Corr}(\theta, \hat{\theta}) = \sigma_\theta^2(1 + rs)/[\sigma_\theta^2\sqrt{1 + 2rs + s^2}]$, which is Equation A5.

Lemma 7 says that the $CATT$ estimates, despite being biased, preserve their ranking of units under two special cases. First, if $s = 0$ (i.e. the contamination is equal across treated units and is unrelated to X), then $\text{Corr} = 1$ and all $CATT$ heterogeneity results are valid. Second, if $r = 1$ (i.e. if the contamination is proportional to the $CATT$), then $\text{Corr} = 1$ again for any s, because $(1 + s)/\sqrt{(1 + s)^2} = 1$. The case where $CATT$ rankings are not perfectly preserved is if both $s > 0$ and $r < 1$ (i.e. the contamination is not equal across units nor is it proportional to the true $CATT$). For example, in the case where the contamination is completely uncorrelated to the true $CATT$ (i.e. where $r = 0$), $\text{Corr} = 1/\sqrt{1 + s^2}$ and the quality of the heterogeneity estimates degrades as the variance in η grows relative to the variance in θ. It also degrades rather slowly: e.g. if $s = 0.5$ (the standard deviation of the error is half as large as the standard deviation of the $CATT$), the correlation between true $CATT$s and estimated $CATT$s is still $\text{Corr} \approx 0.9$.

In the case where the contamination takes on a district-specific linear time trend, then $\delta_i = \zeta_i(t - b_g)$, where ζ_i is the slope specific to district i. In this setting, $\text{Cov}(\delta_i, W_i \mid X_i) = (t - b_g) \cdot \text{Cov}(\zeta_i, W_i \mid X_i)$. In order for Lemma 6 to hold, it must be that each district's linear trend is uncorrelated with its eventual dose in every period e: $\text{Cov}(\zeta_i, W_i \mid X_i = x) = 0$. We can check this by estimating each treated district's own pre-treatment trend, $\hat{\zeta}_i$ and regressing $\hat{\zeta}_i$ on $(1, W_i, X_i)$ to check if these slopes are conditionally correlated with the district's realized dose. We report these results in Table A2. We find no evidence that linear time trends in the pre-treatment years are correlated with dose, conditional on X_i. This provides support that under Conditional Strong Parallel Trends (Assumption 6), our estimates of $CACRT$ (i.e. $\tau(x)$) parameters remain unbiased even in the presence of linear trend divergence.

For the $CATT$ heterogeneity to remain valid under district-specific linear time trends, we require the linear trends to be either homogeneous across X_i neighborhoods or directly proportional to the true $CATT$. Because true $CATT$s are unknown, it is difficult to provide evidence for the latter condition. However, one can seek visual evidence for the former condition by revisiting the neighborhood event-study plots of Figures C10 and B5. Recall, these figures defined neighborhoods of districts based on covariate

similarity using k-means clustering and estimated event-studies within each group. Evidence for homogeneous pre-trend divergence would appear as low variance in $CATT$ estimates during the pre-treatment periods. This is clearly evident in AFGR and ELA scores – the two outcomes where pre-trends seem to be diverging. We view this as strong evidence supporting the validity of the $CATT$ heterogeneity results for both potentially problematic outcomes.

Table A2. Correlation of District-Specific Linear Time Trends and Charter Intensity

Outcome	Event time	N	Dose coef.	SE	t	p
afgr	pooled	5732	0.0092	0.0070	1.32	0.187
math	pooled	7880	0.0085	0.0445	0.19	0.848
ela	pooled	7880	-0.0061	0.0410	-0.15	0.881

Notes: This table reports the results of a regression of district-specific linear trends on that district's charter intensity, conditional on X.

B AFGR Results

B.1 Average Treatment Effect

In this appendix section, we study the main results concerning the average freshman graduation rate (AFGR) outcome. We begin with the ATT as estimated by the CGS-DiD approach. Table B3 displays these results. Charter schools have a small, negative ATT of -0.001 when considering the top 10 VIF moderators (VIF scores come from the DiDCF), indicating that AFGR was about 0.1pp lower at the realized charter share.[33] The DiDCF estimates an $ATT = 0.004$ as seen in Figure B1. However, the $ACRT = 0.041$ indicates that at the average realized charter dose, the marginal effect of additional charters was positive (an additional 10pp of charter intensity would have increased AFGR by an additional 0.4pp). This indicates that districts which adopted charter schools tended to adopt too few for positive impacts on AFGR to be realized.

B.2 Distribution of Treatment Effects

The distribution of $CATT$ estimates per geographic district $\times$ event-time for AFGR can be found in Figure B1 for the DiDCF. Our estimates suggest there exists notable heterogeneity in $CATT$s, with the 5th and 95th percentiles of DiDCF effects ranging from approximately [-0.06, 0.07]. This indicates there exist districts which experienced as much as a 6pp *decrease* in AFGR at their observed charter intensity, while there are simultaneously districts which experienced as much as a 7pp *increase* in AFGR at their own observed charter share.

B.3 Effect Heterogeneity by District Traits

B.3.1 Trait Set #1: Demographics

Like with test scores, we find that demographics are important explainers of charter school effects when considering graduation rates. Each of the six demographic variables score in the top 10 of VIF: Log Enrollment (1), Free-Reduced Lunch (FRL) % (2), Hispanic % (4), Special Ed % (7), White % (8), and Black % (10). Notably, while Log Enrollment is the most important variable for *short run* impacts according to DiDCF, it diminishes quickly in importance when considering long run effects in favor of FRL % and Hispanic %.

The demographic measures generally do not show evidence of linear relationships with the $CATT$ when considering AFGR. Districts with more FRL students seem to be positively impacted more often than not. Specifically, the event study suggests that districts in the top quartile of FRL share are impacted more. The k-means table reinforces this by showing that the two groups most positively impacted by charter schools are also the two groups with the largest FRL share. However, despite these signs that more low income students is associated with larger $CATT$ estimates, the effects seem to be nonlinear beyond the top quartile, and the diff-in-means and BLP tables do not find notable linear relationships between FRL share and the $CATT$.

The ethnic variables (Black %, Hispanic %, and White %) appear to be highly nonlinear in their relationships with the $CATT$. The diff-in-means and BLP for Hispanic % suggest that more Hispanic students is associated with more positive effects, but the event study and k-means profiles suggest the relationship is much more interactive than the linear measures indicate. Specifically, the k-means table suggests

[33] We have a smaller number of treated units for AFGR than for test scores. Because the 2-stage estimator requires us to separately estimate CGS-DiD regressions per cohort-time cell, there are often too few treated units in an estimation cell to obtain reliable moderator coefficients when using all moderators. To gain back statistical power, we use only the top 10 VIF moderators.

that while the 2nd most positively impacted group of districts had a 78% Hispanic share, several other Hispanic-dominated district groupings fared much worse; the event study similarly shows that both the top *and* bottom quartiles of Hispanic % are impacted more positively than the middle 50%. The depth-3 policy tree sheds some light on Hispanic %'s interactivity. While it does not split on Hispanic % directly, it is clear that in order for a district to be both low Black % and low White %, the district is likely high Hispanic % in most cases. For these high Hispanic districts, charter school expansion is likely to be beneficial if the district is sufficiently low performing at baseline.

Black % seems particularly interactive when considering AFGR. The diff-in-means does not find a notable difference between positive vs. negative $CATT$ districts, and the k-means profiles suggest that while the best performing district group had a very high Black share, other districts with a high Black share did not fare as strongly. The depth-3 policy tree sheds some light on the interactivity for Black share. Specifically, high Black share districts should expand charter school presence if they are simultaneously high FRL (i.e. low income) or high Student-Teacher Ratio (i.e. low spending). Conversely, low Black share districts can also expand charter schools if they are simultaneously large and predominantly White %.

District size (in terms of Log Enrollment) seems to also be related to the $CATT$ in an interactive (and important) manner. The various tables and figures suggest different directions. The diff-in-means weakly suggests that smaller districts are more often better off. The BLP finds no relationship. The k-means table seems to suggest that bigger districts are more often positively impacted. The event studies show some evidence that smaller districts are impacted more. These conflicting results are only reconcilable by a highly interactive relationship, which both policy trees corroborate. Enrollment appears to interact with both spending variables (depth-2 tree) and ethnic demographics (depth-3 tree). In particular, larger districts are more often better off but only when the TPS-Charter spending gap is large or if the district is predominantly White.

To summarize, these findings are similar to those of the two test score outcomes, though the relationships are more interactive and nonlinear. While it is more often the case that districts with more low-income and/or ethnic minority students (in larger districts) are more often positively impacted by charters, it is by no means guaranteed that a district with low-incomes or high non-White share will be positively impacted. Like with math scores, the "demographically homogeneous" districts tend to be the ones more positively impacted.

B.3.2 Trait Set # 2: Market Context

Baseline AFGR ranks #6 in VIF score and remains important over time, indicating it is a consistent explainer of heterogeneity. No other market context variable carries notable information according to VIF. Perhaps because indicators like Urban vs. Suburban vs. Rural are better proxied by Log Enrollment according to DiDCF.

The relationship between a district's baseline AFGR and its $CATT$ appears to be more nonlinear and much weaker than with test scores. Our linear measures do not provide much evidence one way or another, but the BLP does weakly suggest that lower Baseline AFGR districts tend to have larger $CATT$ estimates, and the k-means table is fairly clear (though non linear in some groups) that the most positively impacted groups of districts are those with lower Baseline AFGR. Because Baseline AFGR was #6 in VIF, it does not get an event study, and the only instance of Baseline AFGR appearing in the policy trees was the final split of the depth-3 tree where districts with high White and/or Hispanic shares and low baseline AFGRs should expand charter schools.

To summarize: like with test scores, districts with lower Baseline AFGR are more often positively impacted. But this trend is not as clear as it was with test scores.

B.3.3 Trait Set #3: School Funding

School finance variables appear to drive heterogeneity in AFGR effects less than for test scores, but with some important interactions. The TPS-Charter Spending Difference (3), Student-Teacher Ratio (5), and Per-Pupil Revenue (9) each score in the top 10 for VIF.

These variables appear to have largely nonlinear relationships with the $CATT$. As was discussed in the **Demographics** subsection, the TPS-Charter Spending Gap appears to be interactive with both district Enrollment and Per-Pupil Revenue. Specifically, bigger spending gaps are associated with more positive $CATT$ estimates in larger districts, and smaller gaps are associated with positive effects in districts with low revenue per-student.

The other district finance variables do not seem to have linear or particularly meaningful relationships with the $CATT$. The k-means profiles do not exhibit clear trends for either variable, nor do the diff-in-means and BLP for Student-Teacher Ratio (Teacher Salary falls outside the top 10 by VIF and is not reported in those tables). Student-Teacher Ratio does appear in the depth-3 policy tree, interacting with Black student share and FRL. Specifically, a high Student-Teacher ratio can be associated with positive $CATT$s in low-income, high-Black share districts.

B.3.4 Trait Set #4: State Charter Policy

Like with Math and ELA test scores, the policy setting variables generally score low on VIF. Districts where charter schools are automatically exempt from TPS regulations score #11, and districts with Equitable Funding agreements score #14. No other policy variable generates a VIF score greater than 0.01. Because these variables have low VIF scores, they do not appear in the BLP or quartile event studies, and (unsurprisingly) do not show up in either of the learned policy trees.

Our metrics have little to say about how $CATT$s vary with these moderators. They fall outside the top 10 by VIF, so they do not appear in the CGS-DiD, diff-in-means, or BLP tables. The k-means profiles suggest that these variables are likely interactive as (for example) the best and worst performing groups have a similar share of districts which are exempt from state/local regulations. Regardless, these interactions are likely weak explainers of $CATT$ heterogeneity given they do not appear in the policy trees and have low VIFs.

B.3.5 Trait Set #5: Diminishing Returns

We refer the reader to Section 4.3.6 for a review of the methodology for estimating the dose-response and marginal-response functions. Unlike with test scores where diminishing marginal return patterns were fairly clear, we do not find clear evidence of this for AFGR. We generally lack power to estimate precise $ACRT(w)$ parameters at any dose level for AFGR, and Figure B8 suggests there could be a U-shaped pattern in marginal effects. At low charter share (between 0-10%) there are diminishing-but-positive returns to additional charter schools. Marginal returns from 10-75% charter share are negative, before turning positive again at very high doses.

Table B3. Average Treatment Effects of Charter-School Expansion on Graduation: CGS-DiD with Ten Moderators

	Graduation	
	(1)	(2)
ATT	0.006^{**}	-0.001
	(0.003)	(0.006)
ACRT	0.100^{***}	0.041
	(0.037)	(0.052)
Treated observations	5,732	5,732
Moderators	None	Top 10

Notes: This table reports the ATT and ACRT of charter-school enrollment share using a two-stage staggered CGS-DiD estimator for the AFGR outcome on the top 10 VIF moderators according to the DiDCF for AFGR. The first stage uses never-treated districts to estimate counterfactual outcome changes, while the second stage uses treated observations to estimate treatment effects. Column (1) excludes moderators, while Column (2) includes the top 10 moderators according to VIF (from the DiDCF in $e = 0$). The observation counts refer to treated observations. Standard errors are in parentheses. $^{*}p < 0.10$, $^{**}p < 0.05$, and $^{***}p < 0.01$.

Table B4. Heterogeneity in the Effects of Charter-School Expansion on Graduation: CS-DiD with Ten Moderators

		Graduation
Moderator	Component	(1)
Baseline Treatment Effects		
Positive charter exposure	Level	−0.004
		(0.007)
Charter enrollment share	CACRT	0.049
		(0.056)
Student Demographics		
White student share	Level	−0.071
		(0.047)
	CACRT	0.961
		(0.707)
Black student share	Level	−0.011
		(0.049)
	CACRT	−0.448
		(0.773)
Hispanic student share	Level	−0.006
		(0.049)
	CACRT	0.237
		(0.766)
Free or reduced-price lunch share	Level	−0.086***
		(0.027)
	CACRT	0.204
		(0.364)
Special education share	Level	0.194***
		(0.071)
	CACRT	−3.265***
		(1.084)
Market Setting		
Baseline outcome	Level	0.006
		(0.033)
	CACRT	−1.242***
		(0.426)
Log enrollment	Level	−0.006*
		(0.004)
	CACRT	0.120***
		(0.044)
School Revenue and Spending		
Per-pupil spending ($1,000s)	Level	0.004
		(0.004)
	CACRT	0.036
		(0.061)
Teacher salary ($1,000s)	Level	−0.000
		(0.000)
	CACRT	0.004
		(0.007)
Student–teacher ratio	Level	−0.001
		(0.002)
	CACRT	−0.006
		(0.031)
Treated observations		5,732

Notes: The table reports heterogeneity estimates for the graduation rate from the staggered, continuous-dose CS-DiD, estimated separately in each cohort g by calendar-year t cell using charter enrollment share W and $D = 1\{W > 0\}$. In the first stage, the outcome change $Y_t - Y_{g-1}$ of never-treated districts is regressed on the full set of moderators X, which removes the counterfactual trend $\gamma_0 + X'\gamma$ (not reported). In the second stage, the residualized change of treated districts is regressed on $(1, X_2, W, WX_2)$, where X_2 contains the ten moderators with the highest causal-forest variable importance for ELA test scores in the first treated year, plus indicators for their missing values; taking them from ELA means the moderators are not selected on graduation data. Hence $CATT(w, X_2) = \alpha_0 + X_2'\alpha + w(\tau_0 + X_2'\tau)$ and $CACRT(X_2) = \tau_0 + X_2'\tau$, and each coefficient describes how the CATT varies with that moderator holding the other included moderators fixed. Reported coefficients average the cell estimates over post-treatment cells, weighted by each cell's number of treated observations. Moderators are centered at their weighted sample means, so the baseline rows report α_0 and τ_0 at the mean moderator vector. For each moderator k, the Level row reports α_k, the coefficient on $D \times X_k$, and the CACRT row reports τ_k, the coefficient on $W \times X_k$; the gradient of the CATT with respect to X_k at dose w is $\alpha_k + \tau_k w$. Units are districts. Moderators are measured in the first observed year for each district, except the baseline outcome, which is Y_{g-2} centered within each cell. Missing moderator values are set to the sample mean and flagged by missing-value indicators, which are included in the regressions but not reported. Per-pupil spending and teacher salary are expressed in thousands of dollars. Both stages are weighted by the graduation-rate denominator. Standard errors are clustered at the district level and account for estimation of the first stage. Standard errors are shown in parentheses below point estimates. $^{*}p < 0.10$, $^{**}p < 0.05$, and $^{***}p < 0.01$.

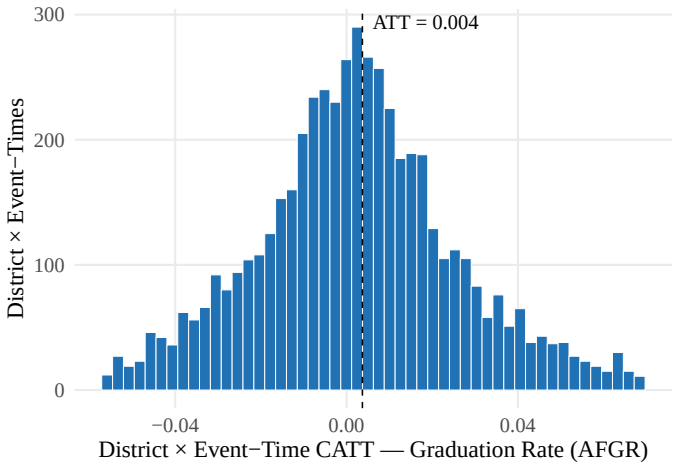


Figure B1. DiDCF CATT Distribution: Graduation Rate (AFGR)

Notes: Distribution of district × event-time CATT estimates from the DiDCF applied to graduation rates (AFGR). The vertical dashed line marks the ATT after fully aggregating across doses, event-times, and covariates. We drop the top and bottom 5% of CATT estimates from this figure.

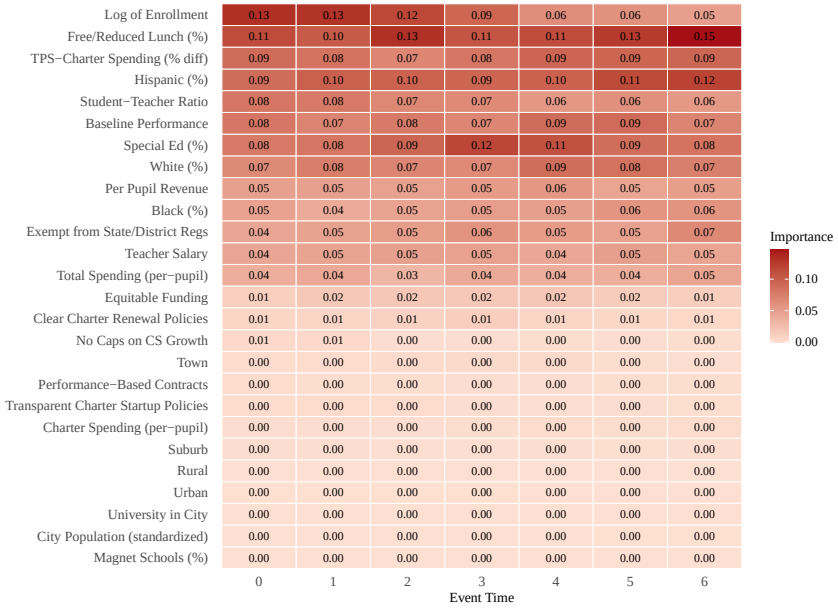
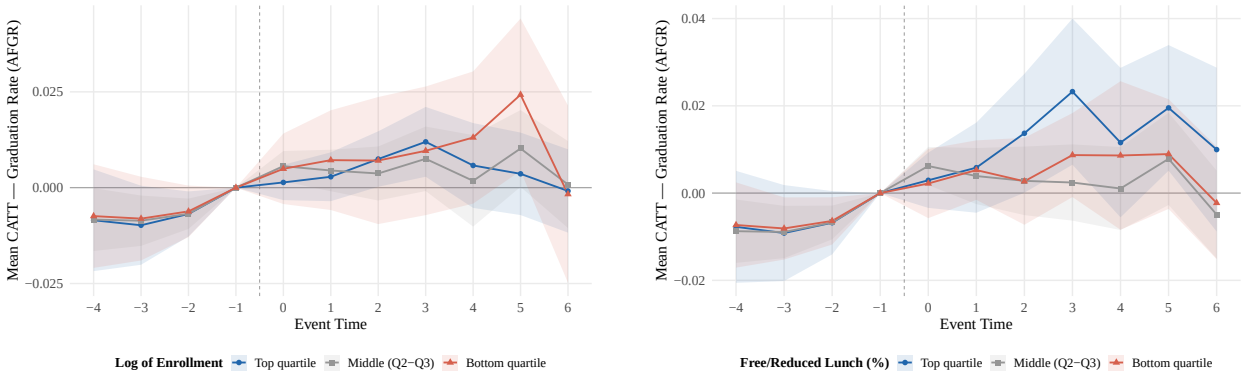


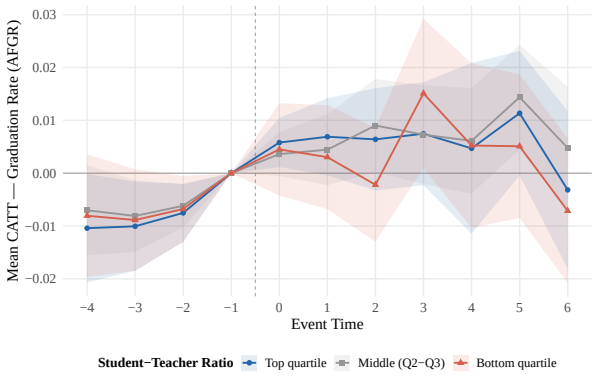
Figure B2. DiDCF Variable Importance Factors: Graduation Rate (AFGR)

Notes: Heatmap of variable importance factors (VIF) across all covariates and post-treatment event-times ($e = 0, \ldots, 6$) from the DiDCF applied to graduation rates. Cell values show the (depth-weighted) share of splits on each covariate at each event-time. Grey cells indicate covariates absent from the data.

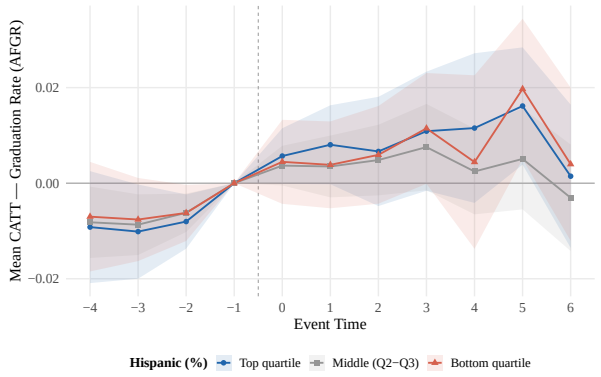


(a) Log Enrollment

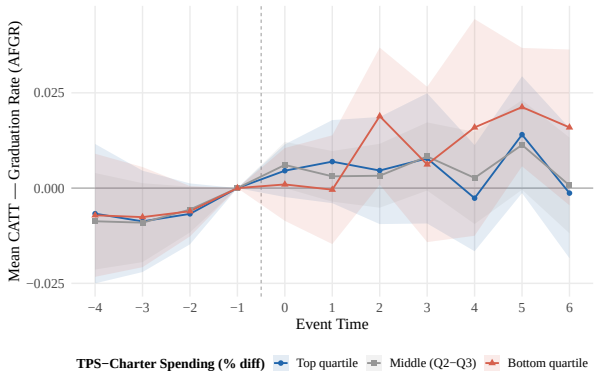
(b) Percent Free/Reduced Lunch



(c) Student-Teacher Ratio



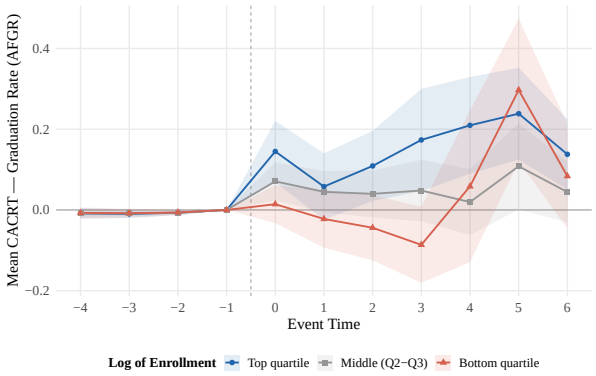
(d) Percent Hispanic



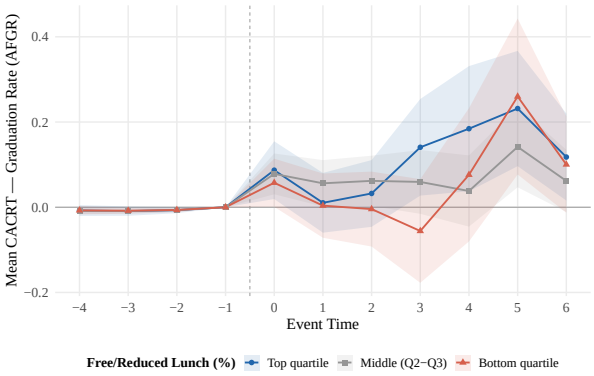
(e) % Difference TPS vs. Charter Spending

Figure B3. DiDCF Quartile Event Studies (CATT): Graduation Rate (AFGR)

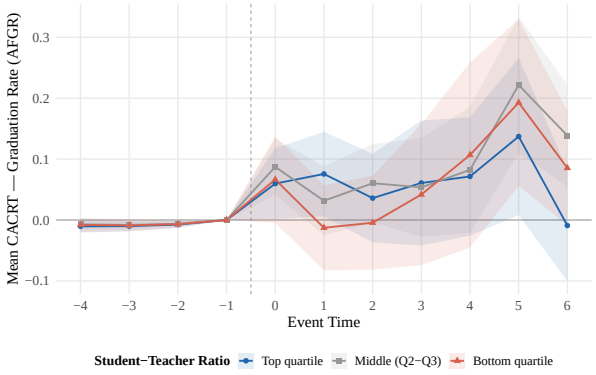
Notes: Each panel shows $\widehat{GATT}^{es}(e, \mathcal{A})$ estimates (Equation 30). These are group average treatment effects in each event-time, where the groups are districts in the top quartile (blue), middle two quartiles (grey), and bottom quartile (red) of the named moderator. Panels correspond to the five highest-ranked covariates by VIF at $e = 0$ for graduation rates.



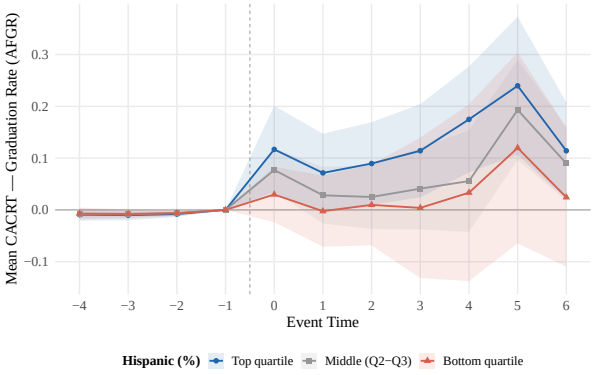
(a) Log Enrollment



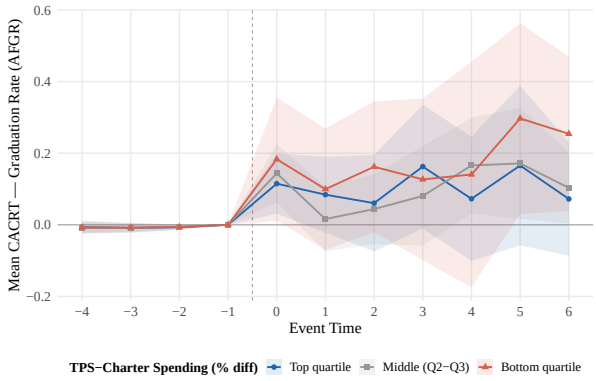
(b) Percent Free/Reduced Lunch



(c) Student-Teacher Ratio



(d) Percent Hispanic



(e) % Difference TPS vs. Charter Spending

Figure B4. DiDCF Quartile Event Studies (CACRT): Graduation Rate (AFGR)

Notes: Each panel shows $\widehat{GACRT}^{es}(e, \mathcal{A})$ estimates (Equation 30). These are group average causal responses in each event-time, where the groups are districts in the top quartile (blue), middle two quartiles (grey), and bottom quartile (red) of the named moderator. Panels correspond to the five highest-ranked covariates by VIF at $e = 0$ for graduation rates.

Table B5. DiDCF Differences-in-Means: Graduation Rate (AFGR)

Variable	Mean (pos. CATT)	Mean (neg. CATT)	Difference	p-value
Log of Enrollment	8.69	8.75	-0.05	0.57
Free/Reduced Lunch (%)	0.24	0.22	0.02	0.18
TPS-Charter Spending (% diff)	0.23	0.19	0.05	0.13
Hispanic (%)	0.13	0.10	0.03	0.01
Student-Teacher Ratio	18.01	17.85	0.16	0.44
Baseline Performance	0.75	0.76	-0.01	0.56
Special Ed (%)	0.10	0.11	-0.01	0.00
White (%)	0.70	0.73	-0.03	0.13
Per Pupil Revenue	6294.47	6311.38	-16.91	0.90
Black (%)	0.12	0.12	0.00	0.86

Notes: For each of the ten moderators with the highest variable importance factor, the table reports the mean value among districts with positive CATT estimates (Mean^{+}), the mean among districts with negative CATT estimates (Mean^{-}), their difference, and the associated p-value of the difference. Rows sorted by variable importance factor.

Table B6. DiDCF Best Linear Predictor: Graduation Rate (AFGR)

	(1) No dose control			(2) Linear dose			(3) Dose-quintile FE		
	Est.	SE	p	Est.	SE	p	Est.	SE	p
(Intercept)	-0.03	0.02	0.21	-0.05	0.02	0.06	-0.04	0.02	0.13
Baseline Performance	-0.01	0.01	0.13	-0.01	0.01	0.25	-0.01	0.01	0.22
Black (%)	0.02	0.01	0.02	0.02	0.01	0.02	0.02	0.01	0.03
Hispanic (%)	0.05	0.01	0.00	0.05	0.01	0.00	0.05	0.01	0.00
Free/Reduced Lunch (%)	0.01	0.01	0.12	0.01	0.01	0.18	0.01	0.01	0.17
White (%)	0.03	0.01	0.00	0.03	0.01	0.00	0.03	0.01	0.00
Special Ed (%)	-0.03	0.02	0.08	-0.03	0.02	0.16	-0.03	0.02	0.14
Log of Enrollment	0.00	0.00	0.63	0.00	0.00	0.72	0.00	0.00	0.91
Per Pupil Revenue ($1,000s)	0.00	0.00	0.59	0.00	0.00	0.53	0.00	0.00	0.52
Student-Teacher Ratio	0.00	0.00	0.81	0.00	0.00	0.98	0.00	0.00	0.79
TPS-Charter Spending (% diff)	0.01	0.01	0.38	0.01	0.01	0.30	0.01	0.01	0.35
TPS-Charter Spending (% diff) (missing)	-0.01	0.00	0.21	-0.01	0.00	0.26	-0.01	0.00	0.21
Mean charter share				0.02	0.01	0.01			

Notes: Best linear predictor (BLP) regression of each district's mean post-treatment $\widehat{CATT}$ estimates on district moderators, applied to graduation rates. Each row reports the OLS coefficient of each covariate from regressing the mean post-treatment $\widehat{CATT}$ on the full covariate set. The "missing" indicator on the second instance of Charter Spending refers to a binary indicator for if the district was missing data for Charter Spending. Group (1) reports estimates without dose controls, Group (2) controls for dose linearly, Group (3) includes dose-quintile fixed effects.

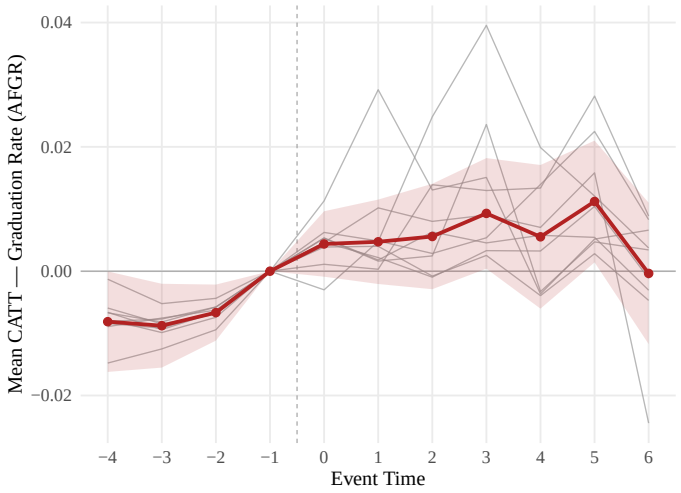


Figure B5. DiDCF K-Means Matched Event Studies (CATT): AFGR

Notes: Each panel shows $\widehat{GATT}^{es}(e, \mathcal{A})$ estimates (Equation 30). These are group average treatment effects in each event-time, where the groups are districts in each of the 10 k-means clusters. Each gray line is a single cluster, the red line is the overall event study line.

Table B7. DiDCF Differences-in-Means (k-means matched groups): AFGR

	Covariate Grouping								
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
Log of Enrollment	8.92	9.40	10.07	8.80	10.57	9.51	7.57	7.05	8.14
Free/Reduced Lunch (%)	0.46	0.58	0.18	0.13	0.42	0.27	0.14	0.38	0.11
TPS-Charter Spending (% diff)	0.27	0.06	0.12	0.40	0.32	0.19	0.19	0.05	0.14
Hispanic (%)	0.05	0.78	0.10	0.04	0.13	0.31	0.02	0.13	0.03
Student-Teacher Ratio	16.08	18.50	17.18	20.11	17.36	22.18	15.71	15.84	18.44
Baseline Performance	0.59	0.63	0.70	0.82	0.52	0.75	0.87	0.74	0.80
Special Ed (%)	0.10	0.10	0.13	0.11	0.09	0.10	0.12	0.15	0.04
White (%)	0.40	0.15	0.69	0.90	0.22	0.49	0.93	0.72	0.91
Per Pupil Revenue	5969.98	5700.17	5733.96	6672.18	7864.86	5222.69	6596.92	8085.67	5422.08
Black (%)	0.46	0.04	0.18	0.03	0.62	0.08	0.02	0.04	0.04
Exempt from State/District Regs	0.72	0.71	0.64	0.49	0.54	0.76	0.50	0.70	0.71
Teacher Salary	54171.63	61972.37	56402.06	73471.05	77147.06	69418.35	55852.93	61532.26	56791.41
Total Spending (per-pupil)	5938.67	5945.22	5754.55	6730.30	7853.69	5361.01	6555.30	7671.86	5419.73
Equitable Funding	0.45	0.75	0.43	0.51	0.50	0.86	0.18	0.58	0.46
Clear Charter Renewal Policies	0.71	0.65	0.64	0.57	0.62	0.73	0.44	0.63	0.75
No Caps on CS Growth	0.78	0.74	0.89	0.80	0.75	0.78	0.80	0.79	0.69
Town	0.19	0.25	0.02	0.23	0.00	0.09	0.33	0.25	0.33
Performance-Based Contracts	0.59	0.58	0.62	0.56	0.56	0.32	0.49	0.51	0.50
Transparent Charter Startup Policies	0.57	0.51	0.57	0.59	0.51	0.51	0.39	0.56	0.43
Charter Spending (per-pupil)	10857.94	9553.39	9526.70	9098.00	12508.40	10032.24	11629.57	13351.44	9586.67
Suburb	0.16	0.22	0.40	0.46	0.16	0.51	0.14	0.14	0.35
Rural	0.29	0.03	0.13	0.09	0.00	0.07	0.45	0.57	0.22
Urban	0.36	0.50	0.44	0.22	0.84	0.33	0.07	0.04	0.10
University in City	0.01	-0.03	-0.03	0.02	-0.11	-0.01	0.00	0.02	0.00
City Population (standardized)	-0.00	-0.05	-0.06	0.00	0.15	0.02	0.00	0.04	0.00
Magnet Schools (%)	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
ATT	0.01	0.01	-0.00	-0.00	0.00	0.00	-0.00	-0.01	-0.01
ACRT	-0.04	-0.05	0.04	0.01	-0.01	-0.02	-0.10	-0.02	-0.03

Notes: Each panel shows $\widehat{GATT}^{es}(e, \mathcal{A})$ estimates (Equation 30). These are group average treatment effects in each event-time, where the groups are districts in each of the 10 k-means clusters. Panels correspond to the five highest-ranked covariates by VIF at $e = 0$ for graduation rates.



Figure B6. Optimal Policy Tree (Depth 2): Graduation Rate (AFGR)

Notes: Depth-2 optimal policy tree estimated via the Athey and Wager (2021) framework applied to DiDCF doubly robust scores for the graduation rate outcome (AFGR). Each internal node shows the split variable and threshold; leaf colors indicate the recommended policy action (blue = expand charter market share; red = do not expand). The sample consists of treated districts with non-missing DR scores averaged across post-treatment event times, restricted to the ten highest-VIF moderators at $e = 0$. The doubly-robust scores in question refer to those associated with CACRT such that the policy prescription refers to the binary decision to "marginally increase" charter intensity or not.

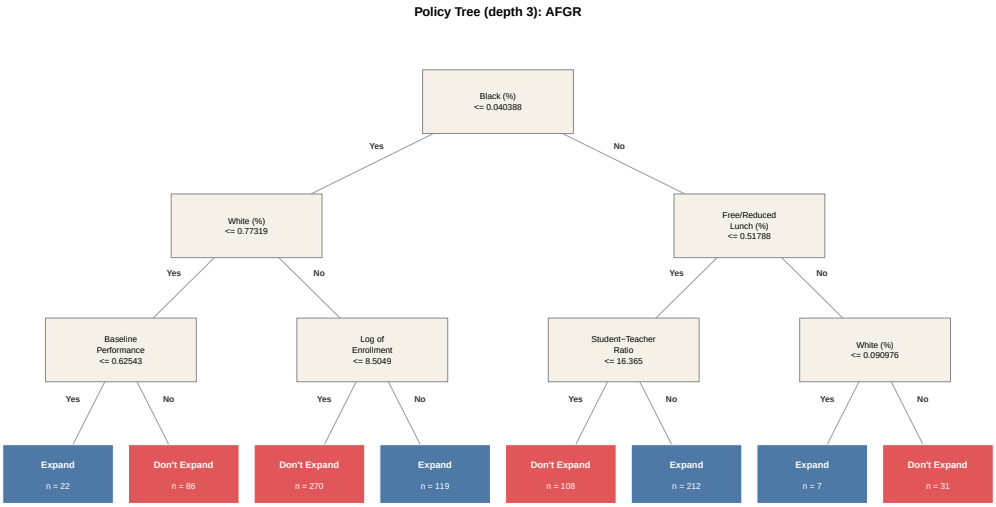


Figure B7. Optimal Policy Tree (Depth 3): Graduation Rate (AFGR)

Notes: Depth-3 optimal policy tree estimated via the Athey and Wager (2021) framework applied to DiDCF doubly robust scores for the graduation rate outcome (AFGR). Each internal node shows the split variable and threshold; leaf colors indicate the recommended policy action (blue = expand charter market share; red = do not expand). The sample consists of treated districts with non-missing DR scores averaged across post-treatment event times, restricted to the ten highest-VIF moderators at $e = 0$. The doubly-robust scores in question refer to those associated with CACRT such that the policy prescription refers to the binary decision to "marginally increase" charter intensity or not.

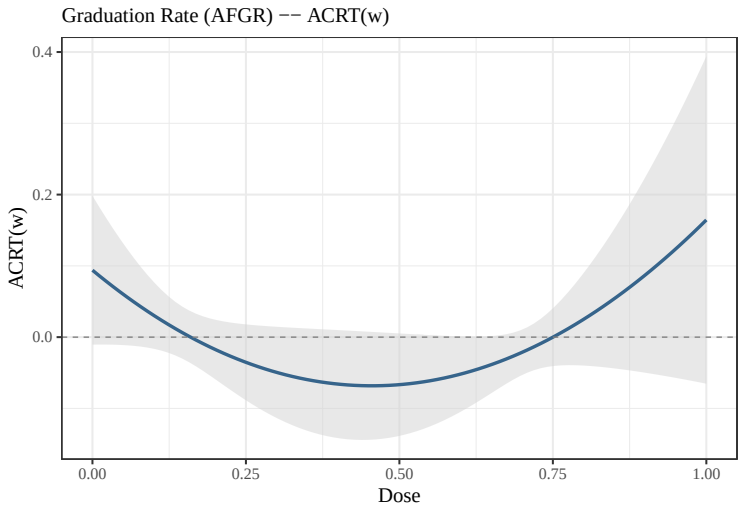


Figure B8. Diminishing Returns to additional Charter Intensity (DiDCF) – AFGR

Notes: This figure reports $ACRT(w)$ parameters (Callaway et al., 2025) across dose levels for the graduation rate outcome. 95% confidence intervals are displayed. This figure is interpreted as the average marginal effect of additional charter schools evaluated at dose w.

C Additional Effect Heterogeneity

C.1 Math vs. ELA Correlation

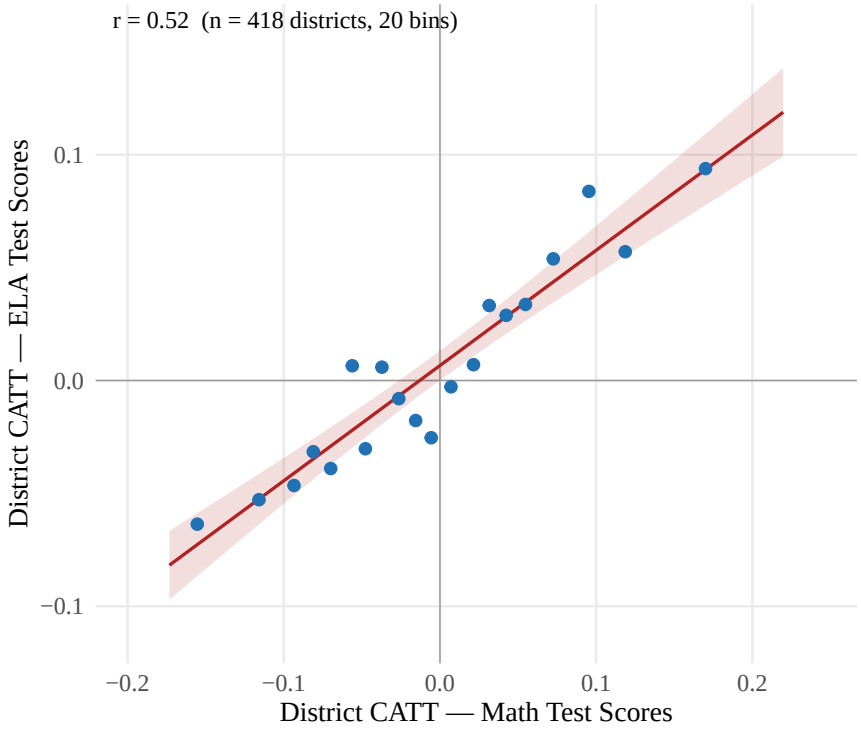
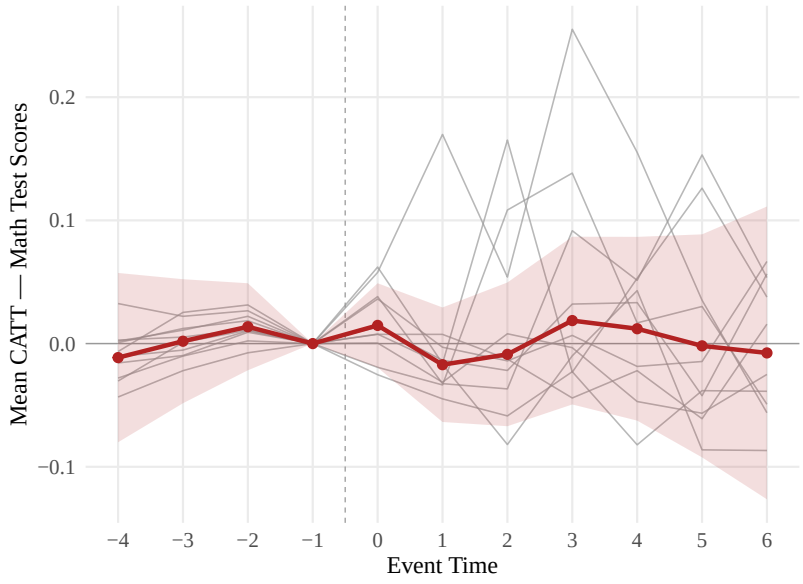


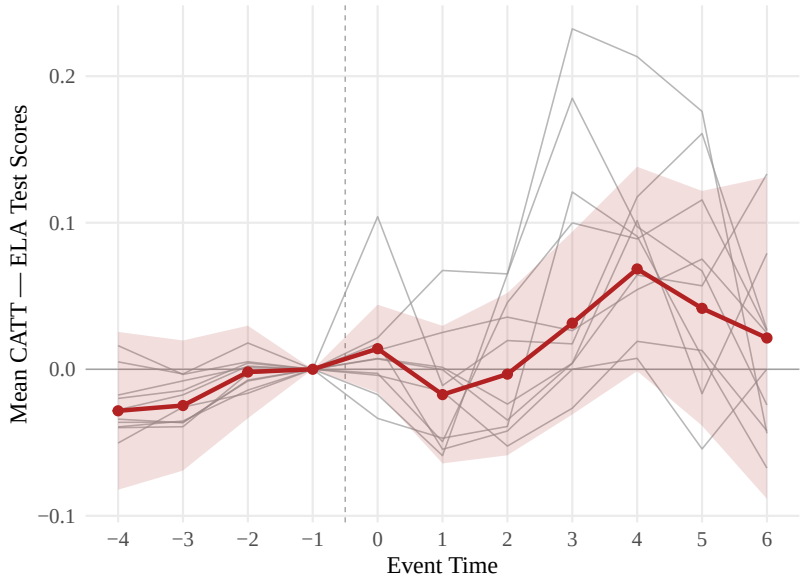
Figure C9. Math $\widehat{CATT}$ vs. ELA $\widehat{CATT}$ Correlation

Notes: This figure reports a binscatter plot of $CATT$ estimates for Math and ELA test scores within district. We also display the linear fit line and the correlation coefficient.

C.2 k-Means Matched Neighborhoods



(a) Math Scores



(b) ELA Scores

Figure C10. DiDCF K-Means Matched Event Studies (CATT): Math and ELA

Notes: Each panel shows $\widehat{GATT}^{es}(e, \mathcal{A})$ estimates (Equation 30). These are group average treatment effects in each event-time, where the groups are districts in each of the 10 k-means clusters. Each gray line is a single cluster, the red line is the overall event study line.

Table C8. DiDCF Differences-in-Means (k-means matched groups): Math Scores

	Covariate Grouping									
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
Baseline Performance	-0.68	-1.86	-0.83	-1.19	-0.72	1.41	-0.36	0.38	0.19	0.61
Free/Reduced Lunch (%)	0.64	0.81	0.49	0.70	0.65	0.19	0.65	0.42	0.40	0.31
Black (%)	0.32	0.74	0.03	0.07	0.47	0.05	0.06	0.17	0.03	0.04
Student-Teacher Ratio	13.17	14.46	18.42	21.92	14.45	15.24	15.64	14.92	14.19	19.93
Log of Enrollment	8.65	8.21	7.50	9.24	9.34	8.63	9.26	9.47	7.48	8.79
Hispanic (%)	0.34	0.05	0.16	0.67	0.10	0.06	0.71	0.16	0.05	0.12
Total Spending (per-pupil)	23236.39	12763.07	10336.27	10976.54	11543.54	13710.96	10271.01	10375.80	12806.54	10516.98
White (%)	0.27	0.17	0.75	0.19	0.39	0.82	0.19	0.63	0.89	0.76
Special Ed (%)	0.16	0.14	0.12	0.10	0.14	0.13	0.10	0.11	0.15	0.12
TPS-Charter Spending (% diff)	0.30	0.11	0.59	0.10	0.14	0.20	0.04	0.11	0.16	0.40
Per Pupil Revenue	23000.65	12292.82	10404.56	10653.91	11596.50	13385.77	9628.86	10091.90	13142.41	10437.44
Teacher Salary	130718.60	94376.71	94000.02	120905.76	88405.78	98787.16	83677.90	84147.72	84027.14	102056.40
Exempt from State/District Regs	0.42	0.70	0.66	0.78	0.69	0.56	0.71	0.67	0.55	0.57
Clear Charter Renewal Policies	0.57	0.68	0.65	0.75	0.71	0.52	0.58	0.65	0.49	0.70
Equitable Funding	0.49	0.45	0.52	0.91	0.52	0.38	0.60	0.45	0.27	0.69
No Caps on CS Growth	0.87	0.92	0.86	0.78	0.91	0.78	0.75	0.85	0.72	0.81
Town	0.00	0.37	0.31	0.17	0.21	0.11	0.17	0.09	0.30	0.21
Charter Spending (per-pupil)	13994.86	12204.52	8151.90	11998.39	10853.28	11420.24	9802.15	9328.18	11814.67	8440.17
Transparent Charter Startup Policies	0.53	0.70	0.55	0.50	0.66	0.51	0.52	0.52	0.44	0.53
Performance-Based Contracts	0.53	0.64	0.50	0.28	0.64	0.57	0.67	0.60	0.57	0.51
Suburb	0.63	0.24	0.16	0.44	0.24	0.62	0.30	0.43	0.21	0.47
Rural	0.13	0.30	0.41	0.12	0.20	0.16	0.12	0.26	0.47	0.07
Urban	0.24	0.10	0.12	0.27	0.35	0.11	0.41	0.21	0.02	0.25
Magnet Schools (%)	0.65	0.46	0.02	0.24	1.11	0.02	0.00	0.35	0.01	0.33
University in City	0.31	-0.02	0.00	-0.02	0.02	0.00	0.00	-0.01	0.00	0.03
City Population (standardized)	0.10	-0.03	0.00	-0.04	0.03	0.00	0.00	-0.02	0.00	0.01
ATT	0.08	0.06	0.04	0.02	0.01	-0.00	-0.02	-0.02	-0.03	-0.05
ACRT	0.19	0.32	0.38	0.39	-0.03	0.37	0.23	0.03	0.21	0.31

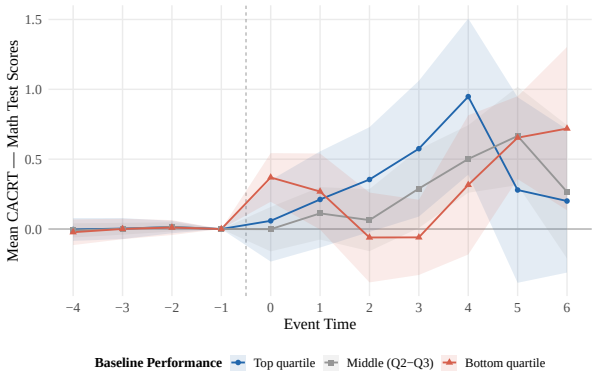
Notes: This table reports the within-group means for each of 10 covariate neighborhoods assigned through VIF-weighted k-means clustering. Rows are sorted by VIF in descending order, columns are sorted by within-group ATT, then by within-group $ACRT$ so that group (1) represents the best performing districts and group (10) are the worst performing districts.

Table C9. DiDCF Differences-in-Means (k-means matched groups): ELA Scores

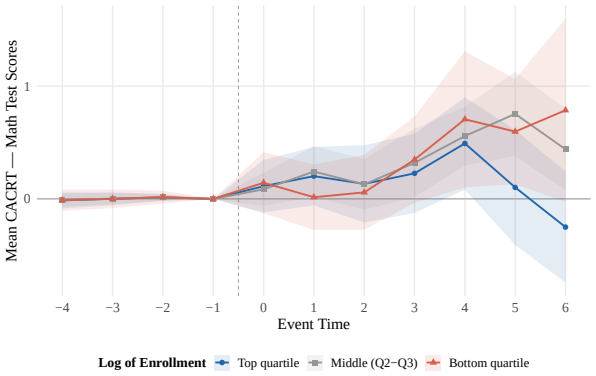
	Covariate Grouping									
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
Baseline Performance	-0.80	-1.64	1.20	-1.44	-0.90	-0.69	0.09	0.59	0.55	-0.40
Free/Reduced Lunch (%)	0.64	0.79	0.19	0.74	0.64	0.50	0.42	0.35	0.32	0.58
Student-Teacher Ratio	12.84	14.33	14.67	21.96	15.88	18.57	13.89	21.32	17.53	14.65
Log of Enrollment	8.60	8.38	8.79	9.32	9.20	7.66	7.29	9.28	8.49	9.62
Black (%)	0.31	0.69	0.07	0.06	0.06	0.04	0.03	0.06	0.04	0.38
Hispanic (%)	0.35	0.06	0.07	0.70	0.68	0.16	0.06	0.27	0.07	0.12
White (%)	0.28	0.20	0.79	0.17	0.21	0.75	0.87	0.52	0.86	0.47
Total Spending (per-pupil)	23652.06	12513.45	14253.22	10888.53	10279.50	10580.29	12320.37	11524.07	10371.49	11082.58
Teacher Salary	129084.33	92837.61	99519.33	123620.81	83535.83	95544.28	79037.68	117136.45	91415.81	87332.28
Special Ed (%)	0.16	0.14	0.13	0.10	0.10	0.13	0.14	0.11	0.13	0.13
TPS-Charter Spending (% diff)	0.31	0.13	0.13	0.11	0.04	0.58	0.17	0.71	0.21	0.12
Per Pupil Revenue	23432.27	12128.30	13704.25	10836.85	9700.59	10562.54	12611.07	10635.49	10616.23	11184.93
Exempt from State/District Regs	0.41	0.70	0.58	0.78	0.73	0.64	0.57	0.69	0.55	0.69
Clear Charter Renewal Policies	0.56	0.70	0.53	0.75	0.60	0.65	0.48	0.69	0.67	0.68
Equitable Funding	0.48	0.44	0.37	0.92	0.59	0.54	0.25	0.77	0.55	0.51
Transparent Charter Startup Policies	0.53	0.70	0.55	0.50	0.51	0.55	0.42	0.55	0.50	0.60
No Caps on CS Growth	0.88	0.92	0.75	0.78	0.76	0.85	0.73	0.78	0.82	0.90
Performance-Based Contracts	0.55	0.63	0.56	0.28	0.66	0.51	0.57	0.31	0.61	0.62
Charter Spending (per-pupil)	13758.42	11750.26	12369.59	12105.31	9759.85	8152.96	11411.80	8505.22	9038.63	10528.67
Town	0.00	0.36	0.06	0.17	0.19	0.33	0.33	0.11	0.22	0.13
Suburb	0.66	0.20	0.69	0.43	0.29	0.18	0.08	0.42	0.49	0.33
Rural	0.16	0.33	0.11	0.12	0.14	0.36	0.58	0.04	0.15	0.20
Urban	0.19	0.12	0.13	0.27	0.37	0.13	0.00	0.43	0.14	0.34
Magnet Schools (%)	0.69	0.37	0.02	0.23	0.00	0.03	0.01	0.69	0.04	1.19
University in City	0.00	0.00	0.00	-0.02	0.00	0.00	0.00	0.27	0.00	-0.01
City Population (standardized)	0.00	-0.01	0.00	-0.05	0.00	0.00	0.00	0.09	0.00	0.00
ATT	0.08	0.07	0.05	0.03	0.03	0.03	0.01	-0.02	-0.02	-0.02
ACRT	0.36	0.25	-0.02	0.55	0.36	0.21	0.12	0.06	0.03	-0.18

Notes: This table reports the within-group means for each of 10 covariate neighborhoods assigned through VIF-weighted k-means clustering. Rows are sorted by VIF in descending order, columns are sorted by within-group ATT, then by within-group $ACRT$ so that group (1) represents the best performing districts and group (10) are the worst performing districts.

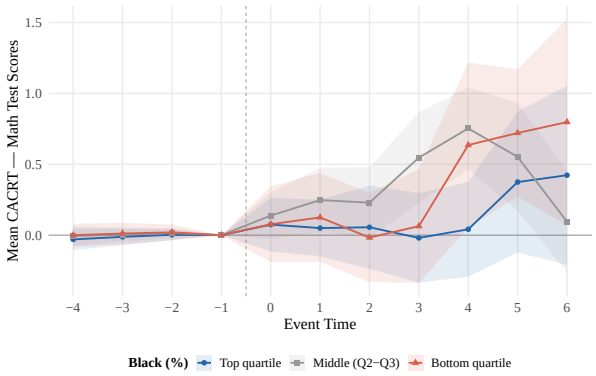
C.3 CACRT Event Studies



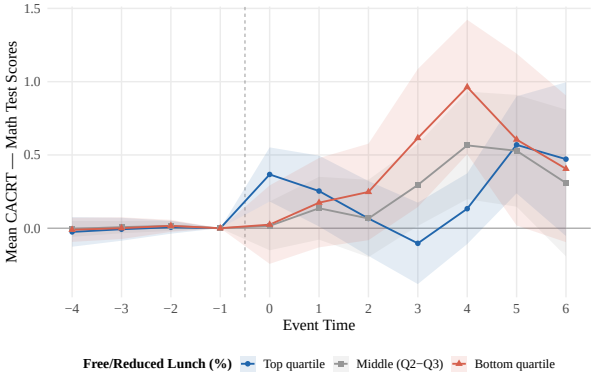
(a) Baseline Math Scores



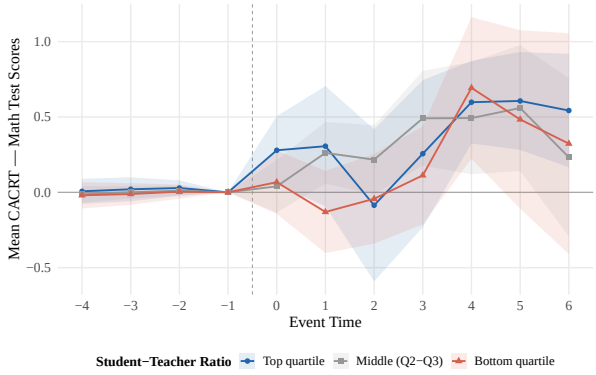
(b) Log Enrollment



(c) Black (%)



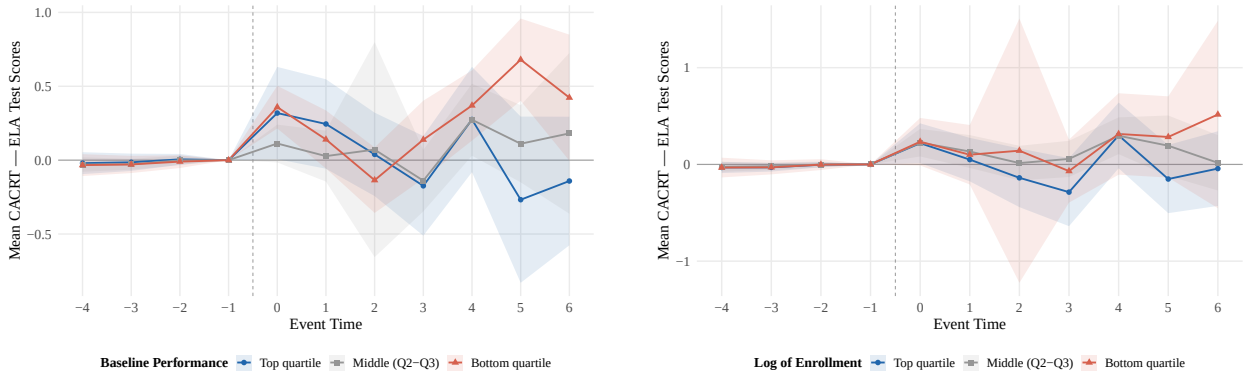
(d) Free/Reduced Price Lunch (%)



(e) Student-Teacher Ratio

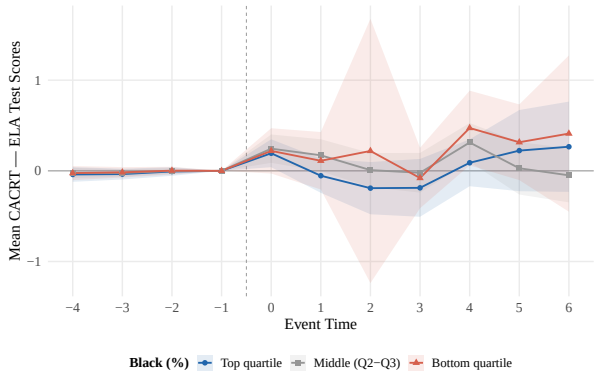
Figure C11. DiDCF Quartile Event Studies (CACRT): Math Scores

Notes: Each panel shows $\widehat{GACRT}^{es}(e, \mathcal{A})$ estimates (Equation 30). These are group average causal responses in each event-time, where the groups are districts in the top quartile (blue), middle two quartiles (grey), and bottom quartile (red) of the named moderator. Panels correspond to the five highest-ranked covariates by VIF at $e = 0$ for math scores.

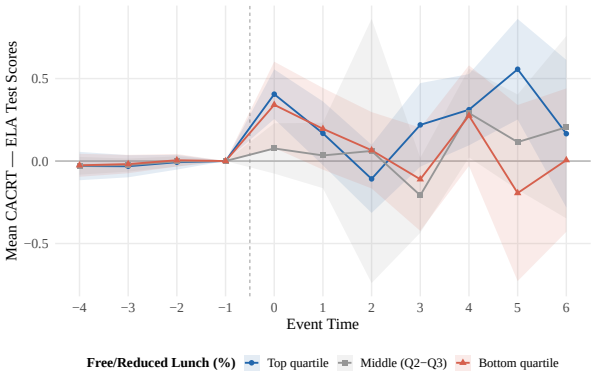


(a) Baseline ELA Scores

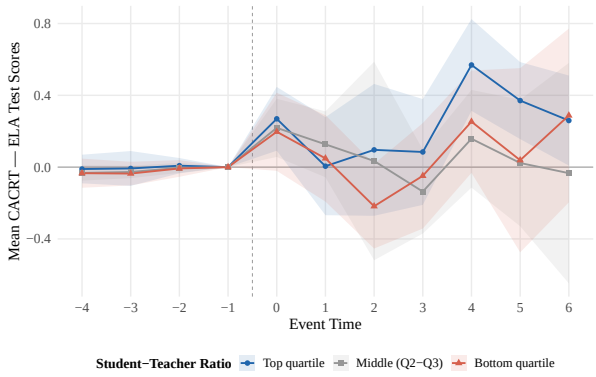
(b) Log Enrollment



(c) Black (%)



(d) Free/Reduced Price Lunch (%)

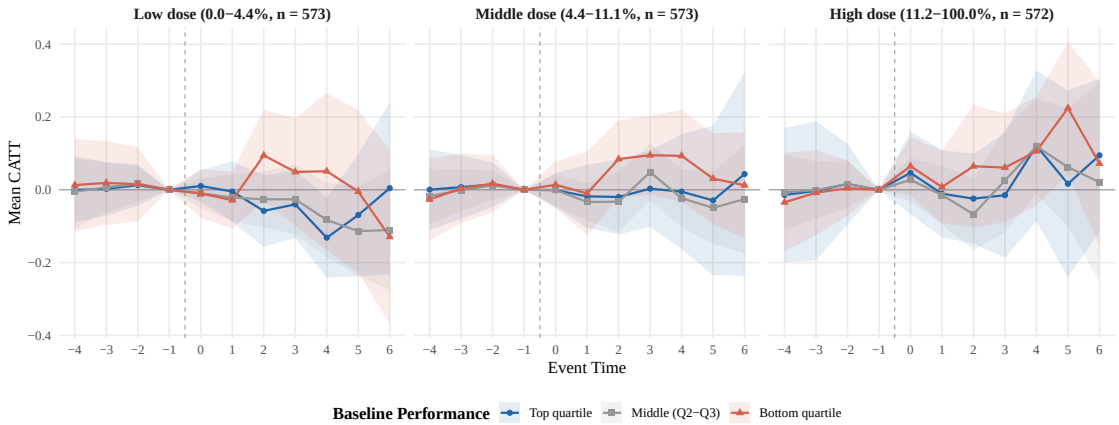


(e) Student-Teacher Ratio

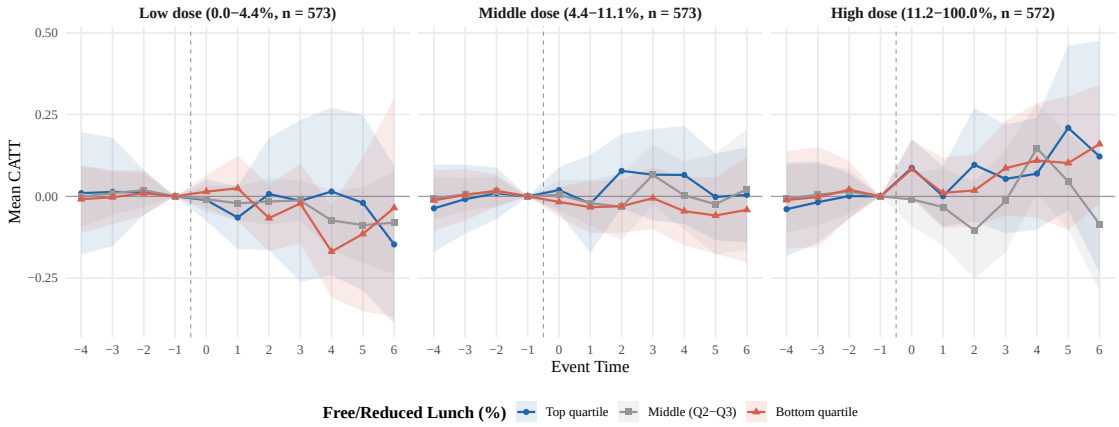
Figure C12. DiDCF Quartile Event Studies (CACRT): ELA Scores

Notes: Each panel shows $\widehat{GACRT}^{es}(e, \mathcal{A})$ estimates (Equation 30). These are group average causal responses in each event-time, where the groups are districts in the top quartile (blue), middle two quartiles (grey), and bottom quartile (red) of the named moderator. Panels correspond to the five highest-ranked covariates by VIF at $e = 0$ for ELA scores.

C.4 Event Studies within Dose Groups



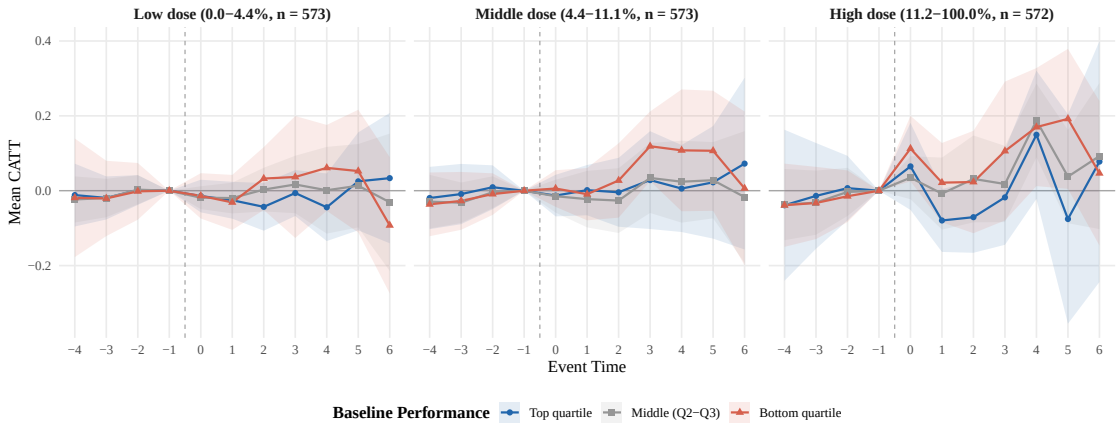
(a) Baseline Math Scores



(b) Free/Reduced Price Lunch (%)

Figure C13. DiDCF Quartile Event Studies (CATT) by Dose Groups: Math Scores

Notes: Each panel shows $\widehat{GATT}^{es}(e, \mathcal{A})$ estimates (Equation 30) for Math test scores. These are group average treatment effects in each event-time, where the groups are districts in the top quartile (blue), middle two quartiles (grey), and bottom quartile (red) of the named moderator. Panels correspond to dose tertile group, e.g. "low dose" corresponds to the bottom 33% of districts by charter share. The top row is for the Baseline Math score and the bottom row is for the FRL share.



(a) Baseline ELA Scores



(b) Free/Reduced Price Lunch (%)

Figure C14. DiDCF Quartile Event Studies (CATT) by Dose Groups: ELA Scores

Notes: Each panel shows $\widehat{GATT}^{es}(e, \mathcal{A})$ estimates (Equation 30) for ELA test scores. These are group average treatment effects in each event-time, where the groups are districts in the top quartile (blue), middle two quartiles (grey), and bottom quartile (red) of the named moderator. Panels correspond to dose tertial group, e.g. "low dose" corresponds to the bottom 33% of districts by charter share. The top row is for the Baseline ELA score and the bottom row is for the FRL share.

D Proofs

Proof of Lemma 1

Proof. Recall Equation 1:

$$CATT(g,t,w,x \mid g,w,x) \equiv \mathbb{E}[Y_{i,t}(g,w) - Y_{i,t}(0) \mid G_i = g,\ W_i = w,\ X_i = x]$$

Add and subtract the base period untreated outcome inside Equation 1:

$$\begin{aligned} CATT(g,t,w,x \mid g,w,x) = \mathbb{E}\big[Y_{i,t}(g,w) - Y_{i,b_g}(0) \mid G_i = g,\ W_i = w,\ X_i = x\big] \\ - \mathbb{E}\big[Y_{i,t}(0) - Y_{i,b_g}(0) \mid G_i = g,\ W_i = w,\ X_i = x\big] \end{aligned}$$

The second term is the counterfactual untreated trend of cohort g. Under Assumption 5, this is equal to the observed trend of never-treated districts with covariates x:

$$\begin{aligned} &\mathbb{E}\big[Y_{i,t}(0) - Y_{i,b_g}(0) \mid G_i = g,\ W_i = w,\ X_i = x\big] \\ &\quad = \mathbb{E}\big[Y_{i,t}(0) - Y_{i,b_g}(0) \mid G_i = \infty,\ W_i = 0,\ X_i = x\big] \end{aligned}$$

From here, we can replace potential outcomes with observed outcomes. For cohort-g districts, consistency gives $Y_{i,t} = Y_{i,t}(g,w)$ in the post-period, and because $b_g < g$, Assumption 3 ensures $Y_{i,b_g} = Y_{i,b_g}(g,w) = Y_{i,b_g}(0)$. For never-treated districts, $Y_{i,t} = Y_{i,t}(\infty,0) \equiv Y_{i,t}(0)$ in every period. Substituting:

$$\begin{aligned} CATT(g,t,w,x \mid g,w,x) = \mathbb{E}\big[Y_{i,t} - Y_{i,b_g} \mid G_i = g,\ W_i = w,\ X_i = x\big] \\ - \mathbb{E}\big[Y_{i,t} - Y_{i,b_g} \mid G_i = \infty,\ W_i = 0,\ X_i = x\big] \end{aligned}$$

which is the statement of Lemma 1.

Proof of Lemma 2

Proof. Recall Equation 2:

$$CACRT(g,t,w,x \mid g,w,x) \equiv \frac{\partial}{\partial l}\,\mathbb{E}[Y_{i,t}(g,l) \mid G_i = g,\ W_i = w,\ X_i = x]\Big|_{l=w}$$

Add and subtract the untreated base-period outcome $Y_{i,b_g}(0)$ inside the expectation. It does not depend on l:

$$\begin{aligned} \mathbb{E}[Y_{i,t}(g,l) \mid G_i = g,\ W_i = w,\ X_i = x] = \mathbb{E}\big[Y_{i,t}(g,l) - Y_{i,b_g}(0) \mid G_i = g,\ W_i = w,\ X_i = x\big] \\ + \mathbb{E}\big[Y_{i,b_g}(0) \mid G_i = g,\ W_i = w,\ X_i = x\big] \end{aligned}$$

The first term is the trend that dose group w would have followed under dose l. Under Assumption 6(a), this is equal to the trend of the districts that actually received dose l:

$$\begin{aligned} &\mathbb{E}\big[Y_{i,t}(g,l) - Y_{i,b_g}(0) \mid G_i = g,\ W_i = w,\ X_i = x\big] \\ &\quad = \mathbb{E}\big[Y_{i,t}(g,l) - Y_{i,b_g}(0) \mid G_i = g,\ W_i = l,\ X_i = x\big] \end{aligned}$$

From here, we can replace potential outcomes with observed outcomes. For cohort-g districts with dose l, consistency gives $Y_{i,t} = Y_{i,t}(g,l)$ in the post-period, and Assumption 3 ensures $Y_{i,b_g} = Y_{i,b_g}(g,l) =$

$Y_{i,b_g}(0)$. Substituting:

$$\mathbb{E}[Y_{i,t}(g,l) \mid G_i = g,\ W_i = w,\ X_i = x] = \mathbb{E}\big[Y_{i,t} - Y_{i,b_g} \mid G_i = g,\ W_i = l,\ X_i = x\big] + \mathbb{E}\big[Y_{i,b_g}(0) \mid G_i = g,\ W_i = w,\ X_i = x\big]$$

Differentiating with respect to l at $l = w$, the second term drops out and

$$CACRT(g,t,w,x \mid g,w,x) = \frac{\partial}{\partial l}\,\mathbb{E}\big[Y_{i,t} - Y_{i,b_g} \mid G_i = g,\ W_i = l,\ X_i = x\big]\Big|_{l=w}$$

which is the statement of Lemma 2. This is the conditional analogue of the second part of Theorem C.1 in Callaway et al. (2025).

Proof of Lemma 3

Proof. Add and subtract $Y_{i,b_g}(0)$ inside Equation 12. The first term, $\mathbb{E}[Y_{i,t}(g, \mathbf{W}_{i,g:t}) - Y_{i,b_g}(0) \mid G_i = g,\ W_{i,t} = w,\ X_i = x]$, equals $\mathbb{E}[Y_{i,t} - Y_{i,b_g} \mid G_i = g,\ W_{i,t} = w,\ X_i = x]$ because $Y_{i,t} = Y_{i,t}(G_i, \mathbf{W}_{i,G_i:t})$ for $t \geq G_i$ and $Y_{i,b_g} = Y_{i,b_g}(0)$ by Assumption 3. The second term, $\mathbb{E}[Y_{i,t}(0) - Y_{i,b_g}(0) \mid G_i = g,\ W_{i,t} = w,\ X_i = x]$, is the never-treated trend by Assumption 5, where $Y_{i,t} = Y_{i,t}(0)$ for all t.

Proof of Lemma 4

Proof. Denote $\mathbf{h} = \mathbf{w}_{g:t-1}$ and drop $G_i = g$ and $X_i = x$ from the notation for simplicity. The base-period outcome $Y_{i,b_g}(0)$ does not depend on the current dose l (Assumption 3), so subtracting it does not change the derivative in Equation 15:

$$CACRT^{path}(g,t,\mathbf{w}_{g:t},x \mid g,\mathbf{w}_{g:t},x) = \frac{\partial}{\partial l}\,\mathbb{E}[Y_{i,t}(g,(\mathbf{h},l)) - Y_{i,b_g}(0) \mid \mathbf{W}_{i,g:t} = (\mathbf{h},w_t)]\Big|_{l=w_t}$$

By Assumption 7(a), this expected change equals the expected change of districts that actually followed $(\mathbf{h}, l)$, for whom it is observed:

$$\mathbb{E}[Y_{i,t}(g,(\mathbf{h},l)) - Y_{i,b_g}(0) \mid \mathbf{W}_{i,g:t} = (\mathbf{h},w_t)] = \mathbb{E}[Y_{i,t} - Y_{i,b_g} \mid \mathbf{W}_{i,g:t-1} = \mathbf{h},\ W_{i,t} = l]$$

Differentiating at $l = w_t$ gives Equation 17.

Derivation of Equation 18

Proof. Drop $G_i = g$ and $X_i = x$ for simplicity. Denote $\Delta Y_{i,g,t} = Y_{i,t} - Y_{i,b_g}$, and assume the expectations below are differentiable in the dose, with derivatives that can be taken inside expectations over histories. By Lemma 3, the never-treated term does not depend on l, so $\partial\, CATT^{path}(g,t,l,x \mid g,l,x)/\partial l =$

$\partial\,\mathbb{E}[\Delta Y_{i,g,t} \mid W_{i,t} = l]/\partial l$. By iterated expectations (first line), then by the chain rule (line 2):

$$\begin{aligned}\frac{\partial}{\partial l}\,\mathbb{E}[\Delta Y_{i,g,t} \mid W_{i,t} = l] &= \frac{d}{dl}\,\mathbb{E}\Big[\mathbb{E}[\Delta Y_{i,g,t} \mid \mathbf{W}_{i,g:t-1},\, W_{i,t} = l] \;\Big|\; W_{i,t} = l\Big] \\ &= \mathbb{E}\left[\frac{\partial\,\mathbb{E}[\Delta Y_{i,g,t} \mid \mathbf{W}_{i,g:t-1},\, W_{i,t} = l]}{\partial l} \;\middle|\; W_{i,t} = l\right] \\ &\quad + \frac{\partial}{\partial l'}\,\mathbb{E}\Big[\mathbb{E}[\Delta Y_{i,g,t} \mid \mathbf{W}_{i,g:t-1},\, W_{i,t} = l] \;\Big|\; W_{i,t} = l'\Big]\Big|_{l'=l}\end{aligned}$$

For the first term, Lemma 4 (first equality) and Equation 16 (second equality) give:

$$\begin{aligned}\mathbb{E}\left[\frac{\partial\,\mathbb{E}[\Delta Y_{i,g,t} \mid \mathbf{W}_{i,g:t-1},\, W_{i,t} = l]}{\partial l} \;\middle|\; W_{i,t} = l\right] &= \mathbb{E}\Big[CACRT^{path}(g,t,\mathbf{W}_{i,g:t},x \mid g,\mathbf{W}_{i,g:t},x) \;\Big|\; W_{i,t} = l\Big] \\ &= CACRT^{path}(g,t,l,x \mid g,l,x)\end{aligned}$$

The second term asks "if we keep the current dose fixed and vary only the dose history, how does the average $\Delta Y_{i,g,t}$ change?"

Consider a path $\mathbf{w}_{g:t}$ with history $\mathbf{w}_{g:t-1}$. Then

$$\begin{aligned}\mathbb{E}[\Delta Y_{i,g,t} \mid \mathbf{W}_{i,g:t} = \mathbf{w}_{g:t}] &= \mathbb{E}[Y_{i,t}(g,\mathbf{w}_{g:t}) - Y_{i,b_g} \mid \mathbf{W}_{i,g:t} = \mathbf{w}_{g:t}] \\ &= \mathbb{E}[Y_{i,t}(g,\mathbf{w}_{g:t}) - Y_{i,b_g}(0) \mid \mathbf{W}_{i,g:t} = \mathbf{w}_{g:t}] \\ &= \mathbb{E}[Y_{i,t}(g,\mathbf{w}_{g:t}) - Y_{i,b_g}(0)]\end{aligned}$$

The third line holds because Assumption 7(a) makes $\mathbb{E}[Y_{i,t}(g,\mathbf{w}_{g:t}) - Y_{i,b_g}(0) \mid \mathbf{W}_{i,g:t} = \mathbf{l}_{g:t}]$ the same for every $\mathbf{l}_{g:t}$, so averaging over the distribution of $\mathbf{W}_{i,g:t}$ returns that value. Under Assumption 8, $\mathbb{E}[Y_{i,t}(g,\mathbf{w}_{g:t})]$ depends on $\mathbf{w}_{g:t}$ only through w_t, and $\mathbb{E}[Y_{i,b_g}(0)]$ does not depend on $\mathbf{w}_{g:t}$. Hence $\mathbb{E}[\Delta Y_{i,g,t} \mid \mathbf{W}_{i,g:t-1},\, W_{i,t} = l]$ is the same value for every history $\mathbf{W}_{i,g:t-1}$, its average over the histories of districts with $W_{i,t} = l'$ does not depend on l', so the second term is zero.

Proof of Lemma 5

Proof. Within $\mathcal{S}_g$, lm_forest solves the local moment condition

$$\mathbb{E}\Big[(1,\mathbf{D}_i')'\big(\Delta Y_{i,g,t} - m_{g,t}(x) - \mathbf{D}_i'\boldsymbol{\beta}_{g,t}(x)\big) \;\Big|\; X_i = x\Big] = 0.$$

This moment identifies the parameters in two steps:

1. For never-treated districts $\mathbf{D}_i = (0,0)'$, so the intercept is $m_{g,t}(x) = \mathbb{E}[\Delta Y_{i,g,t} \mid G_i = \infty,\ X_i = x]$.
2. For cohort-g districts, $\mathbb{E}[\Delta Y_{i,g,t} \mid G_i = g,\ W_i = w,\ X_i = x] - m_{g,t}(x) = \alpha_{g,t}(x) + \tau_{g,t}(x)\,w = CATT(g,t,w,x \mid g,w,x)$, by Lemma 1 and Assumption 10.

Consistency and asymptotic normality follow from Theorems 3 and 5 of Athey et al. (2019), applied to the conditional linear model of their Section 6 with treatment vector $\mathbf{D}_i$ and the linear combinations $(1,w)$ for $\widehat{CATT}(g,t,w,x)$ or $(0,1)$ for $\hat{\tau}_{g,t}(x)$. These results require the conditions of Assumption 9: $\mathbb{E}[\Delta Y_{i,g,t} \mid X_i = x]$, $\mathbb{E}[\mathbf{D}_i \mid X_i = x]$, $\mathrm{Cov}(\Delta Y_{i,g,t}, \mathbf{D}_i \mid X_i = x)$, and $\mathrm{Var}(\mathbf{D}_i \mid X_i = x)$ Lipschitz in x, and $\mathrm{Var}(\mathbf{D}_i \mid X_i = x)$ must be invertible. The convergence is slower than $\sqrt{N}$: with trees grown on subsamples of size $s = N^{\beta}$, their Theorem 5 provides a standard deviation of order $\sqrt{s/N}$ up to a polylogarithmic factor, so the estimates converge at rate $N^{-(1-\beta)/2}$, where $N = |\mathcal{S}_g|$ and $\beta < 1$ is bounded below by a constant that rises toward one as the number of covariates grows.

Now add Assumption 6. Under Assumption 5 alone, the $CATT$ is local to the districts that specifically chose dose w. Therefore, comparing fitted values across doses identifies the causal response *plus* the selection bias term of Equation 25. Assumption 6 ensures the selection bias term is zero. For any $l \in \mathcal{W}_+$, add and subtract the base period outcome:

$$\begin{aligned} CATT(g,t,w,x \mid g,l,x) = \mathbb{E}[Y_{i,t}(g,w) - Y_{i,b_g}(0) \mid G_i = g,\ W_i = l,\ X_i = x] \\ - \mathbb{E}[Y_{i,t}(0) - Y_{i,b_g}(0) \mid G_i = g,\ W_i = l,\ X_i = x] \end{aligned}$$

Under Assumption 6(a) the first term does not depend on l, and under Assumption 5 the second equals the never-treated trend, which does not depend on l either. Hence $CATT(g,t,w,x \mid g,l,x) = CATT(g,t,w,x \mid g,w,x) = CATT(g,t,w,x)$. Lemma 2 identifies the derivative of this function at w as $CACRT(g,t,w,x)$, and under Assumption 10 that derivative is $\tau_{g,t}(x)$. The asymptotic normality of $\hat{\tau}_{g,t}(x)$ established above therefore holds for $\widehat{CACRT}(g,t,w,x) = \hat{\tau}_{g,t}(x)$.

Proof of Lemma 6

Proof. From Equation A1, we know:

$$\operatorname{plim} \hat{\tau}(x) = \frac{\operatorname{Cov}(\Delta\widetilde{Y}_i, W_i \mid X_i)}{\operatorname{Var}(W_i \mid X_i)}$$

Plug Equation A2 into $\operatorname{Cov}(\Delta\widetilde{Y}_i, W_i \mid X_i)$. Noting that in order to move from the second line to the third, we notice that within x, $\alpha(x)$ and $\tau(x)$ are scalars. Therefore $\operatorname{Cov}\Big(\alpha(x), W_i \mid X_i\Big) = 0$ and $\operatorname{Cov}\Big(\tau(x)W_i, W_i \mid X_i\Big) = \tau(x)\operatorname{Cov}\Big(W_i, W_i \mid X_i\Big) = \tau(x)\operatorname{Var}(W_i \mid X_i)$. Also, $\operatorname{Cov}(\varepsilon, W_i \mid X_i) = 0$ because $\mathbb{E}[\varepsilon_i \mid W_i, X_i] = 0$. Going forward, we drop subscripts to avoid clutter:

$$\begin{aligned} \operatorname{Cov}(\Delta\widetilde{Y}, W \mid X) &= \operatorname{Cov}\Big(\alpha(x) + \tau(x)W + \delta + \varepsilon, W \mid X\Big) \\ &= \operatorname{Cov}\Big(\alpha(x), W \mid X\Big) + \operatorname{Cov}\Big(\tau(x)W, W \mid X\Big) + \operatorname{Cov}\Big(\delta, W \mid X\Big) + \operatorname{Cov}\Big(\varepsilon, W \mid X\Big) \\ &= \tau(x)\operatorname{Var}\Big(W \mid X\Big) + \operatorname{Cov}\Big(\delta, W \mid X\Big) \end{aligned}$$

Divide this result by $\operatorname{Var}(W \mid X)$:

$$\begin{aligned} \operatorname{plim} \hat{\tau}(x) &= \frac{\tau(x)\operatorname{Var}(W \mid X) + \operatorname{Cov}(\delta, W \mid X)}{\operatorname{Var}(W \mid X)} \\ &= \tau(x) + \underbrace{\frac{\operatorname{Cov}(\delta, W \mid X)}{\operatorname{Var}(W \mid X)}}_{=0} \\ &= \tau(x) \end{aligned}$$

For the second half of Lemma 6, note that the correct model would be on the true DGP (i.e. the expectation of Equation A2). This value is $\mathbb{E}[\Delta\widetilde{Y} \mid X] = \mathbb{E}[\alpha(x) \mid X] + \mathbb{E}[\tau(x)W \mid X] + \mathbb{E}[\delta \mid X] + \mathbb{E}[\varepsilon \mid X] = \alpha(x) + \mathbb{E}[\tau(x)W \mid X] + \mathbb{E}[\delta \mid X]$. The bias between our DiDCF CATT and the true CATT is therefore:

$$\operatorname{plim} \hat{\alpha}(x) - \alpha(x) = \mathbb{E}[\delta \mid X] - \big(\operatorname{plim} \hat{\tau}(x) - \tau(x)\big)\mathbb{E}[W \mid X]$$

The first part of this proof showed that $\operatorname{plim} \hat{\tau}(x) - \tau(x) = 0$, so $\operatorname{plim} \hat{\alpha}(x) - \alpha(x) = \mathbb{E}[\delta \mid X]$, which is the second claim in Lemma 6.

E Technical Summary of GRF Estimation Procedure

A generalized random forest grows B causal trees indexed by $b = 1, ..., B$. Each tree is grown by the following steps:

1. From the full data sample, draw a random subsample without replacement.
2. Further divide the subsample into a training sample, $\mathcal{J}_1$, and an estimation sample, $\mathcal{J}_2$, which are non-overlapping partitions.
3. With training sample $\mathcal{J}_1$, perform greedy and recursive splitting by grid searching over the covariate space and selecting the single covariate, $j \in \{1, ..., k\}$, and value, c, that maximizes the squared difference in treatment effects across the resulting child nodes. The objective function is given as:

$$\max_{j,c} \quad |C_1||C_2|(\hat{\tau}_{C_1} - \hat{\tau}_{C_2})^2$$

Where C_1 and C_2 denote the resulting child nodes such that $|C_1|$ and $|C_2|$ are the number of units in each resulting child node and $\hat{\tau}_{C_1}$ and $\hat{\tau}_{C_2}$ are the resulting ATEs within each child node. For computational tractability (i.e. in order to avoid estimating linear regressions for each potential split), Athey et al. (2019) show that this process reduces to splitting on a pseudo-outcome corresponding to a first-order gradient-based approximation of the subsample ATEs.

   (a) Denote a parent node, P, as any node that is currently being assessed for an optimal split. Compute $\hat{\alpha}_P$ and $\hat{\tau}_P$ by solving Equation 23, which can be alternatively expressed as:

$$\sum_{i=1}^{|P|}(1, W_i)'(Y_i - \hat{\alpha}_P - \hat{\tau}_P W_i) = 0$$

   (b) Also compute an estimate of the gradient of the expectation of $\psi_{\alpha,\tau}(Y_i, W_i) = (1, W_i)'(Y_i - \alpha - \tau W_i)$, denoted $A_P = \nabla\mathbb{E}[\psi_{\hat{\alpha}_P,\hat{\tau}_P}|X_i \in P]$:

$$A_P = \frac{1}{|\{i : X_i \in P\}|}\sum_{i:X_i \in P} \nabla\psi_{\hat{\alpha}_P,\hat{\tau}_P}(Y_i, W_i)$$

   (c) Compute the pseudo-outcomes as:

$$\rho_i = -\xi' A_P^{-1}\psi_{\hat{\alpha}_P,\hat{\tau}_P}$$

   Where ξ is simply a vector that picks out the τ coordinate from the (α, τ) vector.

   (d) Finally, use the pseudo-outcome in a standard classification and decision tree (CART), which just selects the split that maximizes the sum of the average influence functions between the

resulting child nodes:

$$\widetilde{\Delta}(C_1, C_2) = \sum_{q=1}^{2} \frac{1}{|\{i : X_i \in C_q\}|} \Big(\sum_{\{i: X_i \in C_q\}} \rho_i \Big)^2$$

Intuitively, the influence function captures how much the treatment effect in the child node is likely to differ from the treatment effect estimate in the parent node. In the simplest case of a randomized experiment where the above process refers to OLS-based identification (without a constant, for simplicity), this first-order approximation looks like:

$$\hat{\tau}_{C_q} \approx \hat{\tau}_P + \frac{1}{\sum_{i \in C_q} W_i^2} \sum_{i \in C_q} W_i (Y_i - W_i \hat{\tau}_P)$$

(e) With the two child nodes, repeat steps (a) – (d) on each, treating each node as a new parent node until a full tree, $\mathcal{T}$, is trained.

4. Within tree $\mathcal{T}$, collect all observations in the estimation subsample $\mathcal{J}_2$ that fall in the same leaf as x, $L_b(x)$. Compute the GRF-weights as the share of total trees where unit i falls into the same leaf as x:

$$\omega_i(x) = \frac{1}{B} \sum_{b=1}^{B} \frac{1}{|L_b(x)|} \mathbb{1}\{X_i \in L_b(x)\}$$